

Robust Predictions in Information Acquisition Games*

Tommaso Denti[†]
NYU Stern

Doron Ravid[‡]
University of Chicago

April 18, 2025

Abstract

We derive robust predictions in games preceded by costly information acquisition, valid across all specifications of players' learning abilities. Relative to environments with exogenously given information, costly learning restricts behavior and reduces welfare. These behavioral restrictions exhibit a stark dichotomy: they are either highly stringent or virtually negligible. Stringent restrictions occur in structured environments characterized by global indifference, while in generic games, behavioral predictions under endogenous and exogenous information coincide. The welfare implications, however, differ systematically: optimal policy often depends on whether information is given or acquired. We illustrate the usefulness of our tools for robust policy analysis through applications to regime-change games, informational simplicity, and mechanism design with aftermarkets.

*We thank Alex Bloedel, Benjamin Brooks, Modibo Camara, Marina Halac, Emir Kamenica, Ilia Krasikov, Elliot Lipnowski, Andrew McClellan, Stephen Morris, Wolfgang Pesendorfer, Luciano Pomatto, Larry Samuelson, Philipp Strack, Balazs Szentes, Daniel Rappoport, Phil Reny, and Joe Root for helpful discussions.

[†]td838@nyu.edu, <https://sites.google.com/site/tommasojdenti/home>.

[‡]dravid@uchicago.edu, <http://doronravid.com>.

1. Introduction

Information acquisition is a fundamental economic activity: investors assess financial prospects, employers design interview processes, and voters choose among news sources. A central challenge in studying such environments is that the feasibility and costs of acquiring different pieces of information depend on hard-to-measure factors—such as time, effort, and cognitive resources. As a result, it is difficult to place confidence in any single model of the information acquisition process.

This challenge echoes longstanding difficulties in analyzing settings where agents are endowed with fixed information structures rather than actively choosing what to learn. As Wilson (1987) notes, economists have long struggled with the sensitivity of their conclusions to assumptions about agents’ beliefs. In response to this concern, Bergemann and Morris (2013, 2016) introduced the concept of *Bayes correlated equilibrium* (BCE), which spans the full set of equilibrium outcomes attainable in a simultaneous-move game across all possible *exogenous* information structures. The BCE concept enables predictions that are robust to the exact nature of agents’ beliefs, as any prediction that holds for all BCEs must remain valid regardless of the information available to the agents.¹

In this paper, we develop a framework for making robust predictions in environments where agents endogenously acquire information rather than receive it exogenously. Specifically, we study simultaneous-move games that are preceded by a stage in which players covertly gather costly information. Our approach takes the structure of the second stage as given and characterizes the set of equilibrium outcomes and associated welfare values that can arise under all possible specifications of the players’ information acquisition technologies. We show that the concept of BCE remains applicable for this purpose, provided two key modifications are introduced.

First, welfare predictions must account for the cost of information. When information is provided exogenously, agents’ welfare is simply measured by expected utility. However, when learning is costly, the relevant metric becomes *the net payoff*—expected utility minus the cost of acquiring information. The challenge, of course, is that the precise form of information costs is typically unobservable. We address this

¹Since its introduction, BCE has become a foundational tool in information economics—see Bergemann and Morris (2019) for a review of economic applications of BCE and its role in the broader literature on information design.

issue by characterizing the set of net payoffs that can arise across all cost functions consistent with a given equilibrium behavior.

Second, one must restrict attention to BCEs that are compatible with agents' learning incentives. To see why such a restriction is necessary, consider a single-agent decision problem under uncertainty in which the agent is indifferent among *all* actions across *all* states of nature. In the BCE framework, any correlation between the agent's action and the state can be rationalized by assuming full information is exogenously provided—meaning BCE imposes no constraint on behavior. However, if information is costly, the agent has no incentive to learn and will rationally choose to remain uninformed. Consequently, only behaviors in which the agent's action is independent of the state are compatible with their learning incentives.

We introduce refinements of BCE that incorporate incentive-compatible information acquisition and examine how these refinements compare to standard BCE. We show that the relationship between the two exhibits a stark dichotomy: the BCEs consistent with learning incentives are either *dense* or *nowhere dense* within the full BCE set. In other words, among the behaviors attainable under exogenous information, either virtually all are consistent with costly learning—or very few are.

This sharp dichotomy leads us to the question: When do learning incentives actually reduce the set of BCEs? We show that such reductions are governed by a specific pattern of global indifferences. In single-agent settings, our learning-based refinements significantly narrow the BCE set if and only if the agent has an action that is always optimal but not strictly dominant. In multi-agent games, the relevant indifferences take on a more intricate form, as they can arise endogenously from equilibrium behavior. To capture this broader phenomenon, we provide a general characterization based on Myerson's (1997) notion of *jeopardization*—a generalization of weak dominance tailored to the analysis of correlated equilibria.

It is not clear a priori whether the cases in which learning incentives restrict behavior are common or exceptional. On the one hand, global indifferences in single-agent settings are clearly knife-edge. On the other hand, it is well known that strategic interactions can generate global indifferences in an open set of games (e.g., Viossat, 2008). We resolve this question by showing that our refinements of BCE are generically powerless. Specifically, we prove that in a generic game, the set of behaviors consistent with costly information acquisition coincides with the set implementable under some exogenous information structure.

The aforementioned genericity result applies only to behavior. When one considers the *full* set of predictions—including players’ net payoffs—substantive differences arise between exogenous and acquired information in an open set of environments. We characterize these environments in two complementary ways. The first characterization returns Myerson’s (1997) notion of jeopardization, which emerges as a unifying theme throughout our analysis. Our second characterization considers the implications for welfare: predictions under costly learning differ from those under exogenous information if and only if the two information regimes produce different worst-case scenarios for some weighted-utilitarian social welfare function.

These general results clarify how specific assumptions about agents’ information-acquisition abilities shape model predictions. A few studies derive stark behavioral conclusions based on little more than the hypothesis that information is costly to acquire (e.g., Ravid, Roesler, and Szentes, 2022; Denti, 2024). Our analysis shows that these results arise from economically meaningful indifferences present in certain structured environments—such as bilateral trades or signaling games. However, in settings that lack such structure, the behavioral predictions of models with costly learning depend heavily on the specific assumptions imposed on agents’ information-acquisition technologies, as highlighted by our genericity result.

Our analysis also yields a new toolkit for robust policymaking and design. By precisely characterizing the range of behaviors and welfare values consistent with costly learning, we enable planners who control the payoff environment to design interventions and policies that deliver satisfactory results—regardless of agents’ information-acquisition abilities. To illustrate this, we examine three applications: a regime-change game (e.g., Morris and Shin, 1998), informationally simple mechanisms (Gleyze and Pernoud, 2023), and mechanism design with aftermarkets (e.g., Angeletos and Werning, 2006; Calzolari and Pavan, 2006; Dworczak, 2020).

Finally, our framework broadens the applicability of studies that use BCE to derive predictions that are robust to the nature of agents’ private information (e.g., Bergemann, Brooks, and Morris, 2017; Brooks and Du, 2021; Bergemann, Brooks, and Morris, 2022). In particular, we show that the robustness criteria in these studies continue to apply when information is endogenously acquired—provided the analysis focuses only on moments of the action-state distribution (such as seller revenue in an auction) and our learning-based refinements have no bite. Even when our refinements do restrict the BCE set, our genericity result implies that such restrictions should be

taken seriously only if the analyst has reason to believe that the environment exhibits non-generic features. For studies that focus on welfare outcomes (e.g., Bergemann, Brooks, and Morris, 2015), our framework offers a roadmap for assessing the validity of their conclusions in settings where information is costly to acquire.

Related literature. A few existing papers address specific robust design problems involving costly information acquisition. Carroll (2019a) studies a bilateral trade setting in which a designer seeks to maximize worst-case welfare, accounting for a variety of ways agents might influence the informational environment prior to trade—including information acquisition, signal jamming, costly signaling, and evidence disclosure. He shows optimal design is motivated by the possibility of “informational extortions.” Carroll (2019b) analyzes a principal-agent relationship where the principal offers a contract to incentivize the agent to acquire information. The principal is aware of some of the agent’s information-gathering actions but is concerned the agent may have access to other, unknown experiments. Pei and Strulovici (2025) propose a notion of local ex ante robust implementation in which agents must be incentivized to both acquire and report information truthfully. The designer’s robustness concerns span both the agents’ gross payoffs and their information costs.

A key distinction between our work and these papers is that they focus on design problems, where the principal can manipulate the payoff environment subject to incentive compatibility. In contrast, we develop a framework for deriving robust predictions in a fixed payoff environment. Nonetheless, because our framework characterizes the limits of behavior and welfare under costly learning, it also provides useful tools for robust design exercises of the kind considered in these papers. Our applications—to regime-change games, informationally simple mechanisms, and mechanism design with aftermarkets—illustrate this potential.

More broadly, our work contributes to the literature on information acquisition in games (e.g., Yang, 2015; Ravid, 2020; Morris and Yang, 2022; Denti, 2023). Within this literature, the closely related work of Denti (2024) is the first to consider the question of robust predictions. That paper studies a two-player signaling game in which the receiver must pay a cost to monitor the sender’s action, and shows that, under flexible learning and strictly monotone costs, off-path beliefs are irrelevant in equilibrium. It then characterizes the set of behaviors that can arise under some strictly monotone cost function—a result that corresponds to the single-player ver-

sion of our Theorem 1, part (ii), without welfare values. Our paper extends this analysis in several directions: it allows for multiple players, non-flexible learning, incorporates welfare considerations, and develops novel results on generic games and the comparison between endogenous and exogenous information. These extensions have no counterpart in Denti (2024).

Some of our results can be interpreted as revealed-preference tests for costly information acquisition in strategic settings. A number of papers examine the testable implications of costly learning in single-agent environments (e.g., Caplin and Dean, 2015; Caplin, Dean, and Leahy, 2022; Denti, 2022; Lipnowski and Ravid, 2022). Among these, the most closely related is Caplin and Dean (2015). There are three key differences between their work and ours. First, Caplin and Dean (2015) do not assume that information costs increase with informativeness and therefore do not offer a test specifically tailored to *costly* information acquisition. In particular, their framework does not distinguish between agents who are endowed with information and those who must incur a cost to obtain it. Second, we consider strategic environments rather than single-agent settings. This shift introduces two additional layers of complexity: (i) information acquisition may be correlated across players, and (ii) the value of information is determined endogenously, as it depends on the information and actions of others. Third, while Caplin and Dean (2015) examine comparative predictions across varying payoff environments, our focus is on a fixed base game. Extending our framework to accommodate variation across games is a promising direction for future work (see Section 8).²

Our work also contributes to the literature on the uses and foundations of correlated equilibrium concepts (e.g., Aumann, 1974; Myerson, 1982; Aumann, 1987; Brandenburger and Friedenberg, 2008; Bergemann and Morris, 2016; Doval and Ely, 2020). A closely related early contribution is Lipman and Srivastava (1990), who study games without payoff uncertainty and examine which correlated equilibria can be achieved by allowing players to choose what information to acquire about a common, payoff-irrelevant state. As in Caplin and Dean (2015), their framework does

²de Clippel and Rozen (2022) apply the results of Caplin and Dean (2015) to a communication game. They consider a two-stage setting in which the first-stage player chooses how much to obfuscate the state, and the second-stage player decides what to learn about the state when it is obfuscated. In this framework, the second player effectively faces a single-agent information acquisition problem. Exploiting this structure, de Clippel and Rozen (2022) derive testable predictions that are valid across all information cost functions and evaluate them in a laboratory experiment.

not assume that information acquisition is costly. Our work differs in several key respects: we allow for payoff uncertainty, we require information costs to increase strictly with informativeness, and we establish welfare bounds that have no analog in their analysis.

2. Setup

We analyze games where players face uncertain payoffs and have the option to covertly acquire information before taking action: we call them *information acquisition games*. In this section, we formally define this class of games.

Base Game. Let I be a finite set of players, with typical element i . Each player i has to choose an action a_i from a finite set A_i . As usual, we define $A_{-i} = \prod_{j \neq i} A_j$ and $A = A_i \times A_{-i}$. Accordingly, we use $a_{-i} = (a_j)_{j \neq i}$ to denote the action profile of all players other than i , and $a = (a_i, a_{-i})$ to denote the entire action profile. Throughout the paper we adopt the same notational conventions for all Cartesian products with index sets $I \setminus \{i\}$ and I .

Players are expected utility maximizers who care about each other's actions as well as an exogenous random variable, θ , which is drawn from a finite set Θ according to a probability measure $\pi \in \Delta(\Theta)$. We denote by $u_i : A \times \Theta \rightarrow \mathbb{R}$ player i 's Bernoulli utility function.

Embracing an established terminology (see, e.g., Bergemann and Morris, 2016), we call a tuple $\mathcal{G} = (I, \Theta, \pi, (A_i, u_i)_{i \in I})$ a **base game**.

Information Technologies. Before taking action, each player has the opportunity to acquire information about θ and other exogenous factors of potential interest (e.g., sunspots, noisy public information). We succinctly represent these additional factors by a single random variable, z , which takes values in a finite set Z . A Markov kernel $\zeta : \Theta \rightarrow \Delta(Z)$ details the conditional distribution of z given θ . We refer to the pair (z, θ) as the state of Nature, where z represent the correlation state and θ the payoff state.

Following Blackwell (1951, 1953), we model the acquisition of information using experiments. An experiment for player i is a Markov kernel $\xi_i : Z \times \Theta \rightarrow \Delta(X_i)$, where X_i is a finite space of signal realizations privately observable by player i . The

functions ξ_i details how the distribution of i 's signal, x_i , depends on θ and z .

By construction, players' signals are conditionally independent given θ and z . However, they may be correlated given θ only. Thus, our framework incorporates a form of correlated information acquisition, as in, among others, Hellwig and Veldkamp (2009), Myatt and Wallace (2012), Hébert and La'O (2023), and Denti (2023). Indeed, one can simply view z as a modeling device for situations in which players have access to information sources with correlated noise (e.g., reading the same newspapers or hiring the same consulting firm).

Information acquisition faces two kinds of frictions. First, players are constrained in the kind of experiments they can use: player i can only choose experiments that lie in a given set $\mathcal{E}_i$. Second, experiments come at a cost, where $C_i : \mathcal{E}_i \rightarrow \mathbb{R}_+$ denotes i 's cost function. As a normalization, we assume the existence of an experiment that costs zero; that is, $C_i(\xi_i) = 0$ for some $\xi_i \in \mathcal{E}_i$. When such ξ_i is uninformative, the normalization can also be interpreted as a participation constraint: player i can opt-out from gathering information.

We call a tuple $\mathcal{T} = (Z, \zeta, (X_i, \mathcal{E}_i, C_i)_{i \in I})$ an **information technology**.

Information Acquisition Games. Together, a base game $\mathcal{G}$ and an information technology $\mathcal{T}$ define an **information acquisition game**, which unfolds as follows. First, the state of Nature, z and θ , realizes but remains unobserved by the players. Then, players covertly select experiments from their feasible sets, incurring the associated costs. Finally, each player privately observes the outcome of their experiment and subsequently takes an action. We use $\sigma_i : X_i \rightarrow \Delta(A_i)$ to denote player i 's action plan in this game, and let Σ_i be the set of i 's action plans.

The solution concept we adopt is Nash equilibrium. A strategy for player i consists of an experiment $\xi_i \in \mathcal{E}_i$ and an action plan $\sigma_i \in \Sigma_i$. A strategy profile $(\xi, \sigma) = (\xi_i, \sigma_i)_{i \in I}$ generates for each player i an expected utility of

$$U_i(\xi, \sigma) = \sum_{a, x, z, \theta} u_i(a, \theta) \prod_{j \in I} \sigma_j(a_j | x_j) \xi_j(x_j | z, \theta) \zeta(z | \theta) \pi(\theta).$$

A strategy profile (ξ^*, σ^*) is an equilibrium if for all players i , (ξ_i^*, σ_i^*) maximizes

$$U_i(\xi_i, \xi_{-i}^*, \sigma_i, \sigma_{-i}^*) - C_i(\xi_i)$$

over all $\xi_i \in \mathcal{E}_i$ and $\sigma_i \in \Sigma_i$. The objective function consists of two terms: the benefit of information and the cost of information. As common in applications, costs and benefits are additively separable.

By choosing Nash equilibrium as our solution concept, we take the view that any correlation in player's actions comes from the information they acquire. Consequently, the choice of information has substance even when there is no payoff uncertainty—i.e., when θ is deterministic—since players may benefit from acquiring information about z in order to correlate their actions.

Equilibrium Predictions. We summarize the equilibria of information acquisition games using two statistics: the induced outcome of the base game, and the net value each player obtains from information acquisition.

The induced **outcome** of the base game is the joint distribution $p \in \Delta(A \times \Theta)$ of the players' actions and the payoff-relevant state. Note that the marginal distribution of θ must be the prior π ; we denote by $\Delta_\pi(A \times \Theta)$ the set of probability measures over $A \times \Theta$ whose marginal on Θ is π .

The net **value** player i obtains from information acquisition is simply i 's expected payoff, counting both costs and benefits. We denote by $v = (v_i)_{i \in I} \in \mathbb{R}^I$ the vector of such values. Since v_i includes player i 's information acquisition costs, it may differ from the expectation of u_i under p .

Exogenous, Costly, and Flexible Technologies. Our objective is to derive predictions for information acquisition games that remain valid regardless of the exact nature of the players' learning abilities. To achieve this, we adopt the perspective of an analyst who understands the structure of the base game but lacks detailed knowledge of the information technology available to the players. Specifically, given a base game $\mathcal{G}$ and a class of information technologies, we examine the set of outcome-value pairs that can emerge as equilibria in $(\mathcal{G}, \mathcal{T})$ across all technologies $\mathcal{T}$ within the specified class. The chosen class of information technologies reflects the assumptions the analyst is willing to make about the learning environment.

We focus on three classes of information technologies. The first class consists of all technologies where information is effectively **exogenous** to the game: each player has exactly one feasible experiment, with its cost normalized to zero. Such a technology can be identified by a tuple $\mathcal{S} = (Z, \zeta, (X_i, \xi_i)_{i \in I})$. We refer to such a

tuple as an **information structure**. An information acquisition game $(\mathcal{G}, \mathcal{S})$ is just a classic game of incomplete information, where $\mathcal{G}$ is the base game and $\mathcal{S}$ is a fixed information structure.

The second class encompasses all technologies where gathering more information is costly. We formalize the notion of “more information” using the canonical ranking of experiments due Blackwell (1951, 1953). Given a pair of experiments ξ_i and ξ'_i , we write $\xi_i \succsim \xi'_i$ if there exists a Markov kernel $\gamma : X_i \rightarrow \Delta(X_i)$ such that for every $x_i \in X_i$, $\theta \in \Theta$ with $\pi(\theta) > 0$, and $z \in Z$ with $\zeta(z|\theta) > 0$,

$$\xi'_i(x_i|z, \theta) = \sum_{x'_i \in X_i} \gamma(x_i|x'_i) \xi(x'_i|z, \theta).$$

Thus, $\xi_i \succsim \xi'_i$ if one can generate ξ'_i by garbling the output of ξ_i . As shown by Blackwell (1951, 1953), the relation $\xi_i \succsim \xi'_i$ holds if and only if player i is better off observing the output of ξ_i rather than the output of ξ'_i (holding fixed others’ behavior). In this sense, ξ_i is more informative than ξ'_i . We write $\xi_i \succ \xi'_i$ when $\xi_i \succsim \xi'_i$ and $\xi'_i \not\succsim \xi_i$.

We define technologies with **costly information** as satisfying two conditions for every player i . First, within the feasible set, more informative experiments incur strictly higher costs: if $\xi_i, \xi'_i \in \mathcal{E}_i$ are such that $\xi_i \succ \xi'_i$, then $C_i(\xi_i) > C_i(\xi'_i)$. Second, player i can always opt not to collect any information, meaning $\mathcal{E}_i$ includes an uninformative experiment, with its cost normalized to zero. These minimal conditions are widely satisfied by models of information acquisition in the literature, as they capture the fundamental idea that learning requires effort, and players can opt out of learning. In a classic example, players can either acquire no information at zero cost or perform a fixed experiment at a positive cost.

The third class of technologies we consider is a subclass of costly information technologies that has recently gained significant consideration. This subclass is characterized by flexible learning, meaning all experiments are feasible. Formally, a technology $\mathcal{T}$ is **flexible** if it has costly information and if for each player i , the feasible set of experiments is $\Delta(X_i)^{Z \times \Theta}$. This flexibility assumption originates from the rational inattention literature.³ Compared to other costly learning technology, flexible models allow one to study both the quantity and the type of information agents choose to acquire. Consequently, these models are especially well-suited for understanding the interaction between agents’ material payoffs and the kinds of information they most

³See Maćkowiak, Matějka, and Wiederholt (2023) for a review.

value.

Matějka and McKay (2015) provide a standard specification for flexible information technologies: $C_i(\xi_i)$ equals the expected reduction in uncertainty about θ and z , as measured by Shannon entropy, from observing the output of ξ_i . As is well known, if $\xi_i \succ \xi'_i$ then $C_i(\xi_i) > C_i(\xi'_i)$. More broadly, one could substitute Shannon entropy with any other strictly concave measure of uncertainty (Caplin, Dean, and Leahy, 2022). One could also consider increasing transformations of these costs (Denti, 2022; Zhong, 2022), or any differentiable cost function whose derivative is strictly convex (Lipnowski and Ravid, 2022).⁴

Fixing a base game $\mathcal{G}$, we say that an outcome-value pair (p, v) is **attainable** with exogenous, costly, or flexible learning if a technology $\mathcal{T}$ of said type exists such that $(\mathcal{G}, \mathcal{T})$ admits an equilibrium that induces (p, v) . The set of outcome-value pairs attainable with exogenous information has been studied by Bergemann and Morris (2016). In what follows, we characterize the outcome-value pairs attainable under costly and flexible learning and compare them to the benchmark case of exogenous information.

3. Robust Predictions

This section characterizes the robust predictions of models with costly and flexible learning. Unlike settings with an exogenous information structure—where information is simply used optimally—endogenous learning also requires optimal information acquisition. We precisely identify the behavioral implications of this added constraint. Moreover, the presence of information costs expands the set of attainable payoffs beyond those captured by standard expected utility calculations. We provide a detailed description of these payoffs.

3.1. The Optimal Use of Information

We begin by reviewing the results of Bergemann and Morris (2016) for the case of exogenous information. Bergemann and Morris (2016) characterize the equilibrium outcomes that can arise in a game of incomplete information $(\mathcal{G}, \mathcal{S})$ as we fix the base

⁴Note players' prior beliefs about (θ, z) are exogenously determined by π and ζ . Thus, the issue of experiment-based vs. posterior-based costs that sometimes arise in applications of rational inattention (see, e.g., Ravid, 2020; Denti, Marinacci, and Rustichini, 2022) is irrelevant here.

game $\mathcal{G}$ and range over all information structures $\mathcal{S}$: they call these outcomes Bayes correlated equilibria.

Definition 1. A **Bayes correlated equilibrium (BCE)** of a base game $\mathcal{G}$ is an outcome $p \in \Delta_\pi(A \times \Theta)$ such that, for all $i \in I$ and $a_i, b_i \in A_i$,

$$\sum_{a_{-i}, \theta} (u_i(a_i, a_{-i}, \theta) - u_i(b_i, a_{-i}, \theta)) p(a_i, a_{-i}, \theta) \geq 0. \quad (1)$$

Following standard terminology (Myerson, 1982), we name (1) the **obedience constraint**. When Θ is a singleton, BCE reduces to Aumann's (1987) notion of correlated equilibrium.⁵

The obedience constraint captures players' optimal use of information. To illustrate this, it is helpful to view p as arising from recommendation experiments. A **recommendation experiment** for player i is an experiment ξ_i in which the signal space coincides with the action space, i.e., $X_i = A_i$. The key idea is that such an experiment not only conveys information but also prescribes an action for each player. Obedience says that abiding by this recommendation is optimal.

To clarify the link between recommendation experiments and obedience, suppose an outcome p is generated by players following recommendation experiments. For each player i and action a_i , define

$$p(a_i) := \sum_{a_{-i}, \theta} p(a_i, a_{-i}, \theta)$$

as the probability that player i receives recommendation a_i . Let $\text{supp}_i(p)$ be the set of recommendations player i receives with positive probability:

$$\text{supp}_i(p) := \{a_i : p(a_i) > 0\}.$$

Conditional on receiving a recommendation $a_i \in \text{supp}_i(p)$ and assuming that opponents follow their respective recommendations, the joint probability of (a_{-i}, θ) is

$$p_{a_i}(a_{-i}, \theta) := \frac{p(a_i, a_{-i}, \theta)}{p(a_i)}.$$

⁵Our definition of BCE corresponds to the specialization of Bergemann and Morris's (2016) definition to the case in which players' original type spaces are degenerate. Forges (1993) refers to such outcomes as *universal Bayesian solutions*.

Thus, a recommendation $a_i \in \text{supp}_i(p)$ is followed if and only if it is a best response to the posterior belief $p_{a_i} \in \Delta(A_{-i} \times \Theta)$:

$$a_i \in BR(p_{a_i}) := \operatorname{argmax}_{b_i \in A_i} \sum_{a_{-i}, \theta} u_i(b_i, a_{-i}, \theta) p_{a_i}(a_{-i}, \theta).$$

The obedience constraint ensures that the information conveyed by recommendation experiments is optimally used: $a_i \in BR(p_{a_i})$ for all $i \in I$ and $a_i \in \text{supp}_i(p)$ if and only if (1) holds.

As established by Bergemann and Morris (2016), building on a long tradition of related results, restricting attention to outcomes implementable via recommendation experiments entails no loss of generality when information is exogenous. This result extends to the framework of this paper, where information is endogenous.⁶

3.2. Optimal Information Acquisition

We now outline the behavioral restrictions imposed by costly and flexible learning. An initial observation is that any outcome consistent with either costly or flexible learning must be a BCE. The intuition is straightforward: the obedience constraint, which defines BCE, ensures the optimal use of information. Whether information is exogenous or endogenous, players must utilize it optimally, and so obedience must be satisfied.

In models of endogenous information, information is not only used optimally but also acquired optimally. For Bayesian decision-makers, information has a purely instrumental value—it is worth acquiring only if it leads to a change in actions. We next introduce two refinements of BCE that formalize this basic principle for costly and flexible learning.

As discussed in the context of BCE, the belief profile $(p_{a_i})_{a_i \in \text{supp}_i(p)}$ reflects the information player i receives about others' actions and the payoff state in any implementation of an outcome p via recommendation experiments. In particular, player i receives some positive amount of information about (a_{-i}, θ) if $p_{a_i} \neq p_{b_i}$ for some $a_i, b_i \in \text{supp}_i(p)$. In this case, we say that p is **informative** for player i .

⁶The rational inattention literature frequently presents similar results; see, for instance, Caplin and Dean (2015, Section A) and Matějka and McKay (2015, Corollary 1).

Next we define a refinement of BCE that captures the behavioral implications of costly learning.

Definition 2. A BCE p is **instrumental (iBCE)** if, for every player i for which p is informative,

$$\bigcap_{a_i \in \text{supp}_i(p)} BR(p_{a_i}) = \emptyset.$$

In other terms, a BCE is instrumental if any player who receives some information about other's actions and the payoff state has no action that is optimal in response to all on-path recommendations.

Any BCE that arises from costly learning must be instrumental. The reason is straightforward: if not, a player i could deviate to a free, uninformative experiment and take the action $b_i \in \bigcap_{a_i \in \text{supp}_i(p)} BR(p_{a_i})$, which is a best response to all on-path recommendations. Such a deviation would yield the same expected utility while eliminating information costs, contradicting the principle of optimal information acquisition. As we show in the characterization theorem below, iBCE precisely captures the behavioral constraints imposed by costly learning.

Flexible learning is a sub-case of costly learning, thus any outcome that is consistent with flexible learning must be an iBCE. Flexibility in information acquisition, however, imposes additional restrictions on outcomes. To describe them, consider again an implementation of an outcome p via recommendation experiments. Two recommendations $a_i, b_i \in \text{supp}_i(p)$ provide player i with different information about (a_{-i}, θ) if $p_{a_i} \neq p_{b_i}$. In this case, we say that a_i and b_i are **informationally distinct** with respect to p .

The next definition gives the restrictions flexible learning imposes on behavior.

Definition 3. A BCE p is **separated (sBCE)** if, for each $i \in I$ and $a_i, b_i \in \text{supp}_i(p)$ that are informationally distinct with respect to p ,

$$BR(p_{a_i}) \cap BR(p_{b_i}) = \emptyset.$$

Recall that, by the obedience constraint, $a_i \in BR(p_{a_i})$ and $b_i \in BR(p_{b_i})$. Thus, separation requires that, whenever $p_{a_i} \neq p_{b_i}$, there exists no third action c_i (which may coincide with a_i or b_i) such that $c_i \in BR(p_{a_i}) \cap BR(p_{b_i})$.

Why can flexible technologies only generate separated BCEs? Consider a player i who takes with positive probability a pair of actions a_i and b_i such that $p_{a_i} \neq p_{b_i}$.

Assuming that p arises from recommendation experiments, we can interpret a_i and b_i as signal realizations. With flexible information, a player can save on information costs by substituting a_i and b_i with a single action recommendation, c_i . For this substitution not to be profitable, it must be that either $c_i \notin BR(p_{a_i})$ or $c_i \notin BR(p_{b_i})$. But c_i is arbitrary, and so $BR(p_{a_i}) \cap BR(p_{b_i})$ must be empty.

We demonstrate below that, analogous to how iBCE characterizes outcomes arising from costly learning, sBCE captures the behavioral implications associated with flexible learning.

3.3. Attainable Payoffs

When information is provided exogenously, each player's payoff is simply their expected utility under the equilibrium outcome. We call this quantity the gross value, as it excludes any information acquisition costs.

Definition 4. The **gross value** of an outcome p for a player i is the quantity

$$\bar{v}_i(p) := \sum_{a, \theta} u_i(a, \theta) p(a, \theta).$$

Let $\bar{v}(p) := (\bar{v}_i(p))_{i \in I}$ be the vector of gross values.

When information must be acquired endogenously, the same outcome can arise under multiple cost functions, making it consistent with a range of values. As we demonstrate shortly, this set of attainable values for any player i forms an interval. The top of the interval is given by the gross value $\bar{v}_i(p)$, which is clearly an upper bound on i 's payoff, as acquiring information incurs positive costs. The lower bound of this interval is defined by what we call the outcome's uninformed value, which represents i 's maximal value if they must act before receiving any information.

Definition 5. The **uninformed value** of an outcome p for a player i is the quantity

$$\underline{v}_i(p) := \max_{b_i \in A_i} \sum_{a, \theta} u_i(b_i, a_{-i}, \theta) p(a, \theta).$$

Let $\underline{v}(p) := (\underline{v}_i(p))_{i \in I}$ denote the vector of uninformed values.

A direct calculation reveals that in an equilibrium of an information acquisition game producing an outcome p , the uninformed value $\underline{v}_i(p)$ is exactly the payoff player

i obtains by deviating to learning nothing at zero cost. Such a deviation is always feasible under costly learning. Thus, in any equilibrium inducing outcome p in a game with a costly (or flexible) information technology, player i 's equilibrium payoff must lie within the interval: $[\underline{v}_i(p), \bar{v}_i(p)]$.⁷

In fact, a slightly stronger condition holds. Whenever $\bar{v}_i(p) > \underline{v}_i(p)$, player i 's payoff must lie strictly below the gross value $\bar{v}_i(p)$. The reason is that if $\bar{v}_i(p) > \underline{v}_i(p)$, player i must acquire a strictly informative signal to generate p , and so must pay a strictly positive cost under every costly technology.

3.4. Characterization Theorem

The above paragraphs outline necessary conditions for (p, v) to be attainable with a costly or a flexible technology. The main result of this section shows these conditions are also sufficient.

Theorem 1. *Fix a base game $\mathcal{G}$ and an outcome-value pair (p, v) .*

- (i) *(p, v) is attainable with a costly technology if and only if p is an $iBCE$, and for all i , $\underline{v}_i(p) = \bar{v}_i(p) = v_i$ or $v_i \in [\underline{v}_i(p), \bar{v}_i(p))$.*
- (ii) *(p, v) is attainable with a flexible technology if and only if p is an $sBCE$, and for all i , $\underline{v}_i(p) = \bar{v}_i(p) = v_i$ or $v_i \in [\underline{v}_i(p), \bar{v}_i(p))$.*

Thus, treating information as acquired rather than endowed alters predictions in two key ways. First, outcomes attainable through information acquisition must satisfy additional constraints that capture players' learning incentives. This gives rise to new solution concepts: instrumental and separated BCE. Second, acquiring information incurs a cost, which reduces net payoffs. When this cost is minimal, gross and net payoffs are nearly indistinguishable. But when the cost is high, it can fully offset the value of the information, leaving a player no better off than if they had remained uninformed.

The theorem also reveals that the sole effect of assuming flexible information acquisition, as opposed to merely costly learning, is the requirement that BCEs be

⁷The uninformed value is related to the notion of *coarse correlated equilibrium* (see, e.g., Roughgarden, 2016, Definition 13.5). Coarse correlated equilibrium relaxes the obedience constraint, requiring instead that no player can benefit from deviating to a single action from all of the mediator's recommendation; that is, $\bar{v}_i(p) \geq \underline{v}_i(p)$ for every i .

separated rather than just instrumental. This distinction reflects the broader range of informational deviations available to players under flexible learning. Notably, however, sBCE and iBCE coincide in binary action games. This yields the following corollary.

Corollary 1. *If $|A_i| \leq 2$ for all players i , an outcome-value pair is attainable with a costly technology if and only if it is attainable with a flexible technology.*

We conclude the section by sketching the proof of Theorem 1. Having already discussed the necessity of the theorem’s conditions, we now focus on their sufficiency.

The first step is to construct an information structure $\mathcal{S} = (Z, \zeta, (X_i, \xi_i)_{i \in I})$ and an equilibrium $\sigma = (\sigma_i)_{i \in I}$ of the incomplete information game $(\mathcal{G}, \mathcal{S})$ that implements a given instrumental BCE p . In the subsequent steps of the proof, we embed $\mathcal{S}$ and σ into a game with endogenous learning, in which each player i selects ξ_i as their equilibrium experiment.

An important observation is that this embedding cannot be carried out using an arbitrary information structure consistent with p . In particular, we cannot simply invoke the construction in Bergemann and Morris (2016), as the information structures they produce may implicitly encode players’ private randomization devices into the state space—rendering them costly to use and violating incentive compatibility when learning is costly.

To illustrate this issue, consider the classic matching pennies game with no payoff uncertainty. Let p denote the BCE corresponding to the game’s Nash equilibrium, in which players independently randomize uniformly over their actions. Suppose we attempt to implement p in an information acquisition setting using the canonical construction from Aumann (1987) or Bergemann and Morris (2016), where the state space is $Z = A$, and each player i observes their own coordinate of the public correlation device with certainty; that is, $X_i = A_i$ and $\xi_i(a_i \mid a_i, a_{-i}) = 1$ for all $a \in A$. In this construction, each player receives a perfectly revealing signal about a component of the state. Under any costly information technology, such experiments entail a strictly positive cost. Yet in the target equilibrium, players are indifferent among actions and can replicate the same outcome through costless randomization. As a result, no player would be willing to pay for the signal, making the canonical construction incompatible with information acquisition.

To address this issue, we construct $\mathcal{S}$ and σ in a different manner. Specifically, we partition each player’s action set into equivalence classes, where two actions a_i

and b_i are equivalent if they induce the same belief over (θ, a_{-i}) under the target outcome p —that is, if $p_{a_i} = p_{b_i}$. Each player then chooses an experiment that reveals only the equivalence class of their recommended action. The equal-beliefs condition ensures that the relative likelihood of actions within the same equivalence class is independent of both the state θ and the opponent’s action a_{-i} . As a result, conditional on observing an equivalence class, there exists a mixture over actions within that class—independent of θ and a_{-i} —that replicates p . Obedience further guarantees that this mixture is incentive compatible, since all actions in a given class induce the same belief and thus yield the same expected payoff.

With this construction in hand, we now prove part (i) of Theorem 1 as follows. Fix an iBCE outcome p and a profile of values v that lie between the gross and uninformed values associated with p , as specified in the theorem. Let $\mathcal{S}$ and σ be the implementation of p constructed above, and define an information acquisition technology $\mathcal{T} = (Z, \zeta, (X_i, \{\xi_i, \xi'_i\}, C_i)_{i \in I})$ with the following properties. First, it preserves the correlation structure of $\mathcal{S}$. Second, each player i is offered a choice between two experiments: the experiment ξ_i from $\mathcal{S}$ and a fully uninformative experiment ξ'_i . The cost of ξ_i is set to $\bar{v}_i(p) - v_i$, while ξ'_i is costless.

Given this setup, it is optimal for each player i to select ξ_i , since doing so yields a net payoff of v_i , which exceeds the payoff $\underline{v}_i(p)$ the player would receive from deviating to the uninformative experiment. It follows that (σ, ξ) constitutes an equilibrium of the information acquisition game $(\mathcal{G}, \mathcal{T})$, with outcome p and value profile v .

It remains to verify that $\mathcal{T}$ satisfies the properties of costly learning—i.e., within the feasible set, more informative experiments are strictly more costly. There are two cases to consider. First, if $\bar{v}_i(p) > \underline{v}_i(p)$, then the cost of player i ’s experiment, $\bar{v}_i(p) - v_i$, is strictly positive, as required. Second, if $\bar{v}_i(p) = \underline{v}_i(p) = v_i$, then p being instrumental means that ξ_i is uninformative in our constructed information structure, and so the monotonicity restriction holds trivially.

To prove part (ii) of Theorem 1, we build on a result from Denti (2024). Denti (2024) examines a two-player signaling game in which the receiver incurs a cost to monitor the sender’s actions. Among other results, Denti (2024) demonstrates that when the sender takes every feasible action with positive probability, any receiver behavior that satisfies both obedience and separation can be supported with a flexible technology. By interpreting the correlation state as analogous to the sender’s action, we can apply Denti’s (2024) result iteratively, player by player, to construct a profile

of cost functions that sustain p as an equilibrium outcome.⁸

To complete the proof of part (ii), we adjust the cost function delivered by Denti’s (2024) result so that any payoff between the uninformed and the gross value can be generated in equilibrium. This step is needed because Denti’s (2024) result is silent about achievable payoffs.

4. Examples

In this section, we present simple illustrative examples to clarify the implications of Theorem 1 before proceeding to a more systematic analysis in later sections. The first three examples demonstrate the similarities and differences between BCE, iBCE, and sBCE, while the final example focuses on the distinction between gross and uninformed values and suggests its relevance for welfare analysis.

In the first example, based on the game *Battle of the Sexes with an Outside Option*, we highlight a feature that can drive a wedge between BCE, iBCE, and sBCE: the presence of weakly dominated actions.

Example 1. Consider the following two-player game without payoff uncertainty:

	B	S
O	2, 3	2, 3
B	4, 1	0, 0
S	0, 0	1, 4

This payoff matrix represents the reduced normal form of a *Battle of the Sexes with an Outside Option*. In the underlying extensive form, the row player chooses between an outside option—providing a payoff of 2 to themselves and 3 to the column player—and entering a coordination game. Within this game, coordinating on (B, B) favors the row player, while coordinating on (S, S) favors the column player.

Since there is no payoff uncertainty, information acquisition matters solely as a

⁸Some readers may wonder why we need Denti’s (2024) result. In particular, one may try to take any weakly monotone cost function that implements the outcome p , and approximate it using a strictly monotone cost function. There are many difficulties with this approach. One difficulty is that separation only says that a particular class of deviations are suboptimal, whereas flexibility requires the suboptimality of many other deviations. A related difficulty is that, under flexibility, each player has an infinite set of feasible experiments at their disposal. Consequently, one cannot use upper hemicontinuity to guarantee that the each player is willing to choose their equilibrium experiment.

way to generate correlation between players' actions. We next argue that non-trivial correlation is feasible when information is exogenous—as in BCE, which is equivalent to Aumann's correlated equilibrium in this example. By contrast, correlation becomes unsustainable when information acquisition is costly; that is, players' strategies must be independent in every iBCE (and therefore in every sBCE as well). As a result, we show that fair outcomes—where players receive equal payoffs through correlated action—are ruled out under costly information acquisition.

Formally, the BCE outcomes consist of all distributions $p \in \Delta(A)$ satisfying:

$$p(S, B) = p(S, S) = p(B, S) = 0 \quad \text{and} \quad p(O, B) \leq p(O, S).$$

Action S is never chosen by the row player because it is dominated by the outside option O . Given that the row player never selects S , the outcome (B, S) must also have zero probability: the column player chooses action S only when certain that the row player selects O . Lastly, the inequality $p(O, B) \leq p(O, S)$ ensures the row player prefers action O over B whenever it is prescribed. All other incentive constraints can easily be verified as satisfied. A concrete example of non-trivial correlation is the BCE given by $p(B, B) = 1/4$ and $p(O, S) = 3/4$. In this BCE, both players achieve the same gross payoff, $5/2$.

We claim that a BCE p is instrumental (or separated) if and only if

$$p(B, B) \in \{0, 1\}.$$

To put it differently, the iBCE set consists of all mixed Nash equilibria of the game: (B, B) and $(O, tB + (1 - t)S)$ with $t \leq 1/2$. In this case, the row player either exclusively chooses action B or exclusively chooses action O , leading to independence in players' actions under costly information acquisition. Notably, all instrumental BCEs result in unequal gross payoffs for the players—either $(4, 1)$ or $(2, 3)$. Moreover, since no information is acquired, the gross and net payoffs coincide in these scenarios.

Intuitively, if $p(B, B) \in \{0, 1\}$, then the BCE provides no new information to either player, making it trivially instrumental. Conversely, if $p(B, B) \in (0, 1)$, then the BCE provides information to the column player: since $p_B(B) > 0 = p_S(B)$, the recommendations B and S are informationally distinct. However, since action B weakly dominates S for the column player (given that the row player never takes action S), we have $B \in BR(p_B) \cap BR(p_S)$. Hence, the information acquired by the

column player has no instrumental value, rendering the BCE non-instrumental.

As Example 1 illustrates, one situation in which information acquisition incentives have clear behavioral consequences is in the presence of weakly dominated actions. However, these incentives can matter even when no action is weakly dominated. The following example demonstrates this fact.

Example 2. The payoff matrix below illustrates a Matching Pennies game with state-independent utilities, where the states θ_L and θ_R are equally likely.

	H	T		H	T
H	1, -1	-1, 1	H	1, -1	-1, 1
T	-1, 1	1, -1	T	-1, 1	1, -1
	θ_L			θ_R	

This configuration serves as a simplified model of a competitive environment characterized by non-fundamental volatility, akin to the scenario examined by Hébert and La'O (2023) in their study of economies with rationally inattentive agents.

In this game, no action is weakly dominated, yet the set of BCEs is much larger than the iBCE and sBCE sets, which are a singleton.⁹ The BCE set includes all outcomes p satisfying

$$p_{a_i}(a_{-i}, \theta_L) + p_{a_i}(a_{-i}, \theta_R) = 1/2$$

for all $i \in I$, $a_i \in A_i$, and $a_{-i} \in A_{-i}$. In all these BCEs, the players do not have any information about their opponent's action. Information about the state, on the other hand, could be substantial. For example, a BCE exists where the row player selects H if the state is θ_L and T if the state is θ_R , while the column player randomizes independently of the state. In this equilibrium, the row player fully learns the state.

When information acquisition incentives are accounted for, no learning can occur in equilibrium. In the unique iBCE (a fortiori, also the unique sBCE), the players randomize independently of each other and of the state, resulting in

$$p(a, \theta) = 1/8$$

for all $a \in A$ and $\theta \in \Theta$. This single outcome arises because, in all BCEs, the players remain indifferent between playing H and T after receiving any recommendation.

⁹Since each player has two actions, the iBCE and sBCE sets coincide.

Consequently, information holds no instrumental value, and the players cannot be incentivized to acquire it, even if the cost of doing so is negligible.

Examples 1 and 2 illustrate scenarios where BCE differs from iBCE and sBCE. In Section 5, we identify the class of games in which such differences arise and relate them to the concept of *jeopardization*—a generalization of weak dominance introduced by Myerson (1997) in the study of correlated equilibria.

In Examples 1 and 2, each player has only two relevant actions. As a result, there is no distinction between costly and flexible information acquisition (Corollary 1). Our next example shows that this equivalence breaks down beyond the binary action case: the sBCE and iBCE sets can differ significantly when agents have more than two actions.

Example 3. The payoff matrix below represents a single-agent decision problem under uncertainty, where rows correspond to actions and columns to states:

	L	R
L	1	0
R_1	0	1
R_2	0	1

Action L yields a positive payoff only in state L , while actions R_1 and R_2 are both optimal in state R —with R_2 being a replica of R_1 . The two states are equally likely. This structure may represent a stylized political choice in a three-candidate election along a Left-Right spectrum. In one state of the world (e.g., a booming economy), the left-leaning candidate is most attractive, whereas in the other state (e.g., an economic downturn), either of the two right-leaning candidates is equally appealing.

In this example, an outcome $p \in \Delta_\pi(A \times \Theta)$ is a BCE if and only if

$$p(L, L) \geq p(L, R), \quad p(R_1, L) \leq p(R_1, R), \quad p(R_2, L) \leq p(R_2, R).$$

Moreover, every such BCE is instrumental—that is, each can be supported by a costly information acquisition technology. To see why, note that any informative BCE must involve a non-trivial splitting of the prior: a recommendation of L implies that state L is strictly more likely than state R , whereas a recommendation of R_1 or R_2 implies the reverse. Because no single action is a best response under all these beliefs— L is uniquely optimal in the first case, and R_1 or R_2 in the second—the BCE must be

instrumental. Since uninformative BCEs are automatically instrumental, we conclude that all BCEs in this example are instrumental.

However, only a small subset of these outcomes are separated. Specifically, a BCE is separated if and only if one of the following conditions holds: (i) $p(R_1) = 0$, (ii) $p(R_2) = 0$, or (iii) the conditional distributions satisfy $p_{R_1} = p_{R_2}$. This characterization follows from the logic above. If the BCE is uninformative, then it is trivially separated. If it is informative, then L is the unique best response to its associated belief, while the best-response set to the belief associated with R_1 or R_2 is $\{R_1, R_2\}$. The only way these best-response sets are disjoint—required for separation—is if R_1 and R_2 are never used together or are associated with the same belief.

Intuitively, this situation is one where every informative equilibrium provide the agent with at least one valuable signal realization. Consequently, the agent would be willing to pay a positive amount in order to receive that signal compared to remaining uninformed. However, because R_1 and R_2 are duplicates, the agent does not benefit from having multiple beliefs in which these actions are optimal, and so would pool these beliefs together under any flexible technology.

We next use a simple investment game to illustrate the differences in welfare analysis under exogenous versus endogenous information.

Example 4. The payoff matrices below represent a two-player, two-action, two-state investment game:

	I	N		I	N
I	1, 1	2, 0	I	0, 0	-1, 0
N	0, 2	0, 0	N	0, -1	0, 0
	θ_G			θ_B	

Each player chooses between investing (I) and not investing (N) in a project with uncertain returns. In the good state, θ_G , investment is profitable, and the players' decisions act as strategic substitutes—each prefers to be the sole investor. In the bad state, θ_B , investment is unprofitable, and the decisions become strategic complements—players prefer to share the burden of a poor investment outcome. The two states occur with equal probability.

To illustrate the difference in welfare analysis under exogenous and endogenous information, consider the following outcome: players invest if the state is good and

refrain from investing if the state is bad:

$$p(I, I, \theta_G) = 1/2 \quad \text{and} \quad p(N, N, \theta_B) = 1/2.$$

This outcome is a BCE: in the good state, investing is a dominant strategy; in the bad state, not investing is the unique optimal response when the other player also does not invest. Moreover, it is a separated BCE: the two recommendations, I and N , are informationally distinct and instrumentally valuable—they lead players to change their behavior based on the information received.

Whether p is desirable from a utilitarian social planner’s perspective crucially depends on whether information is treated as exogenous or endogenous to the game. We next observe that, under exogenous information, p reflects the most *optimistic* welfare evaluation; under endogenous information, it reflects the most *pessimistic*.

When information is exogenous, information acquisition costs are not considered, and the value of p corresponds to its expected gross payoff, which is $1/2$ for each player. Moreover, a straightforward calculation reveals that p separately maximizes the sum of the two players’ payoffs for each θ . Thus, if information is freely available, p represents the best-case scenario from a welfare standpoint.

Under endogenous information, however, the payoff from p ranges from its *gross* value, $1/2$, to its *uninformed* value, which is 0. This lower bound reflects the situation when players must choose actions without receiving informative recommendations but still anticipate the opponent’s strategy: best responding to the belief that the other player invests in the good state and does not invest in the bad state. Given this belief, each player is indifferent between investing and not investing, as both actions yield an expected payoff of zero. Importantly, a player’s payoff can never fall below zero, as this is guaranteed by unilaterally choosing not to invest. Therefore, if acquiring information is costly, p represents the most pessimistic welfare assessment of the investment game.

A more detailed treatment of welfare analysis with and without information acquisition is provided in Section 6.

5. Instrumental and Separated BCE

The concepts of instrumental and separated BCE encapsulate the behavioral consequences of costly and flexible learning, as established by Theorem 1. This section examines the properties of these refinements of BCE. A preliminary observation is that, while the BCE set is always closed, the iBCE and sBCE sets may lack this property. Nevertheless, we regard elements within their closures as reasonable predictions under costly and flexible learning, as they cannot be distinguished from true instrumental and separated BCEs based on finite data. Accordingly, our analysis centers on comparing the BCE set to the *closures* of the iBCE and sBCE sets.

To preview our findings, we characterize the class of games in which the behaviors predicted by costly and flexible learning differ significantly from those predicted under exogenous information. Specifically, we characterize the games where the iBCE and sBCE sets are not dense in the BCE set. Additionally, we show that these refinements manifest in an “all-or-nothing” manner: the sets of instrumental and separated BCEs are either *dense* or *nowhere dense* within the BCE set. Lastly, we demonstrate that the density case is generic in the utility space. Collectively, these results suggest that information acquisition incentives can have a dramatic impact on behavioral predictions, but only in structured environments.

We begin by characterizing the games in which costly and flexible learning significantly reduce the BCE set. As we establish in Theorem 2 below, the key condition involves Myerson’s (1997) notion of jeopardization, a generalization of weak dominance relevant for correlated equilibria. Formally, an action a_i **jeopardizes** another action b_i if $a_i \in BR(p_{b_i})$ for every BCE p where $b_i \in \text{supp}_i(p)$. In other words, a_i jeopardizes b_i if player i , when expected to take b_i in a BCE, would not suffer any loss by deviating to a_i .¹⁰

Myerson (1997) identifies two primary forms of jeopardization. The first occurs when a_i weakly dominates b_i . For instance, in Example 1, action B weakly dominates—and therefore jeopardizes—action S for the column player, once action S is deleted for the row player. The second form emerges when a_i and b_i are “inextricably connected” (Myerson, 1997, p. 188) such as H and T in Example 2, where the two actions jeopardize each other. Let $J(b_i)$ be the set of actions that jeopardize

¹⁰Myerson (1997) focuses on games without payoff uncertainty. Here, we extend his definition to games in which Θ is not a singleton in a natural way.

b_i . Note that the set $J(b_i)$ is non-empty since b_i jeopardizes itself.

The concept of jeopardization enables us to extend the properties of being instrumental and separated from individual outcomes to the entire BCE set. Recall that a BCE p is instrumental if, for each player i such that p is informative,

$$\bigcap_{a_i \in \text{supp}_i(p)} BR(p_{a_i}) = \emptyset.$$

We say that the BCE set is **globally instrumental** if, for each player i such that *some* BCE p is informative,

$$\bigcap_{a_i \in A_i} J(a_i) = \emptyset.$$

In simpler terms, the BCE set is globally instrumental if any player who uses some information in some BCE has no single action that jeopardizes all other actions.

Recall that a BCE p is separated if, for each player i and pair of actions a_i and b_i that are informationally distinct with respect to p ,

$$BR(p_{a_i}) \cap BR(p_{b_i}) = \emptyset.$$

We say that the BCE set is **globally separated** if, for each player i and pair of actions a_i and b_i that are informationally distinct in *some* BCE,

$$J(a_i) \cap J(b_i) = \emptyset.$$

In other words, the BCE set is globally separated if any two actions that induce distinct beliefs in some BCE do not share a common jeopardizing action. Note that if the BCE set is globally separated, it is also globally instrumental. The converse holds when every player has only two actions but is generally false otherwise (see Example 3).

We are now ready to characterize the games in which information acquisition incentives significantly reduce the BCE set.

Theorem 2. *Fix a base game $\mathcal{G}$. Then:*

- (i) *The iBCE set is dense in the BCE set if and only if the BCE set is globally instrumental.*
- (ii) *The sBCE set is dense in the BCE set if and only if the BCE set is globally separated.*

Thus, the behaviors predicted by costly and flexible learning differ from those predicted under exogenous information when the BCE set fails to be globally instrumental or globally separated, respectively.

In the simple case where $\mathcal{G}$ is a two-player, binary-action game without payoff uncertainty, it is easy to identify when the BCE set is not globally instrumental—or, equivalently, not globally separated. First, no player can have a strictly dominant action. Second, at least one player i must have an action a_i that is no worse for them than their other action b_i regardless of what others do; that is, $u_i(a_i, a_{-i}) \geq u_i(b_i, a_{-i})$ for all a_{-i} .¹¹ Note this requirement is slightly weaker than a_i being weakly dominant, since a_i may never be strictly better than the alternative action. As Example 2 demonstrates, this characterization no longer holds outside the class of two-player, binary-action games without payoff uncertainty.

Let us briefly review the proof of the theorem. The “only if” statements are straightforward to establish. For example, suppose by way of contradiction that the BCE set is not globally instrumental. This means there exists a BCE p , a player i , and actions $a_i, b_i \in \text{supp}_i(p)$ such that $p_{a_i} \neq p_{b_i}$, yet $\bigcap_{c_i \in A_i} J(c_i) \neq \emptyset$. This BCE p is not instrumental, nor are any BCEs sufficiently close to it. Consequently, the iBCE set cannot be dense within the BCE set. A similar argument shows that if the BCE is not globally separated, then the sBCE is not dense within the BCE set.

Demonstrating the “if” statements require more effort. Our proof relies on a technical lemma that characterizes the properties of being globally instrumental and separated through the existence of a specific class of BCEs. To state this lemma, say a BCE p is **maximally supported** if its support contains the support of any other BCE. Furthermore, a maximally supported BCE p is **maximally differentiated** if every pair of actions that are informationally distinct with respect to some BCE

¹¹The key to obtaining this characterization is to use the results of Calvó-Armengol (2006) to show that the BCE set is globally instrumental when there are no ties and no dominant actions. To complete the proof, one can then make a tedious yet straightforward case-based argument.

are also informationally distinct with respect to p .

Lemma 1. *Fix a base game $\mathcal{G}$. Then:*

- (i) *The BCE set is globally instrumental if and only if there exists a BCE that is instrumental and maximally differentiated.*
- (ii) *The BCE set is globally separated if and only if there exists a BCE that is separated and maximally differentiated.*

With Lemma 1 in hand, the reader can readily verify the “if” statements of Theorem 2. For instance, suppose the BCE set is globally instrumental. Then, any BCE p can be approximated by an iBCE of the form $tp + (1 - t)q$, where $t \in (0, 1)$ is small, and q is an iBCE that is maximally differentiated. A similar reasoning applies to show that if the BCE set is globally separated, then the sBCE set is dense in the BCE set.

Building on Theorem 2 and Lemma 1, we now demonstrate that iBCE and sBCE serve as all-or-nothing refinements of BCE, with “nothing” being the generic case. We adopt the following notion of genericity: We fix a finite set of players I , a finite set of payoff states Θ , a full-support prior $\pi \in \Delta(\Theta)$, and a finite set of actions A_i for each player $i \in I$. To obtain a base game, it remains to specify a profile of utility function $u = (u_i)_{i \in I}$. We identify u with an element of the Euclidean space $\mathbb{R}^{I \times A \times \Theta}$, and say a statement is true for **generic** u if the closure of the subset in $\mathbb{R}^{I \times A \times \Theta}$ for which it is false has Lebesgue measure zero.

Theorem 3. *The iBCE and sBCE sets are either dense or nowhere dense in the BCE set, with the former being the case for generic u .*

Therefore, while incentives for information acquisition can dramatically alter behavioral predictions, this effect arises primarily within structured environments. Such structure might originate from an underlying game tree, which commonly generates weakly dominated strategies in its normal-form representation, or it could result from specific institutional details that shape strategic interactions. For example, Ravid, Roesler, and Szentes (2022) and Denti (2024) analyze structured settings—bilateral trade and signaling games, respectively—in which economically meaningful indifference create significant divergence between predictions under endogenous and exogenous information. In Section 7.3, we further illustrate that games where iBCE

and sBCE meaningfully refine the set of BCE can naturally arise as solutions to mechanism-design problems involving aftermarket considerations.

To conclude the section, we provide a brief discussion of the proof of Theorem 3. To demonstrate the all-or-nothing nature of iBCE as a refinement of BCE, we draw upon Theorem 2 and Lemma 1. By definition, the iBCE set is either nowhere dense or it is dense in some open subset P of BCEs. Suppose the latter holds. By Theorem 2 and Lemma 1, one can show the iBCE set is dense in the whole BCE set by finding an iBCE that is maximally differentiated. In the appendix, we show the set of BCEs that are maximally differentiated is open and dense within the BCE set. Therefore, P includes a non-empty and open subset $Q \subseteq P$ of maximally differentiated BCEs. But the iBCE set is dense in P , meaning some BCE in Q —a set that contains only BCEs that are maximally differentiated—must be instrumental. Thus, we have shown that if iBCE is not nowhere dense, it must be dense in the entire BCE set. A similar reasoning applies to the sBCE set, showing that it, too, is either dense or nowhere dense within the BCE set.

To prove the second part of Theorem 3, we combine two independent lemmas. The first lemma shows that for any BCE p and utility profile u , there exists a perturbation of u that makes p separated. In other words, every BCE of a given game is the limit of separated BCEs of nearby games.

The second lemma shows that for generic games, any BCE that is a limit of separated BCE in nearby games can also be approximated by separated BCEs when the game is held fixed. Specifically, the second lemma shows that the correspondence $u \mapsto \text{cl}(sBCE(u))$, which takes utilities to the closure of the sBCE set, generically is upper hemicontinuous (in fact, continuous). To prove it, we employ ideas from Blume and Zame (1994), who study the algebraic geometry of Nash, perfect, and sequential equilibria. In particular, we use the Tarsky-Sierberg Theorem to show that $u \mapsto \text{cl}(sBCE(u))$ has semi-algebraic graph, and so must be continuous for all utility profiles outside a closed low-dimensional manifold.

6. Welfare Analysis

In generic games, models based on exogenous information produce the same behavioral predictions as those incorporating acquired information, whether obtained through costly or flexible learning (Theorem 3). However, when the *entire* set of

predictions is considered—including players’ equilibrium net payoffs—significant differences arise in an open class of environments. In what follows, we identify these differences and highlight their consequences for welfare analysis.

As shown by Bergemann and Morris (2016), the set of outcome-value pairs attainable under exogenous information is given by

$$E_{ex} := \{(p, v) : p \in BCE, v = \bar{v}(p)\},$$

where BCE denotes the set of Bayes correlated equilibria, and $\bar{v}(p) = (\bar{v}_i(p))_{i \in I}$ is the profile of gross values associated with outcome p . In this section, we compare E_{ex} with the set

$$E_{ac} := \{(p, v) : p \in BCE, v \in [\underline{v}(p), \bar{v}(p)]\},$$

where $\underline{v}(p) = (\underline{v}_i(p))_{i \in I}$ represents the profile of uninformed values associated with outcome p .

Theorems 1 and 3 establish that, in generic games, the sets of outcome-value pairs attainable under costly and flexible learning share the same closure, given by E_{ac} . By focusing on generic games in this section, we highlight that models of exogenous and acquired information differ across a broad range of environments, provided that players’ welfare is taken into account.

By construction, $E_{ex} \subseteq E_{ac}$. The main result of this section characterizes the conditions under which $E_{ex} \subset E_{ac}$ and demonstrates the relevance of this difference for a social planner with robustness concerns.

Specifically, we consider a planner who values both the net utility of the players in the set I and the welfare of an outside observer $0 \notin I$. Although the observer does not directly participate in the game, their well-being is influenced by the game’s outcome, as captured by a utility function $u_0 : A \times \Theta \rightarrow \mathbb{R}$. This utility may represent a societal objective, the planner’s own payoff, or the aggregate externalities generated by the game for non-participating agents.

The planner trades off the payoffs of different individuals using a weighted-utilitarian welfare function. To elaborate, let $I_0 = I \cup \{0\}$ be the set of agents (including the observer), and let $w \in \mathbb{R}_+^{I_0}$ be a vector of weights expressing the degree the planner cares about each agent’s utility. Then the planner’s value from (p, v) is

$$W(p, v; u_0, w) := w_0 \sum_{a \in A, \theta \in \Theta} u_0(a, \theta) p(a, \theta) + \sum_{i \in I} w_i v_i.$$

The planner knows the physical environment as described by a generic base game, but expresses extreme uncertainty about players' information technology. We consider two scenarios: one where the planner assumes players have a fixed information structure, and another where costly or flexible learning occurs endogenously. In both scenarios, the planner takes a worst-case approach, evaluating the game according to the information technology and equilibrium that minimizes their welfare criterion:

$$W_{ex}(u_0, w) := \min_{(p,v) \in E_{ex}} W(p, v; u_0, w) = \min_{p \in BCE} W(p, \bar{v}(p); u_0, w)$$

$$W_{ac}(u_0, w) := \min_{(p,v) \in E_{ac}} W(p, v; u_0, w) = \min_{p \in BCE} W(p, \underline{v}(p); u_0, w).$$

Obviously, $W_{ex}(u_0, w) \geq W_{ac}(u_0, w)$ because $E_{ex} \subseteq E_{ac}$. We are ready to state the main result of this section.

Theorem 4. *The following statements are equivalent:*

- (i) *Exogenous and acquired information generate the same predictions, $E_{ex} = E_{ac}$.*
- (ii) *Each player i has an action that jeopardizes every other action, $\bigcap_{a_i \in A_i} J(a_i) \neq \emptyset$.*
- (iii) *Every $u_0 \in \mathbb{R}^{A \times \Theta}$ and every $w \in \mathbb{R}_+^{I_0}$ has $W_{ex}(u_0, w) = W_{ac}(u_0, w)$.*

The equivalence of (i) and (ii) characterizes when models of exogenous and acquired information yield the same predictions, relying on the concept of jeopardization (Myerson, 1997; see Section 5 for a precise definition). Specifically, $E_{ex} = E_{ac}$ holds if and only if a particular pattern of globally binding obedience constraints is satisfied: each player must possess an action that is always as advantageous as following the equilibrium prescription, irrespective of the BCE being played. Theorem 4, thus, complements Theorem 2 by further establishing a connection between the notion of jeopardization and learning incentives.

A key corollary of this result is that models of exogenous and acquired information yield different predictions across a broad range of environments. For instance, condition (ii) fails whenever every action is played in some strict Nash equilibrium—a common occurrence in many coordination games studied in applications (see, for instance, Example 4).

Finally, condition (iii) highlights the policy relevance of the gap between E_{ex} and E_{ac} : a gap emerges if and only if there exists a robust social welfare criterion that distinguishes between acquired and endowed information.

Theorem 4 demonstrates that differences in predictions between models with exogenous and acquired information are policy-relevant for at least some welfare criteria. However, the theorem does not address whether these differences are relevant for a specific welfare criterion in a given setting. In the following, we offer such a characterization for symmetric binary-action games.

A base game has **binary actions** if every player has exactly two feasible actions. It is **symmetric** if for every permutation $\phi : I \rightarrow I$, player i , action profile a , and payoff state θ ,

$$u_i(a_\phi, \theta) = u_{\phi(i)}(a, \theta),$$

where $a_\phi = (a_{\phi(j)})_{j \in I}$.¹² Symmetric binary-action games are commonly used in applications, such as team effort, regime change, and firm entry.¹³

A pair of weights and observer utility (w, u_0) is **symmetric** if w_i does not depend on i and $u_0(a, \theta)$ is invariant to permutations of the action profile. To characterize when $W_{ac}(u_0, w) < W_{ex}(u_0, w)$ for a fixed symmetric (u_0, w) , we adopt the following definition from Myerson (1997): a BCE p has **elementary incentives** if for all $i \in I$ and distinct $a_i, b_i \in A_i$,

$$\sum_{a_{-i}, \theta} (u_i(a_i, a_{-i}, \theta) - u_i(b_i, a_{-i}, \theta)) p(a_i, a_{-i}, \theta) > 0.$$

Thus, a BCE has elementary incentives if the obedience constraint always holds with strict inequality. Note that every player takes every action with positive probability in a BCE with elementary incentives. Let $\Delta_\pi^{sy}(A \times \Theta)$ be the set of symmetric outcomes.¹⁴

Proposition 1. *Let $\mathcal{G}$ be a symmetric, binary-action base game. For a symmetric social planner (u_0, w) , the following statements are equivalent:*

- (i) $W_{ex}(u_0, w) > W_{ac}(u_0, w)$.
- (ii) Every solution of the optimization problem

$$\min_{p \in \Delta_\pi^{sy}(A \times \Theta)} W(p, \underline{v}(p); u_0, w) \tag{2}$$

¹²A fortiori, $A_i = A_j$ for all $i, j \in I$ in a symmetric game.

¹³See Morris and Shin (2003) and Angeletos and Lian (2016) for surveys.

¹⁴Given $\mathcal{G}$ symmetric, p is **symmetric** if $p(a, \theta) = p(a_\phi, \theta)$ for every permutation ϕ and (a, θ) .

is a BCE with elementary incentives.

The proposition suggests a two-step recipe to determine if $W_{ac}(u_0, w) > W_{ex}(u_0, w)$ in symmetric binary-action games. First, minimize the planner’s welfare criterion, assuming players get their uninformed values, across all symmetric outcomes p while disregarding the obedience constraint. Then, verify whether all these minimizers are BCEs with elementary incentives. A direct corollary offers a simplified formula for computing $W_{ac}(u_0, w)$: whenever $W_{ac}(u_0, w) < W_{ex}(u_0, w)$, the value of (2) is $W_{ac}(u_0, w)$. The significant aspect of this result is that the minimization in (2) spans *all* symmetric outcomes, not only BCEs.

The core of Proposition 1 lies in the same observation that underlies the equivalence between parts (i) and (ii) of Theorem 4: a player’s uninformed and gross values coincide only if a player has single deviation that is as advantageous as following all of the equilibrium recommendations. With binary actions, each recommendation allows only one meaningful deviation, so a player’s uninformed and gross values diverge if and only if the player has no binding obedience constraint—in other words, if the player has elementary incentives.

7. Applications

7.1. Regime Change Games

This section demonstrates the practical relevance of our framework for welfare analysis in economies with information frictions. We examine regime change games, where the status quo collapses once a critical mass of agents opposes it. These well-studied games represent various social phenomena, including currency crises, bank runs, debt crises, and political revolts. We characterize utilitarian welfare and show that the policies a cautious planner would adopt depend on whether they treat agents’ information as an exogenous endowment or the outcome of endogenous learning.

In our application, there are n identical investors, $i \in I = \{1, \dots, n\}$, each deciding whether to speculate against (i.e., attack) a distressed financial institution ($a_i = 1$) or not ($a_i = 0$). Speculating costs $k \in (0, 1)$. If enough investors speculate, the institution fails (i.e., the attack succeeds), generating a profit of 1 to the speculators, and an externality of $-x$ to all passive investors, where $x \in (0, \infty)$. The payoff state, $\theta \in \Theta = \{2, \dots, n-1\}$, determines the number of speculators required for the attack

to succeed. The assumption that θ is always between 2 and $n - 1$ means that no single investor can go against the will of all others. We summarize these payoffs in the matrix below:

	$\sum_j a_j \geq \theta$	$\sum_j a_j < \theta$
$u_i(1, a_{-i}, \theta)$	$1 - k$	$-k$
$u_i(0, a_{-i}, \theta)$	$-x$	0

We adopt the perspective of a planner who assesses various policy interventions, represented by changes in the parameters of the game: k , x , and π . The planner's objective is a special case of the robust welfare functions considered in Section 6, with equal weighting across players and no external observer. As in that analysis, the planner faces uncertainty about how investors acquire information and thus employs a worst-case evaluation criterion. We compare the scenario where the planner assumes information is exogenously provided with the one where it is acquired via costly learning.

Under the assumption of exogenous information, the planner evaluates a policy vector (k, x, π) according to the following criterion:

$$W_{ex}(k, x, \pi) := \min_{p \in BCE} \sum_{i=1}^n \bar{v}_i(p),$$

where BCE is the set of Bayes correlated equilibria and $\bar{v}_i(p)$ is the gross value of outcome p for investor i . In the case of endogenous information, the planner's assessment is given by:

$$W_{ac}(k, x, \pi) := \min_{p \in BCE} \sum_{i=1}^n \underline{v}_i(p),$$

where $\underline{v}_i(p)$ is the uninformed value of outcome p for investor i . In this particular game, the sets iBCE and sBCE are dense within the BCE set, a fact easily verified by applying Theorem 2.¹⁵ Consequently,

$$W_{ac}(k, x, \pi) = \inf_{p \in iBCE} \sum_{i=1}^n \underline{v}_i(p) = \inf_{p \in sBCE} \sum_{i=1}^n \underline{v}_i(p).$$

¹⁵The game admits a strict BCE p where all players take both actions with positive probability—such as a convex combinations of the pure Nash equilibria in which everyone speculates or no-one speculates. Consequently, the BCE set is globally instrumental and separated.

The proposition below shows optimal policy depends on whether information is acquired or exogenously given. Part (i) gives the worst-case utilitarian welfare under information acquisition. This expression implies that a cautious planner who thinks information is acquired values policies that decrease the size of the externality x and the cost of speculating k . More pertinently, such a planner does not care about the institution's fundamentals π . Part (ii) shows the latter does not hold when information is exogenous: when the number of investors is relatively large ($n \geq 6$), one can increase the worst-case welfare under exogenous information by changing π .

Proposition 2. *The regime-change game has the following properties:*

(i) *The worst-case welfare under information acquisition is*

$$W_{ac}(k, x, \pi) = -\frac{nkx}{1+x}.$$

In particular, $W_{ac}(k, x, \pi)$ is independent of π .

(ii) *Suppose $n \geq 6$. Then, for all k and x , there exist π and π' such that*

$$W_{ex}(k, x, \pi) = W_{ac}(k, x, \pi) = W_{ac}(k, x, \pi') < W_{ex}(k, x, \pi').$$

A rough sketch of the proposition's proof is helpful for intuition. To prove part (i), we solve the relaxed program (2) detailed in Proposition 1. The planner's utilitarian objective, combined with the focus on symmetric outcomes, makes the program equivalent to solving a zero sum game between nature and a generic investor i^* . In this game, investor i^* chooses whether to attack or not without receiving any information. Nature on the other hand, has complete control of the other investors, subject to the restriction that their behavior is symmetric.

We show nature's optimal decision in this zero sum game is characterized by two features. First, investor i^* is never pivotal: whether the attack succeeds or fails cannot depend on i^* 's action. This requirement comes from nature wishing to maximize the miscoordination between i^* action and the institution's failure, and the fact that investor i^* always takes the correct action when pivotal. The second feature is that i^* is indifferent between attacking and not attacking; otherwise, nature could predict i^* 's action, perfectly miscoordinate that action with the institution's failure, thereby making that action suboptimal.

We then solve (2) by observing that nature can achieve both of these features by creating perfect coordination among investors: they either all attack or none do. For an explanation, note such coordination means neither investor i^* nor θ has any influence over the attack's fate, and so investor i^* is not pivotal. Moreover, since i^* is uninformed, all that matters for his incentives is the probability all other investors attack. In particular, i^* is indifferent between attacking and not whenever this probability is $k/(1+x)$. Hence, the outcome p^* in which all investors attack together with probability $k/(1+x)$ and none of them attack otherwise—independent of θ —solves the program (2).

Armed with the above, we obtain part (i) of Proposition 2 as follows. First, note that p^* is a BCE, and so must also minimize welfare across all symmetric BCEs. And second, we show symmetry of the game and the planner's objective implies that minimizing welfare across all symmetric BCEs is without loss. Hence, p^* is the worst BCE for the planner under information acquisition.

Notice, however, that p^* cannot be the worst case under exogenous information. The reason is that perfect coordination makes the signal about others' actions highly valuable, and in the absence of information costs, players do not have to pay for this value. The consequence is a potential wedge between the worst case welfare under exogenous information and costly learning. In the online appendix, we build on Proposition 1 to fully characterize the situations in which such a wedge occurs. We then use this wedge to show that when $n \geq 6$, one can construct a π where this wedge exists, and a π' where it does not. Part (ii) of Proposition 2 follows.

Thus, Proposition 2 demonstrates that the planner's policy choices may differ depending on whether she views information as exogenous or endogenous. To explain, suppose the planner has the option of switching from prior π to prior π' at a cost. Since $W_{ac}(k, x, \pi') = W_{ac}(k, x, \pi)$, a planner who believes investors' information is the result of endogenous learning would never opt to pay for this switch. By contrast, there is a cost the planner will pay to make this change if she treats information as exogenous, since $W_{ex}(k, x, \pi') > W_{ex}(k, x, \pi)$.

For clarity, we have focused on worst-case analysis. However, the full range of possible utilitarian welfare values under information acquisition is straightforward to characterize. The maximum achievable welfare is $n(1-k)$, corresponding to a BCE where all investors coordinate on attacking. Proposition 2 provides the worst-case welfare: $-nkx/(1+x)$. Thus, the range of utilitarian welfare values under information

acquisition spans the interval $[-nkx/(1+x), n(1-x)]$.

7.2. Informational Simplicity

Consider an auction for a single good of uncertain quality. Before bidding, each buyer can *flexibly* choose what information to acquire about both their own valuation and the valuations of others. What auction mechanisms are *informationally simple*—meaning they induce buyers to learn only about their own value?

Gleyze and Pernoud (2023) raise this question and identify a class of flexible information technologies under which every informationally simple mechanism is *separable*, meaning it is strategically equivalent to a collection of individual decision problems. Their result highlights a fundamental tension between flexible learning and the private values assumption, a cornerstone of classic mechanism design.

In this section, we expand on Gleyze and Pernoud (2023) and ask: which separable mechanisms remain informationally simple across *all* flexible information technologies? We show that relaxing all assumptions about information gathering—except for flexibility—leads to a stark result: informational simplicity requires not only separability but also the complete absence of learning incentives, resulting in no information acquisition in equilibrium. This finding further underscores the conflict between flexible learning and the private values assumption.

Before stating our result, we need a few definitions. A base game is **separable** if:

- The state space is a Cartesian product, $\Theta = \prod_{i \in I} \Theta_i$.
- The prior is a product measure, $\pi = \prod_{i \in I} \pi_i$.
- For each player i there exists a function $\bar{u}_i : A_i \times \Theta_i \rightarrow \mathbb{R}$ such that

$$u_i(a_i, a_{-i}, \theta) - u_i(b_i, a_{-i}, \theta) = \bar{u}_i(a_i, \theta_i) - \bar{u}_i(b_i, \theta_i)$$

for all $a_i, b_i \in A_i$, $a_{-i} \in A_{-i}$, and $\theta \in \Theta$.

In other words, a separable game consists of independent individual decision problems with payoff externalities but no strategic externalities.

An outcome $p \in \Delta(A \times \Theta)$ of a separable base game is **informationally simple** if $p = \prod_{i \in I} p_i$ is a product measure, with $p_i \in \Delta(A_i \times \Theta_i)$. It is straightforward to show that an outcome is informationally simple if and only if it is induced by an

information structure in which every player only learns about their own payoff state. One obvious example of an informationally simple outcome is one in which the players do not learn anything. Formally, an outcome $p \in \Delta(A \times \Theta)$ **involves no learning** if a_i is independent of (a_{-i}, θ) for every player i . The main result of this section shows a separable base game is informationally simple for all flexible technology if and only if all separated BCEs involve no learning.

Proposition 3. *Let $\mathcal{G}$ be a separable base game where $|\text{supp}(\pi_i)| \geq 2$ for at least two distinct players i . Then, the following conditions are equivalent:*

- (i) *Every sBCE is informationally simple.*
- (ii) *Every player i has an action a_i that is always a best response, $\bar{u}_i(a_i, \theta_i) \geq \bar{u}_i(b_i, \theta_i)$ for all b_i and θ_i .*
- (iii) *Every sBCE involves no learning.*

Thus, the only separable games that guarantee players have no incentive to acquire information about others' valuations under all flexible information technologies (condition (i)) are those in which information has no value (condition (ii)). Consequently, in these games, the only sustainable outcomes under flexible information acquisition are those where players remain entirely uninformed (condition (iii)).

Intuitively, Proposition 3 stems from the fact that an experiment incorporating information about both θ_i and θ_j is not necessarily more informative (in Blackwell's sense) than an experiment that conveys the same information about θ_i while revealing nothing about θ_j . Consequently, the monotonicity of player i 's cost function does not preclude the possibility that learning about θ_i and θ_j together is less costly than acquiring information solely about θ_i . In fact, under certain monotone cost functions, player i may opt to acquire non-instrumental information about θ_j as a means of improving their knowledge of θ_i . The proposition establishes that in separable games, the only way to prevent player i from acquiring information about θ_j across all such cost functions is to ensure they derive no benefit from learning.

The intuition above connects to the structure that Gleyze and Pernoud (2023) impose on their information technologies. Specifically, while Gleyze and Pernoud (2023) show that the separability of the environment is necessary for informational simplicity, they establish sufficiency only under the additional assumption that learning about θ_i alone is cheaper than learning about θ_i and θ_j jointly. Proposition 3 strengthens this

result by showing that, without such an assumption, the only separable games that guarantee informational simplicity are those in which information has no value.¹⁶

7.3. Mechanism Design with Aftermarkets

Aftermarkets are a concern in many mechanism design problems. Principals often fear that information elicited from agents might spill over, negatively affecting subsequent interactions outside their control (see, for example, Angeletos and Werning, 2006; Calzolari and Pavan, 2006; Dworczak, 2020). This section applies our results to a design problem where the agents' information derives from costly information acquisition. We demonstrate that aftermarket concerns can drive principals to design non-generic games in which instrumental BCEs are nowhere dense in the set of BCEs.

We consider an environment featuring a principal and a finite number of agents. The agents engage in an information acquisition game, $(\mathcal{G}, \mathcal{T})$. The principal can modify the agents' incentives in the information acquisition game by committing to a payment scheme, $t(a, \theta) = (t_i(a, \theta))_{i \in I}$. The transfers are player-specific and may depend on all agents' actions as well as on the state of fundamentals. Agents' preferences are quasi-linear: the utility that a player i derives from an action profile a in a state θ is $u_i(a, \theta) + t_i(a, \theta)$.

The transfers must satisfy two feasibility constraints. First, the agents have limited liability, meaning transfers are non-negative: $t_i(a, \theta) \geq 0$ for all $i \in I$, $a \in A$, and $\theta \in \Theta$. Second, the principal has a budget constraint. Specifically, the principal cannot promise more than an amount of $b \geq 0$ to each agent. We assume the budget constraint must hold for every realization of actions and fundamental states: $t_i(a, \theta) \leq b$ for all $i \in I$, $a \in A$, and $\theta \in \Theta$.

The principal is concerned that the agents' actions in this game, $a = (a_i)_{i \in I}$, might become public, revealing information about the state of fundamentals, θ , that adversely affect the functioning of an aftermarket. An example of such an aftermarket could be akin to the beauty contest game described by Morris and Shin (2002), though we do not model it explicitly. Instead, we assume the principal aims to minimize the informativeness of a regarding θ as captured by a function $\mathbb{I} : \Delta(A \times \Theta) \rightarrow \mathbb{R}_+$.

¹⁶We note that Proposition 3 remains valid when sBCE is replaced with iBCE. However, separable environments should not be expected to be necessary for informational simplicity under costly learning alone. When information choice is constrained, a player may only be able to learn about their own valuation by acquiring information about others' valuations.

The quantity $\mathbb{I}(p)$ measures the information conveyed by the agents' actions about the fundamental state, based on their joint distribution p . We assume $\mathbb{I}$ is a lower semicontinuous function¹⁷ with the property that $\mathbb{I}(p) \geq 0$, with equality if and only if a and θ are independent random variables. Many statistical measures of information satisfy these properties, including Shannon's mutual information.

The principal is confident in their understanding of the base game $\mathcal{G}$, but is unsure about the information technology $\mathcal{T}$. The only assumption the principal is willing to make about $\mathcal{T}$ is that learning is costly. Hence, $\mathcal{T}$ does not a priori prevent the agents from acquiring any particular piece of information, but agents only obtain information if it is valuable for their decisions. Subject to this assumption, the principal takes a worst-case approach. Thus, the principal's problem is to design a feasible transfer scheme $t : I \times A \times \Theta \rightarrow [0, b]$ to minimize

$$\sup \{ \mathbb{I}(p) : p \in iBCE(u + t) \}.$$

Notice our principal is interested only in reducing the amount of information that may leak to the aftermarket. Whereas we focus on this case for clarity, it is straightforward to generalize our analysis to allow the principal to directly care about the material consequences of the agents' behavior.

When the principal operates under a loose budget constraint, the mechanism design problem admits an easy solution. When b is high compared to the stakes in the original base game $\mathcal{G}$, the principal can attach a large payment to a fixed action for each agent, making such an action strictly dominant. Consequently, each agent plays a deterministic action, regardless of the opponent's behavior and the fundamental state. Thus, in any iBCE (in fact, in any BCE), the players' actions are independent of each other and of the state, which minimizes the transmission of information to the aftermarket.

The design problem is more intricate when the principal's budget constraint is tight. If b is small, the principal might not be able to curtail the informational content of the agents' behavior simply by making certain actions unambiguously attractive. Instead, the principal must rely on more subtle transfer schemes to reduce the agents' incentives to gather information. The following proposition, the main result of this section, shows that in such situations the planner may choose to exploit

¹⁷Thus, every convergent sequence $(p^n)_{n=1}^\infty$ in $\Delta(A \times \Theta)$ has $\liminf_{n \rightarrow \infty} \mathbb{I}(p^n) \geq \mathbb{I}(\lim_{n \rightarrow \infty} p^n)$.

agents' learning incentives.

Proposition 4. *Suppose at least two players have two or more actions, and that Θ has at least two positive-probability elements. For each $b > 0$ sufficiently small, there is an open set of $u \in \mathbb{R}^{I \times A \times \Theta}$ such that, for each optimal transfer scheme t ,*

$$\sup \{\mathbb{I}(p) : p \in iBCE(u + t)\} = 0 < \sup \{\mathbb{I}(p) : p \in BCE(u + t)\}.$$

In particular, $iBCE(u + t)$ is nowhere dense in $BCE(u + t)$.

For an intuition, suppose u^* is the state-independent matching pennies game from Example 2. In this example, the unique instrumental BCE has both players randomizing in a way that is independent of the underlying state. However, the game also contains many non-instrumental BCEs, including ones where one player's actions are highly correlated with θ . Our proof shows that such BCEs exist in all nearby games as well. Thus, if the budget is small and u is close enough to u^* , the only way the principal can preclude the players from leaking information about the state is by inducing a game where instrumental BCEs are not dense in the BCE set. Moreover, whenever u is within b from u^* , the principal could prevent the players from leaking such information by making the game equal to u^* . Since the set of utilities for which such a transformation is feasible is open, the proposition follows.¹⁸

8. Discussion

We conclude by discussing some of our assumptions and future research directions.

Posterior separable costs. Applications may suggest additional assumptions about information technologies beyond costly and flexible learning. A natural direction for future research is to refine this paper's predictions by incorporating further considerations on information costs.

For instance, one could impose *posterior separability* (Caplin, Dean, and Leahy 2022), a common assumption in models of flexible information acquisition. A preliminary analysis reveals that assuming posterior separability alongside flexible learning would not alter the predictions regarding attainable outcomes: sBCE would remain

¹⁸We note that Proposition 4 remains valid if we replace $iBCE$ with $sBCE$.

the relevant solution concept. This follows from a result in Denti (2024), which states that if an experiment can be rationalized through a strictly monotone cost function, it can also be rationalized via a strictly monotone cost function that is posterior separable. However, it is unclear whether the welfare bounds established in Theorem 1 would be affected; in proving Theorem 1, we construct cost functions that are not posterior separable to achieve these bounds. Investigating the welfare implications of posterior separability thus remains an intriguing avenue for future research.

Correlation devices. Throughout this paper, we allow players unrestricted (albeit costly) access to correlation devices. However, in practice, there may be constraints on the extent to which players can share information sources. The most restrictive case eliminates all possibilities for correlation, requiring that players’ experiments pertain only to the fundamental state—i.e., the set Z is a singleton in every information technology. Adapting our analysis to this scenario is straightforward: Theorem 1 remains valid with iBCE and sBCE replaced by *conditionally independent* iBCE and sBCE, meaning that recommendations are conditionally independent across players given θ .

A less restrictive limitation on correlation would be to assume that a non-trivial correlation device is available but fixed across all information technologies—that is, all information technologies share the same Z and ζ , which are known to the analyst. This case is, in fact, a special instance of the scenario just discussed: it suffices to redefine the payoff-relevant state to include the fixed correlation device. In our framework, while θ is nominally payoff-relevant, actual utilities may be independent of it.

All information technologies. Theorem 1 characterizes the predictions consistent with technologies that are costly or flexible. One might wonder what predictions are consistent without these restrictions. A simple argument shows that an outcome-value pair (p, v) is attainable in equilibrium in an information acquisition game for some *arbitrary* technology if and only if p is a BCE and for all players i , $v_i \in [\underline{v}_i(p), \bar{v}_i(p)]$. In other words, if an analyst seeks robustness in information acquisition games without making any assumptions about players’ learning abilities, BCE is the appropriate solution concept.

Predictions across games. A natural next step in the analysis is to explore robust predictions *across* information acquisition games. Specifically, we have in mind the following question: Given a finite collection of base games $\mathcal{G}_1, \dots, \mathcal{G}_n$ that share the same set of players I and the same payoff state (Θ, π) , what are the set of outcome-value profiles $(p_1, v_1), \dots, (p_n, v_n)$ that can arise as equilibria of $(\mathcal{G}_1, \mathcal{T}), \dots, (\mathcal{G}_n, \mathcal{T})$ for some common information technology $\mathcal{T}$ that accounts for costly or flexible learning? This paper focuses on the case $n = 1$, while Bergemann, Brooks, and Morris (2022) analyze the setting where information is exogenous, and Caplin and Dean (2015) study the single-agent case without restricting the information technology.

A preliminary analysis suggests that this question remains meaningful even when all information technologies are considered. For example, comparing the results of Bergemann, Brooks, and Morris (2022) and Caplin and Dean (2015) in the single-agent case indicates that endogenous learning generally expands the set of attainable outcome profiles $p_1, \dots, p_n$ under exogenous information. In contrast, when $n = 1$, the behavioral predictions under both knowledge regimes coincide, yielding the set of Bayes correlated equilibria, as discussed above.

Exploring this direction further will be worthwhile.

Sequential games. Interpreting the base game as the normal form of a sequential game allows our analysis to apply to dynamic strategic interactions in which all information is gathered at the outset. Ravid, Roesler, and Szentes (2022) provides a recent study of a setting with this timeline. However, an important caveat is that our solution concept for information acquisition games is standard Nash equilibrium, which does not incorporate additional requirements typically imposed in dynamic games, such as sequential rationality. Incorporating these refinements could offer an interesting extension of our analysis.

Another promising direction in the context of sequential games is to consider scenarios where information acquisition occurs as the game unfolds. This would naturally lead to study costly learning not only about Nature’s moves but also about other players’ past actions. Denti (2024) provides a recent study of flexible monitoring in signaling games.

Both Ravid, Roesler, and Szentes (2022) and Denti (2024) highlight economically relevant environments where flexible information acquisition yields sharper behavioral predictions than exogenous information. This suggests that sequential games

provide a compelling avenue for further research on robust predictions in information acquisition games. This paper serves as a foundation for this broader research agenda.

References

- C. D. Aliprantis and K. C. Border. *Infinite Dimensional Analysis*. Springer, third edition, 2006. ISBN 9783540295860.
- G.-M. Angeletos and C. Lian. Incomplete information in macroeconomics: Accommodating frictions in coordination. In *Handbook of macroeconomics*, volume 2, pages 1065–1240. Elsevier, 2016.
- G.-M. Angeletos and I. Werning. Crises and prices: Information aggregation, multiplicity, and volatility. *American Economic Review*, 96(5):1720–1736, 2006.
- R. J. Aumann. Subjectivity and correlation in randomized strategies. *Journal of Mathematical Economics*, 1:67–96, 1974.
- R. J. Aumann. Correlated equilibrium as an expression of Bayesian rationality. *Econometrica*, 55:1–18, 1987.
- D. Bergemann and S. Morris. Robust predictions in games with incomplete information. *Econometrica*, 81:1251–1308, 2013.
- D. Bergemann and S. Morris. Bayes correlated equilibrium and the comparison of information structures in games. *Theoretical Economics*, 11(2):487–522, 2016.
- D. Bergemann and S. Morris. Information design: A unified perspective. *Journal of Economic Literature*, 57:44–95, 2019.
- D. Bergemann, B. Brooks, and S. Morris. The limits of price discrimination. *American Economic Review*, 105(3):921–957, 2015.
- D. Bergemann, B. Brooks, and S. Morris. First-price auctions with general information structures: Implications for bidding and revenue. *Econometrica*, 85:107–143, 2017.
- D. Bergemann, B. Brooks, and S. Morris. Counterfactuals with latent information. *American Economic Review*, 112(1):343–368, 2022.

- D. Blackwell. Comparison of experiments. In J. Neyman, editor, *Proceedings of the 2nd Berkeley Symposium on Mathematical Statistics and Probability*, pages 93–102. University of California Press, 1951.
- D. Blackwell. Equivalent comparisons of experiments. *Annals of Mathematical Statistics*, 24:265–272, 1953.
- L. E. Blume and W. R. Zame. The algebraic geometry of perfect and sequential equilibrium. *Econometrica*, 62:783–794, 1994.
- A. Brandenburger and A. Friedenberg. Intrinsic correlation in games. *Journal of Economic Theory*, 141(1):28–67, 2008.
- B. Brooks and S. Du. Optimal auction design with common values: An informationally robust approach. *Econometrica*, 89:1313–1360, 2021.
- A. Calvó-Armengol. The set of correlated equilibria of 2×2 games. *mimeo*, 2006.
- G. Calzolari and A. Pavan. On the optimality of privacy in sequential contracting. *Journal of Economic theory*, 130(1):168–204, 2006.
- A. Caplin and M. Dean. Revealed preference, rational inattention, and costly information acquisition. *American Economic Review*, 105:2183–2203, 2015.
- A. Caplin, M. Dean, and J. Leahy. Rationally inattentive behavior: Characterizing and generalizing Shannon entropy. *Journal of Political Economy*, 130:1676–1715, 2022.
- G. Carroll. Information games and robust trading mechanisms. *mimeo*, 2019a.
- G. Carroll. Robust incentives for information acquisition. *Journal of Economic Theory*, 181:382–420, 2019b.
- G. de Clippel and K. Rozen. Communication, perception, and strategic obfuscation. 2022. URL <https://sites.google.com/brown.edu/kareenrozen>.
- T. Denti. Posterior separable cost of information. *American Economic Review*, 112:3215–59, 2022.
- T. Denti. Unrestricted information acquisition. *Theoretical Economics*, 18(3):1101–1140, 2023.

- T. Denti. Costly monitoring in signaling games. 2024. URL <https://sites.google.com/site/tommasojdenti>.
- T. Denti, M. Marinacci, and A. Rustichini. Experimental cost of information. *American Economic Review*, 112:3106–23, 2022.
- L. Doval and J. C. Ely. Sequential information design. *Econometrica*, 88:2575–2608, 2020.
- P. Dworczak. Mechanism design with aftermarkets: Cutoff mechanisms. *Econometrica*, 88(6):2629–2661, 2020.
- F. Forges. Five legitimate definitions of correlated equilibrium in games with incomplete information. *Theory and Decision*, 35:277–310, 1993.
- S. Gleyze and A. Pernoud. Informationally simple incentives. *Journal of Political Economy*, 131(3):802–837, 2023.
- B. Hébert and J. La’O. Information acquisition, efficiency, and nonfundamental volatility. *Journal of Political Economy*, 131(10):2666–2723, 2023.
- C. Hellwig and L. Veldkamp. Knowing what others know: Coordination motives in information acquisition. *Review of Economic Studies*, 76:223–251, 2009.
- B. L. Lipman and S. Srivastava. Computation as a correlation device. *Games and Economic Behavior*, 2:154–172, 1990.
- E. Lipnowski and D. Ravid. Predicting choice from information costs. *arXiv preprint arXiv:2205.10434*, 2022.
- B. Maćkowiak, F. Matějka, and M. Wiederholt. Rational inattention: A review. *Journal of Economic Literature*, 61:226–273, 2023.
- F. Matějka and A. McKay. Rational inattention to discrete choices: A new foundation for the multinomial logit model. *American Economic Review*, 105:272–298, 2015.
- S. Morris and H. S. Shin. Unique Equilibrium in a Model of Self-Fulfilling Currency Attacks. *American Economic Review*, 88(3):587–597, 1998.
- S. Morris and H. S. Shin. Social value of public information. *American Economic Review*, 92(5):1521–1534, 2002.

- S. Morris and H. S. Shin. Global games: theory and applications. In M. Dewatripont, L. P. Hansen, and S. Turnovsky, editors, *Advances in Economics and Econometrics (Proceedings of the Eighth World Congress of the Econometric Society)*, pages 56–114. Cambridge University Press, 2003.
- S. Morris and M. Yang. Coordination and continuous stochastic choice. *The Review of Economic Studies*, 89(5):2687–2722, 2022.
- D. P. Myatt and C. Wallace. Endogenous information acquisition in coordination games. *Review of Economic Studies*, 79:340–374, 2012.
- R. B. Myerson. Optimal coordination mechanisms in generalized principal–agent problems. *Journal of mathematical economics*, 10(1):67–81, 1982.
- R. B. Myerson. Dual reduction and elementary games. *Games and Economic Behavior*, 21:183–202, 1997.
- H. Pei and B. Strulovici. Robust implementation with costly information. *Review of Economic Studies*, 92(1):476–505, 2025.
- D. Ravid. Ultimatum Bargaining with Rational Inattention. *American Economic Review*, 110:2948–63, 2020.
- D. Ravid, A.-K. Roesler, and B. Szentes. Learning before trading: on the inefficiency of ignoring free information. *Journal of Political Economy*, 130:346–387, 2022.
- R. T. Rockafellar. *Convex analysis*. Princeton University Press, 1970.
- T. Roughgarden. *Twenty lectures on algorithmic game theory*. Cambridge University Press, 2016.
- Y. Viossat. Is having a unique equilibrium robust? *Journal of Mathematical Economics*, 44:1152–1160, 2008.
- R. Wilson. *Game-theoretic analyses of trading processes*, page 33–70. Econometric Society Monographs. Cambridge University Press, 1987.
- M. Yang. Coordination with flexible information acquisition. *Journal of Economic Theory*, 158:721–738, 2015.

W. Zhong. Optimal dynamic information acquisition. *Econometrica*, 90:1537–1582, 2022.

Online Appendix

A. Proof of Theorem 1

As outlined in the main text, the proof of Theorem 1 builds on two auxiliary results. The first is an implementation of BCE with exogenous information that departs from the canonical construction in Bergemann and Morris (2016). The second is a single-agent lemma from Denti (2024) concerning flexible information acquisition. We present these two results first, before proceeding to the proof of Theorem 1 itself.

A.1. An Implementation of BCE

Let p be a BCE. We now define an information structure $\mathcal{S} = (Z, \zeta, (X_i, \xi_i)_{i \in I})$ and an equilibrium $\sigma = (\sigma_i)_{i \in I}$ of the incomplete information game $(\mathcal{G}, \mathcal{S})$ that induce p .

For every player i and pair of actions $a_i, b_i \in \text{supp}_i(p)$, define $a_i \sim_i b_i$ by $p_{a_i} = p_{b_i}$. Note that $\sim_i$ is an equivalence relation on $\text{supp}_i(p)$. Let Z_i be the corresponding set of equivalence classes. We take $Z = \prod_{i \in I} Z_i$. To simplify the exposition, we write $a \in z$ if $a_i \in z_i$ for all $i \in I$. Given this convention, we define $\zeta : \Theta \rightarrow Z$ by $\zeta(z|\theta) = \sum_{a \in z} p(a, \theta) / \pi(\theta)$.¹⁹

For every player i , we take $X_i = Z_i$, and we define $\xi_i : Z \times \Theta \rightarrow \Delta(X_i)$ by $\xi_i(x_i|z, \theta) = 1$ if $x_i = z_i$, and $\xi_i(x_i|z, \theta) = 0$ otherwise. We also define $\sigma_i : X_i \rightarrow \Delta(A_i)$ by $\sigma_i(a_i|x_i) = p(a_i) / \sum_{b_i \in x_i} p(b_i)$ if $a_i \in x_i$, and $\sigma_i(a_i|x_i) = 0$ otherwise.

Lemma 2. *The outcome induced by (ξ, σ) is p , that is, for all $a \in A$ and $\theta \in \Theta$,*

$$p(a, \theta) = \sum_z \left[\prod_{i \in I} \sigma_i(a_i|z_i) \right] \zeta(z|\theta) \pi(\theta). \quad (3)$$

To prove this lemma, we need an intermediate result.

Claim 1. For all $z \in Z$, $a \in z$, and $\theta \in \Theta$,

$$p(a, \theta) = \left[\prod_{i \in I} \sigma_i(a_i|z_i) \right] \sum_{b \in z} p(b, \theta).$$

¹⁹If $\pi(\theta) = 0$, set $\zeta(z|\theta) = \sum_{a \in z, \theta \in \Theta} p(a, \theta)$; this case will not play any role in what follows.

Proof. It suffices to show that for all $i \in I$,

$$p(a, \theta) = \sigma_i(a_i | z_i) \sum_{b_i \in z_i} p(b_i, a_{-i}, \theta). \quad (4)$$

The desired result then follows from reasoning by induction on the number of players.

So, fix some player i . By Bayes rule, $p(a, \theta) = p(a_i) p_{a_i}(a_{-i}, \theta)$. Recall that $b_i \in z_i$ if and only if $p_{b_i} = p_{a_i}$; in addition, $\sum_{b_i \in z_i} \sigma_i(b_i | z_i) = 1$. Therefore,

$$\begin{aligned} p(a_i, a_{-i}, \theta) &= p(a_i) p_{a_i}(a_{-i}, \theta) = p(a_i) \sum_{b_i \in z_i} \sigma_i(b_i | z_i) p_{a_i}(a_{-i}, \theta) \\ &= p(a_i) \sum_{b_i \in z_i} \sigma_i(b_i | z_i) p_{b_i}(a_{-i}, \theta). \end{aligned}$$

Substituting in the definition of σ_i , one obtains (4). $\square$

Proof of Lemma 2. Fix $a \in A$ and $\theta \in \Theta$. If $p(a_i) = 0$ for some player i , then $\sigma(a_i | x_i) = 0$ for all $x_i \in X_i$. Thus, both sides of (3) are equal to zero.

Suppose now that $p(a_i) > 0$ for all players i . Take $z^* \in Z$ such that $a \in z^*$. Using Claim 1 we get that

$$p(a, \theta) = \left[\prod_{i \in I} \sigma_i(a_i | z_i^*) \right] \sum_{b \in z^*} p(b, \theta).$$

The equality (3) follows from noting that $\prod_{i \in I} \sigma_i(a_i | z_i) > 0$ if and only if $z = z^*$. $\square$

Next we prove a property of best-response sets that we will use to embed $\mathcal{S}$ and σ into an information acquisition game. Given $\Omega := Z \times \Theta$, we define $\rho \in \Delta(\Omega)$ and $w_i : A_i \times \Omega \rightarrow \mathbb{R}$ by $\rho(z, \theta) = \zeta(z | \theta) \pi(\theta)$ and

$$w_i(a_i, z, \theta) = \sum_{a_{-i}} u_i(a_i, a_{-i}, \theta) \prod_{j \neq i} \sigma_j(a_j | z_j).$$

Let $\mu(x_i)$ be the ex-ante probability that ξ_i generates x_i , and let $\mu_{x_i} \in \Delta(\Omega)$ be i 's posterior beliefs about $\omega = (z, \theta)$ when x_i is observed. Note that μ_{x_i} is different from p_{a_i} , which is i 's posterior beliefs about (a_{-i}, θ) given recommendation a_i . The next

result relate the following two best-response sets:

$$\begin{aligned} BR(\mu_{x_i}) &= \arg \max_{b_i \in A_i} \sum_{\omega} w_i(b_i, \omega) \mu_{x_i}(\omega), \\ BR(p_{a_i}) &= \arg \max_{b_i \in A_i} \sum_{\theta, a_{-i}} u_i(b_i, a_{-i}, \theta) p_{a_i}(a_{-i}, \theta). \end{aligned}$$

Lemma 3. *For all $x_i \in X_i$ and $a_i \in x_i$, $BR(p_{a_i}) = BR(\mu_{x_i})$.*

Proof. It is enough to show that for all $a_{-i} \in A_{-i}$ and $\theta \in \Theta$,

$$p_{a_i}(a_{-i}, \theta) = \sum_z \left[\prod_{j \neq i} \sigma_j(a_j | z_j) \right] \mu_{x_i}(z, \theta). \quad (5)$$

Towards this goal, we first observe that for all $a_{-i} \in A_{-i}$ and $\theta \in \Theta$,

$$\begin{aligned} p(a_i, a_{-i}, \theta) &= \sum_{x'_i, x_{-i}} \left[\sigma_i(a_i | x'_i) \prod_{j \neq i} \sigma_j(a_j | x_j) \right] \zeta(x'_i, x_{-i} | \theta) \pi(\theta) \\ &= \sigma_i(a_i | x_i) \sum_{x_{-i}} \left[\prod_{j \neq i} \sigma_j(a_j | x_j) \right] \zeta(x_i, x_{-i} | \theta) \pi(\theta) \\ &= \frac{p(a_i)}{\sum_{b_i \in x_i} p(b_i)} \sum_{x_{-i}} \left[\prod_{j \neq i} \sigma_j(a_j | x_j) \right] \zeta(x_i, x_{-i} | \theta) \pi(\theta), \end{aligned} \quad (6)$$

where the first equality holds by Lemma 2, the second equality because $\sigma_i(a_i | x'_i) > 0$ if and only if $x'_i = x_i$, and the third equality by definition of $\sigma_i(a_i | x_i)$. Summing both sides of (6) over a_{-i} and θ , we obtain that

$$p(a_i) = \frac{p(a_i) \mu(x_i)}{\sum_{b_i \in x_i} p(b_i)}.$$

Thus, $\mu(x_i) = \sum_{b_i \in x_i} p(b_i)$. We plug this finding in (6) and obtain that for all $a_{-i} \in A_{-i}$ and $\theta \in \Theta$,

$$p(a_i, a_{-i}, \theta) = \frac{p(a_i)}{\mu(x_i)} \sum_{x_{-i}} \left[\prod_{j \neq i} \sigma_j(a_j | x_j) \right] \zeta(x_i, x_{-i} | \theta) \pi(\theta).$$

The equality (5) follows from observing that $\mu_{x_i}(z, \theta) = \zeta(z | \theta) \pi(\theta) / \mu(x_i)$ if $z_i = x_i$, and $\mu_{x_i}(z, \theta) = 0$ otherwise. $\square$

A consequence of this lemma is that, by obedience, the profile of action plan σ is an equilibrium of the incomplete information game $(\mathcal{G}, \mathcal{S})$.

A.2. A Single-Agent Lemma on Flexible Learning

We take the perspective of an individual i who has to choose an action $a_i \in A_i$ whose utility $w_i(a_i, \omega)$ depends on an uncertain state of nature $\omega \in \Omega$. Both A_i and Ω are finite sets. Let $\rho \in \Delta(\Omega)$ be the prior distribution of the state. Before choosing an action, the decision maker can run an experiment $\xi_i : \Omega \rightarrow \Delta(X_i)$ at a cost $C_i(\xi_i) \in \mathbb{R}_+$, where X_i is a finite signal space. Any experiment with signal space X_i is feasible: $\mathcal{E}_i = \Delta(X_i)^\Omega$. Overall, the decision maker faces the following information acquisition problem:

$$\max_{\xi_i \in \mathcal{E}_i, \sigma_i \in \Sigma_i} \left[\sum_{\omega, x_i, a_i} w_i(a_i, \omega) \sigma_i(a_i | x_i) \xi_i(x_i | \omega) \rho(\omega) \right] - C_i(\xi_i) \quad (7)$$

where Σ_i is the set of all action plans $\sigma_i : X_i \rightarrow \Delta(A_i)$.

In accordance with the terminology used in the main text, $\xi_i \succsim \xi'_i$ if there is a Markov kernel $\gamma : X_i \rightarrow \Delta(X_i)$ such that for every $x_i \in X_i$ and $\omega \in \Omega$ with $\rho(\omega) > 0$,

$$\xi'_i(x_i | \omega) = \sum_{x'_i \in X_i} \gamma(x_i | x'_i) \xi(x'_i | \omega).$$

A cost function is **strictly increasing** if $\xi_i \succ \xi'_i$ implies $C_i(\xi_i) > C_i(\xi'_i)$.

Among other results, Denti (2024) characterizes the pairs (ξ_i, σ_i) that are optimal solutions of (7) for some strictly increasing C_i . To describe the characterization, let $\mu(x_i)$ be the ex-ante probability that ξ_i generates x_i . For every x_i such that $\mu(x_i) > 0$, we denote by $\mu_{x_i} \in \Delta(\Omega)$ the agent's posterior belief. Finally, let $BR(\mu_{x_i})$ be the set of best response given beliefs μ_{x_i} .

Lemma 4 (Denti, 2024). *A pair (ξ_i, σ_i) is an optimal solution of (7) for some strictly increasing C_i if and only if the following conditions hold:*

- (i) *For all x_i with $\mu(x_i) > 0$, $\sigma_i(BR(\mu_{x_i}) | x_i) = 1$.*
- (ii) *For all x_i and x'_i with $\mu(x_i)\mu(x'_i) > 0$, $\mu_{x_i} \neq \mu_{x'_i}$ implies $BR(\mu_{x_i}) \cap BR(\mu_{x'_i}) = \emptyset$.*

A.3. Proof of Theorem 1, Part (i): “If” Direction

Let p be an iBCE and let v be a profile values such that, for every player i , $\underline{v}_i(p) = \bar{v}_i(p) = v_i$ or $v_i \in [\underline{v}_i(p), \bar{v}_i(p))$. We desire to show that (p, v) can arise in equilibrium in an information acquisition game with costly learning.

Take the information structure $\mathcal{S} = (Z, \zeta, (X_i, \xi_i)_{i \in I})$ and the profile of action plans $\sigma = (\sigma_i)_{i \in I}$ constructed in Section A.1. We now define an information acquisition technology $\mathcal{T} = (Z, \zeta, (X_i, \{\xi_i, \xi'_i\}, C_i)_{i \in I})$ with the following properties. First, it preserves the correlation structure of $\mathcal{S}$. Second, each player i is offered a choice between two experiments: the experiment ξ_i from $\mathcal{S}$ and a fully uninformative experiment ξ'_i . The cost of ξ_i is set to $\bar{v}_i(p) - v_i$, while ξ'_i is costless.

Claim 2. The strategy profile (ξ, σ) is an equilibrium of $(\mathcal{G}, \mathcal{T})$.

Proof. Take the perspective of an arbitrary player i . Since σ is an equilibrium of the incomplete-information game $(\mathcal{G}, \mathcal{S})$ (Lemma 3), the action plan σ_i is optimal for player i in $(\mathcal{G}, \mathcal{T})$ when fixing their experiment ξ_i and the strategy profile of the co-players, (ξ_{-i}, σ_{-i}) . Therefore, the only deviation left to consider for player i in $(\mathcal{G}, \mathcal{T})$ is a switch to the uninformative experiment ξ'_i . Such a deviation yields a payoff of $\underline{v}_i(p)$. In contrast, sticking with the strategy (ξ_i, σ_i) yields a payoff of $\bar{v}_i(p) - (\bar{v}_i(p) - v_i) = v_i$. Since $v_i \geq \underline{v}_i(p)$, the deviation is not profitable. We conclude that (ξ_i, σ_i) is a best response to (ξ_{-i}, σ_{-i}) . Because player i was arbitrary, it follows that (ξ, σ) constitutes an equilibrium of $(\mathcal{G}, \mathcal{T})$. $\square$

Claim 3. The information technology $\mathcal{T}$ exhibits costly learning.

Proof. Take the perspective of a player i . To show that C_i is strictly increasing, we consider two cases.

Case (i): p is informative for player i . In this case, $\underline{v}_i(p) < \bar{v}_i(p)$, since p is instrumental. Hence,

$$C(\xi_i) = \bar{v}_i(p) - v_i > 0 = C(\xi'_i).$$

Moreover, by construction (see Section A.1), we have $\xi_i \succ \xi'_i$. It follows that C_i is strictly increasing when p is informative.

Case (ii): p is uninformative for player i . In this case, $\underline{v}_i(p) = \bar{v}_i(p)$, so

$$C(\xi_i) = \bar{v}_i(p) - v_i = \bar{v}_i(p) - \underline{v}_i(p) = 0 = C(\xi'_i).$$

Again, by construction, $\xi_i \sim \xi'_i$. Thus, C_i is strictly increasing in this case as well.

In both cases, the cost function C_i respects strict monotonicity. $\square$

Taken together, Claims 2 and 3 establish that (p, v) can arise in equilibrium in an information acquisition game with costly learning, as desired.

A.4. Proof of Theorem 1, Part (i): “Only If” Direction

Let $\mathcal{T} = (Z, \zeta, (X_i, \mathcal{E}_i, C_i)_{i \in I})$ be an information technology that exhibits costly learning, and let (ξ, σ) be an equilibrium of the information acquisition game $(\mathcal{G}, \mathcal{T})$. Denote by (p, v) the induced outcome-value pair.

The fact that p is a BCE follows immediately from the observation that σ is an equilibrium of the incomplete-information game $(\mathcal{G}, \mathcal{S})$, where $\mathcal{S} = (Z, \zeta, (X_i, \xi_i)_{i \in I})$.

To show that p is instrumental, fix an arbitrary player i and suppose that p is informative for i . Then, the equilibrium experiment ξ_i must reveal some information about the state of nature (θ, z) . Since learning is costly, it follows that $C_i(\xi_i) > 0$. Moreover, because ξ_i is chosen optimally, the player must strictly prefer it to the uninformative experiment. This implies that $\bar{v}_i(p) > \underline{v}_i(p)$, which in turn implies that $\bigcap_{a_i \in \text{supp}_i(p)} BR(p_{a_i}) = \emptyset$. Since i was arbitrary, p is an instrumental BCE.

The availability of the uninformative experiment as a deviation also implies that, for every i , we have $v_i \geq \underline{v}_i(p)$. In addition, non-negativity of information costs ensures that $v_i \leq \bar{v}_i(p)$. Finally, because information is strictly costly, the inequality $\bar{v}_i(p) > \underline{v}_i(p)$ implies that $v_i < \bar{v}_i(p)$.

This completes the proof of the “only if” direction of Theorem 1, Part (i).

A.5. Proof of Theorem 1, Part (ii): “If” Direction

Let p be an sBCE and let v be a profile values such that, for every player i , $\underline{v}_i(p) = \bar{v}_i(p) = v_i$ or $v_i \in [\underline{v}_i(p), \bar{v}_i(p))$. We desire to show that (p, v) can arise in equilibrium in an information acquisition game with flexible learning.

First, we focus on generating outcome p .

Lemma 5. *There exist a flexible technology $\mathcal{T}$ and an equilibrium (ξ, σ) of $(\mathcal{G}, \mathcal{T})$ whose outcome is p .*

Proof. Take the information structure $\mathcal{S} = (Z, \zeta, (X_i, \xi_i)_{i \in I})$ and the profile of action plans $\sigma = (\sigma_i)_{i \in I}$ constructed in Section A.1.

We now use Lemma 4 to construct a monotone cost function $C_i : \Delta(X_i)^{Z \times \Theta} \rightarrow \mathbb{R}$ for every player i . To verify Lemma 4-(i), take $x_i \in X_i$ and $a_i \in A_i$ such that $\sigma_i(a_i|x_i) > 0$. By the construction of the action plan, $a_i \in x_i$. By the obedience constraint for p , $a_i \in BR(p_{a_i})$. By Lemma 3, $BR(p_{a_i}) = BR(\mu_{x_i})$. Therefore, $a_i \in BR(\mu_{x_i})$. We deduce that Lemma 4-(i) holds.

To verify Lemma 4-(ii), take $x_i, x'_i \in X_i$ such that $\mu_{x_i} \neq \mu_{x'_i}$. Let $a_i \in x_i$ and $b_i \in x'_i$. Since $\mu_{x_i} \neq \mu_{x'_i}$, we must have $x_i \neq x'_i$ and, therefore, $p_{a_i} \neq p_{b_i}$. By the separation constraint for p , $BR(p_{a_i}) \cap BR(p_{b_i}) = \emptyset$. By Lemma 3, $BR(p_{a_i}) = BR(\mu_{x_i})$ and $BR(p_{b_i}) = BR(\mu_{x'_i})$. We deduce $BR(\mu_{x_i}) = BR(\mu_{x'_i}) = \emptyset$: Lemma 4-(ii) holds.

In sum, for every player i , we can invoke Lemma 4 to find a strictly increasing cost function $C_i : \Delta(X_i)^{Z \times \Theta} \rightarrow \mathbb{R}$ such that (ξ_i, σ_i) is optimal given (ξ_{-i}, σ_{-i}) . This shows that $\mathcal{T} = (Z, \zeta, (X_i, \mathcal{E}_i, C_i)_{i \in I})$, with $\mathcal{E}_i = \Delta(X_i)^{Z \times \Theta}$ for all i , is a flexible technology such that (ξ, σ) is an equilibrium of $(\mathcal{G}, \mathcal{T})$ with outcome p . $\square$

Next we consider the case in which $v = \underline{v}(p)$:

Lemma 6. *There exist a flexible technology $\mathcal{T}$ and an equilibrium (ξ, σ) of $(\mathcal{G}, \mathcal{T})$ whose outcome-value pair is $(p, \underline{v}(p))$.*

Proof. By Lemma 5, there is a flexible technology $\mathcal{T} = (Z, \zeta, (X_i, \mathcal{E}_i, C_i)_{i \in I})$ and an equilibrium (ξ, σ) of $(\mathcal{G}, \mathcal{T})$ whose outcome is p . For every player i , we pick $t_i \in \mathbb{R}$ such that $\bar{v}_i(p) - C_i(\xi_i) = \underline{v}_i(p) + t_i$. Since (ξ_i, σ_i) is a best response to (ξ_{-i}, σ_{-i}) , we have $t_i \geq 0$. Then, we define the cost function C'_i as follows: $C'_i(\xi'_i) = C_i(\xi'_i) + t_i$ if ξ'_i is informative, and $C'_i(\xi'_i) = 0$ if ξ'_i is uninformative.²⁰ It is easy to check that the information technology $\mathcal{T}' = (Z, \zeta, (X_i, \mathcal{E}_i, C'_i)_{i \in I})$ is flexible. Moreover, (ξ, σ) is an equilibrium of $(\mathcal{G}, \mathcal{T}')$ whose outcome-value pair is $(p, \underline{v}(p))$. $\square$

Now we consider the general case in which v may differ from $\underline{v}(p)$. As a starting point, take $\mathcal{T} = (Z, \zeta, (X_i, \mathcal{E}_i, C_i)_{i \in I})$ and (ξ, σ) as in the statement of Lemma 6.

For each $i \in I$, take $\lambda_i \in (0, 1]$ and define $C_{\lambda_i} : \mathcal{E}_i \rightarrow \mathbb{R}_+$ by $C_{\lambda_i}(\xi'_i) = \lambda_i C_i(\xi'_i)$ if $\xi_i \succsim \xi'_i$, and $C_{\lambda_i}(\xi'_i) = C_i(\xi'_i)$ if $\xi_i \not\succsim \xi'_i$. Since C_i is increasing, C_{λ_i} is increasing. Given $\lambda = (\lambda_i)_{i \in I}$, consider the flexible technology $\mathcal{T}_\lambda = (Z, \zeta, (X_i, \mathcal{E}_i, C_{\lambda_i})_{i \in I})$.

Lemma 7. *The strategy profile (ξ, σ) is an equilibrium of $(\mathcal{G}, \mathcal{T}_\lambda)$.*

²⁰An experiment ξ'_i is **uninformative** if $\xi'_i(x_i|z, \theta) = \xi'_i(x_i|z', \theta')$ for all $x_i \in X_i$, $\theta, \theta' \in \text{supp}(\pi)$, and $z, z' \in Z$ with $\zeta(z|\theta) > 0$ and $\zeta(z'|\theta') > 0$. It is **informative** if it is not uninformative.

Proof. For a player i , consider an alternative strategy (ξ'_i, σ'_i) . Suppose first that $\xi_i \succsim \xi'_i$. Since (ξ, σ) is an equilibrium of $(\mathcal{G}, \mathcal{T})$,

$$\bar{v}_i(p) - C_i(\xi_i) \geq U_i(\xi'_i, \sigma'_i, \xi_{-i}, \sigma_{-i}) - C_i(\xi'_i).$$

In addition, $\bar{v}_i(p) \geq U_i(\xi'_i, \sigma'_i, \xi_{-i}, \sigma_{-i})$ because $\xi_i \succsim \xi'_i$. It follows that

$$\bar{v}_i(p) - \lambda_i C_i(\xi_i) \geq U_i(\xi'_i, \sigma'_i, \xi_{-i}, \sigma_{-i}) - \lambda_i C_i(\xi'_i).$$

We deduce that deviating to (ξ'_i, σ'_i) is not profitable in $(\mathcal{G}, \mathcal{T}_\lambda)$.

Suppose instead that $\xi_i \not\succsim \xi'_i$. Then,

$$\bar{v}_i(p) - \lambda_i C_i(\xi_i) \geq \bar{v}_i(p) - C_i(\xi_i) \geq U_i(\xi'_i, \sigma'_i, \xi_{-i}, \sigma_{-i}) - C_i(\xi'_i)$$

where we use again the fact that (ξ, σ) is an equilibrium of $(\mathcal{G}, \mathcal{T})$. Thus, also in this case, deviating to (ξ'_i, σ'_i) is not profitable in $(\mathcal{G}, \mathcal{T}_\lambda)$. We conclude that (ξ, σ) is an equilibrium of $(\mathcal{G}, \mathcal{T}_\lambda)$. $\square$

Thus, (ξ, σ) is an equilibrium of $(\mathcal{G}, \mathcal{T}_\lambda)$ whose outcome is p . In addition,

$$\bar{v}_i(p) - C_{\lambda_i}(\xi_i) = \bar{v}_i(p) - \lambda_i C_i(\xi_i) = \bar{v}_i(p) - \lambda_i(\bar{v}_i(p) - \underline{v}_i(p)) = (1 - \lambda_i)\bar{v}_i + \lambda_i \underline{v}_i(p).$$

Hence, by appropriately choosing the vector λ , we can be sure that v is the value of (ξ, σ) . This concludes the proof of the “if” direction of Theorem 1, part (ii).

A.6. Proof of Theorem 1, Part (ii): “Only If” Direction

Let (p, v) be the outcome-value pair induced by an equilibrium (ξ, σ) of an information acquisition game $(\mathcal{G}, \mathcal{T})$ where $\mathcal{T}$ is flexible.

First we show that p is a separated BCE. For every player i , by the definition of equilibrium, the equilibrium strategy (ξ_i, σ_i) maximizes

$$\left[\sum_{a, x, z, \theta} u_i(a, \theta) \sigma'_i(a_i | x_i) \xi'_i(x_i | z, \theta) \prod_{j \neq i} \sigma_j(a_j | x_j) \xi_j(x_j | z, \theta) \zeta(z | \theta) \pi(\theta) \right] - C_i(\xi_i)$$

over all $\xi'_i \in \mathcal{E}_i$ and $\sigma'_i \in \Sigma_i$. Since $\mathcal{E}_i$ is unrestricted and C_i is monotone, we can apply Lemma 4 with appropriate definitions for Ω , ρ , and w_i . Specifically, we take

$\Omega = Z \times \Theta$, and we define $\rho \in \Delta(\Omega)$ and $w_i : A_i \times \Omega \rightarrow \mathbb{R}$ by

$$\rho(z, \theta) = \zeta(z|\theta)\pi(\theta),$$

$$w_i(a_i, z, \theta) = \sum_{x_{-i}, a_{-i}} u_i(a, \theta) \left[\prod_{j \neq i} \sigma_j(a_j|x_j) \xi_j(x_j|z, \theta) \right].$$

From Lemma 4-(i), we obtain $\sigma_i(BR(\mu_{x_i})|\mu_{x_i}) = 1$ for all x_i with $\mu(x_i) > 0$. From Lemma 4-(ii), we get that for all x_i and x'_i such that $\mu(x_i) > 0$ and $\mu(x'_i) > 0$, if $\mu_{x_i} \neq \mu_{x'_i}$ then $BR(\mu_{x_i}) \neq BR(\mu_{x'_i})$.

Next, we relate $BR(\mu_{x_i})$ and $BR(p_{a_i})$. Towards this goal, for each player i and action a_i , let X_{a_i} be the set of positive-probability signals that makes player i take action a_i : $X_{a_i} = \{x_i : \mu(x_i) > 0 \text{ and } \sigma(a_i|x_i) > 0\}$.

Lemma 8. *For all $i \in I$ and $a_i \in \text{supp}_i(p)$, $BR(p_{a_i}) = \bigcap_{x_i \in X_{a_i}} BR(\mu_{x_i})$.*

Proof. First, take $b_i \in BR(p_{a_i})$: we wish to show that $b_i \in BR(\mu_{x_i})$ for all $x_i \in X_{a_i}$. By Lemma 4-(i), $a_i \in BR(\mu_{x_i})$ for all $x_i \in X_{a_i}$. Hence,

$$\sum_{z, \theta} w_i(a_i, z, \theta) \mu_{x_i}(z, \theta) \geq \sum_{z, \theta} w_i(b_i, z, \theta) \mu_{x_i}(z, \theta). \quad (8)$$

For each $x_i \in X_{a_i}$, define $\mu(x_i|a_i) = \sigma_i(a_i|x_i)\nu(x_i)/p(a_i)$. Simple algebra shows that

$$p_{a_i}(a_{-i}, \theta) = \sum_{x_i \in X_{a_i}} \mu(x_i|a_i) \sum_{x_{-i}, z} \left[\prod_{j \neq i} \sigma_j(a_j|x_j) \xi_j(x_j|z, \theta) \right] \mu_{x_i}(z, \theta). \quad (9)$$

Therefore,

$$\sum_{x_i \in X_{a_i}} \mu(x_i|a_i) \left(\sum_{z, \theta} w_i(a_i, z, \theta) \mu_{x_i}(z, \theta) \right) = \sum_{a_{-i}, \theta} u_i(a_i, a_{-i}, \theta) p_{a_i}(a_{-i}, \theta),$$

$$\sum_{x_i \in X_{a_i}} \mu(x_i|a_i) \left(\sum_{z, \theta} w_i(b_i, z, \theta) \mu_{x_i}(z, \theta) \right) = \sum_{a_{-i}, \theta} u_i(b_i, a_{-i}, \theta) p_{a_i}(a_{-i}, \theta).$$

As a consequence, since $b_i \in BR(p_{a_i})$, we have that

$$\sum_{x_i \in X_{a_i}} \mu(x_i|a_i) \left(\sum_{z, \theta} w_i(a_i, z, \theta) \mu_{x_i}(z, \theta) \right) \leq \sum_{x_i \in X_{a_i}} \mu(x_i|a_i) \left(\sum_{z, \theta} w_i(b_i, z, \theta) \mu_{x_i}(z, \theta) \right).$$

It follows that (8) holds with equality for all $x_i \in X_{a_i}$. Therefore, since $a_i \in BR(\mu_{x_i})$ for all $x_i \in X_{a_i}$, we deduce that $b_i \in BR(\mu_{x_i})$ for all $x_i \in X_{a_i}$.

Next, take $b_i \in A_i$ such that $b_i \in BR(\nu_{x_i})$ for all $x_i \in X_{a_i}$: we wish to show that $b_i \in BR(p_{a_i})$. By hypothesis, for all $x_i \in X_{a_i}$ and $c_i \in A_i$,

$$\sum_{z, \theta} w_i(c_i, z, \theta) \mu_{x_i}(z, \theta) \leq \sum_{z, \theta} w_i(b_i, z, \theta) \mu_{x_i}(z, \theta).$$

Averaging across inequalities over $x_i \in X_{a_i}$, we obtain that for all $c_i \in A_i$,

$$\sum_{x_i \in X_{a_i}} \mu(x_i | a_i) \left(\sum_{z, \theta} w_i(c_i, z, \theta) \mu_{x_i}(z, \theta) \right) \leq \sum_{x_i \in X_{a_i}} \mu(x_i | a_i) \left(\sum_{z, \theta} w_i(b_i, z, \theta) \mu_{x_i}(z, \theta) \right).$$

Using (9), we deduce that

$$\sum_{a_{-i}, \theta} u_i(c_i, a_{-i}, \theta) p_{a_i}(a_{-i}, \theta) \leq \sum_{a_{-i}, \theta} u_i(b_i, a_{-i}, \theta) p_{a_i}(a_{-i}, \theta).$$

We conclude that $b_i \in BR(p_{a_i})$. □

We are ready to show that p is a separated BCE. For the obedience constraint, take $i \in I$ and $a_i \in \text{supp}_i(p)$. By Lemma 4-(i), $a_i \in BR(\mu_{x_i})$ for all $x_i \in X_{a_i}$. By Lemma 8, $a_i \in BR(p_{a_i})$. For the separation constraint, take $i \in I$ and $a_i, b_i \in \text{supp}_i(p)$ such that $p_{a_i} \neq p_{b_i}$. Since $p_{a_i} \neq p_{b_i}$, there are $x_{a_i} \in X_{a_i}$ and $x_{b_i} \in X_{b_i}$ such that $\mu_{x_{a_i}} \neq \mu_{x_{b_i}}$ (this should be obvious, but for confirmation, see (9)). By Lemma 4-(ii), $BR(\mu_{x_{a_i}}) \cap BR(\mu_{x_{b_i}}) = \emptyset$. By Lemma 8, $BR(p_{a_i}) \cap BR(p_{b_i}) = \emptyset$.

To complete the proof of the “only if” direction of Theorem 1, part (ii). we need to show that for every player i , either $v_i = \underline{v}_i(p) = \bar{v}_i(p)$ or $v_i \in [\underline{v}_i(p), \bar{v}_i(p))$. This follows from part (i), since flexible technologies are costly technologies.

B. Proof of Theorem 2

As a first step in proving Theorem 2, we demonstrate a basic lemma about best responses. In what follows, for $p \in \Delta(A \times \Theta)$, $a_i \in \text{supp}_i(p)$, and $b_i \in A_i$, take $u_i(b_i, p_{a_i}) \in \mathbb{R}$ to be

$$u_i(b_i, p_{a_i}) = \sum_{a_{-i}, \theta} u_i(b_i, a_{-i}, \theta) p_{a_i}(a_{-i}, \theta).$$

Lemma 9. For every $t \in (0, 1)$, $p, q \in BCE$, $i \in I$, and $a_i \in \text{supp}_i(p)$,

$$BR\left((tp + (1 - t)q)_{a_i}\right) \subseteq BR(p_{a_i}).$$

Proof. Take $b_i \in BR\left((tp + (1 - t)q)_{a_i}\right)$. If $a_i \notin \text{supp}_i(q)$, then $(tp + (1 - t)q)_{a_i} = p_{a_i}$, which immediately implies the desired result.

Suppose now that $a_i \in \text{supp}_i(q)$. Since $p, q \in BCE$, we have

$$u_i(a_i, p_{a_i}) \geq u_i(b_i, p_{a_i}) \quad \text{and} \quad u_i(a_i, q_{a_i}) \geq u_i(b_i, q_{a_i}).$$

Simple algebra shows that there exists $s \in (0, 1)$ such that

$$(tp + (1 - t)q)_{a_i} = sp_{a_i} + (1 - s)q_{a_i}.$$

Since $b_i \in BR\left((tp + (1 - t)q)_{a_i}\right)$, we obtain that

$$\begin{aligned} su_i(b_i, p_{a_i}) + (1 - s)u_i(b_i, q_{a_i}) &= u_i(b_i, sp_{a_i} + (1 - s)q_{a_i}) \\ &\geq u_i(a_i, sp_{a_i} + (1 - s)q_{a_i}) \\ &= su_i(a_i, p_{a_i}) + (1 - s)u_i(a_i, q_{a_i}). \end{aligned}$$

We conclude that $u_i(a_i, p_{a_i}) = u_i(b_i, p_{a_i})$ and $u_i(a_i, q_{a_i}) = u_i(b_i, q_{a_i})$. It follows from $p \in BCE$ that $b_i \in BR(p_{a_i})$. $\square$

Next, we show that taking convex combinations of BCEs usually preserve the set of action recommendations that lead to different beliefs.

Lemma 10. For every $p, q \in \Delta(A \times \Theta)$, $i \in I$, and $a_i, b_i \in \text{supp}_i(p)$ with $p_{a_i} \neq p_{b_i}$, there are at most two $t \in (0, 1)$ such that

$$(tp + (1 - t)q)_{a_i} = (tp + (1 - t)q)_{b_i}. \quad (10)$$

Proof. Note that $t \in (0, 1)$ is a solution of (10) if and only if for every $a_{-i} \in A_{-i}$ and $\theta \in \Theta$,

$$\begin{aligned} &(tp(a_i, a_{-i}, \theta) + (1 - t)q(a_i, a_{-i}, \theta))(tp(b_i) + (1 - t)q(b_i)) \\ &= (tp(b_i, a_{-i}, \theta) + (1 - t)q(b_i, a_{-i}, \theta))(tp(a_i) + (1 - t)q(a_i)). \end{aligned} \quad (11)$$

Each equation (11) is polynomial in t , with degree at most two. Since $p_{a_i} \neq p_{b_i}$, at least one such polynomial equation does not have degree zero and, therefore, has at most two solutions. We deduce that (10) has at most two solutions for $t \in (0, 1)$. $\square$

Our next goal is to prove properties of maximally differentiated BCEs.

Lemma 11. *A maximally differentiated BCE exists.*

Proof. For every $p \in BCE$, define the set

$$X(p) = \bigcup_{i \in I} \{(a_i, b_i) : a_i, b_i \in \text{supp}_i(p) \text{ and } p_{a_i} \neq p_{b_i}\}.$$

Note that $p \in BCE$ is maximally differentiated if and only if it has maximal support and for every $q \in BCE$, $X(q) \subseteq X(p)$.

Since the set $A \times \Theta$ is finite and BCE is a convex set, we can find a maximal support $p \in BCE$ such that for every maximal support $q \in BCE$, the cardinality of $X(p)$ is larger than the cardinality of $X(q)$.

We now show that p is maximally differentiated. Fix an arbitrary $q \in BCE$. For every $t \in (0, 1)$, define $p^t = tp + (1 - t)q$, which is a BCE because BCE is convex. Since p has maximal support, the same is true for p^t . Thus, the cardinality of $X(p)$ is larger than the cardinality of $X(p^t)$. By Lemma 10, we can find $t \in (0, 1)$ such that $X(p) \subseteq X(p^t)$ and $X(q) \subseteq X(p^t)$. This shows that $X(q) \subseteq X(p)$; otherwise, the cardinality of $X(p^t)$ would be strictly larger than the cardinality of $X(p)$. We conclude that p is maximally differentiated. $\square$

Lemma 12. *The set of maximally differentiated BCEs is open in the BCE set.*

Proof. The set of maximally differentiated BCEs can be written as the finite intersection of sets of the form:

$$\{p \in BCE : p(a_i) > 0\}, \quad \text{and} \quad \{p \in BCE : p(a_i)p(b_i) > 0 \text{ and } p_{a_i} \neq p_{b_i}\}.$$

Since these sets are open in the BCE set, the set of maximally differentiated BCEs is also open in the BCE set. $\square$

We now proceed to prove Theorem 2 and Lemma 1.

Lemma 13. *The following statements are equivalent:*

(i) The *iBCE* set is dense within the *BCE* set.

(ii) There exists an instrumental and maximally differentiated *BCE*.

(iii) The *BCE* set is globally instrumental.

Proof. (i) implies (ii). This implication follows from Lemmas 11 and 12.

(ii) implies (iii). Let q be a maximally differentiated *iBCE*. Fix any $p \in BCE$ and $i \in I$ such that $p_{a_i} \neq p_{b_i}$ for some $a_i, b_i \in \text{supp}_i(p)$. Since q is maximally differentiated, $a_i, b_i \in \text{supp}_i(q)$ and $q_{a_i} \neq q_{b_i}$. Thus, q is informative for player i . Given that q is instrumental, we obtain:

$$\emptyset = \bigcap_{c_i \in \text{supp}_i(q)} BR(q_{a_i}) \supseteq \bigcap_{c_i \in A_i} J(c_i),$$

where we use the fact that $BR(p_{c_i}) \subseteq J(c_i)$ for all $c_i \in A_i$. This proves that the *BCE* set is globally instrumental.

(iii) implies (i). Fix any $p \in BCE$. Because A is finite and *BCE* is convex, it follows from Lemma 9 that we can find $q \in BCE$ such that q has maximal support and $BR(q_{a_i}) = J(q_{a_i})$ for all $i \in I$ and $a_i \in \text{supp}_i(q)$.

For $t \in (0, 1)$, let $p^t = tp + (1 - t)q$. We claim that $p^t \in iBCE$. That $p^t \in BCE$ follows from convexity of the *BCE* set. To see p^t is an *iBCE*, take any $i \in I$ and for which p^t is informative. Since the *BCE* set is globally instrumental, we obtain $\bigcap_{a_i \in A_i} J(a_i) = \emptyset$. Moreover, since q has maximal support, $\bigcap_{a_i \in \text{supp}_i(q)} J(a_i) = \bigcap_{a_i \in A_i} J(a_i)$. Then,

$$\bigcap_{a_i \in \text{supp}_i(p^t)} BR(p_{a_i}^t) \subseteq \bigcap_{a_i \in \text{supp}_i(q)} BR(q_{a_i}) = \bigcap_{a_i \in \text{supp}_i(q)} J(a_i) = \emptyset,$$

where first we use Lemma 9, then $BR(q_{a_i}) = J(a_i)$ for all $a_i \in \text{supp}_i(q)$. We conclude $p^t \in iBCE$ for all $t \in (0, 1)$. This shows the *iBCE* set is dense within the *BCE* set. $\square$

Lemma 13 implies Theorem 2, part (i), and Lemma 1, part (i). The next lemma implies Theorem 2, part (ii), and Lemma 1, part (ii).

Lemma 14. *The following statements are equivalent:*

(i) The *sBCE* set is dense within the *BCE* set.

(ii) *There exists a separated and maximally differentiated BCE.*

(iii) *The BCE set is globally separated.*

Proof. (i) implies (ii). This implication follows from Lemmas 11 and 12.

(ii) implies (iii). Let q be a maximally differentiated sBCE. We demonstrate that the BCE set is globally separated; that is, if a_i and b_i are informationally distinct, then $J(a_i) \cap J(b_i) = \emptyset$. Fix any $p \in BCE$, $i \in I$ and $a_i, b_i \in \text{supp}_i(p)$ such that $p_{a_i} \neq p_{b_i}$. Since q is maximally differentiated, $q_{a_i} \neq q_{b_i}$. Thus,

$$\emptyset = BR(q_{a_i}) \cap BR(q_{b_i}) \supseteq J(a_i) \cap J(b_i),$$

where we use the separation constraint, and then the fact that $BR(q_{a_i}) \subseteq J(a_i)$ and $BR(q_{b_i}) \subseteq J(b_i)$. We conclude that $J(a_i) \cap J(b_i) = \emptyset$, as desired.

(iii) implies (i). Fix any $p \in BCE$. Because A is finite and BCE is convex, it follows from Lemma 9 that we can find $q \in BCE$ such that q has maximal support and $BR(q_{a_i}) = J(q_{a_i})$ for all $i \in I$ and $a_i \in \text{supp}_i(q)$.

For $t \in (0, 1)$, let $p^t = tp + (1 - t)q$. We claim that $p^t \in sBCE$. That $p^t \in BCE$ follows from convexity of the BCE set. To see p^t is a sBCE, take any $i \in I$ and $a_i, b_i \in \text{supp}_i(p^t)$ such that $p_{a_i}^t \neq p_{b_i}^t$. Hence, a_i and b_i are informationally distinct, and so $J(a_i) \cap J(b_i) = \emptyset$ (since the BCE set is globally separated by hypothesis). Since q has maximal support, $a_i, b_i \in \text{supp}_i(q)$. Then,

$$BR(p_{a_i}^t) \cap BR(p_{b_i}^t) \subseteq BR(q_{a_i}) \cap BR(q_{b_i}) = J(a_i) \cap J(b_i) = \emptyset,$$

where first we use Lemma 9, then $BR(q_{a_i}) = J(a_i)$ and $BR(q_{b_i}) = J(b_i)$, and finally the fact that $J(a_i) \cap J(b_i) = \emptyset$. We conclude $p^t \in sBCE$ for all $t \in (0, 1)$. This shows that the sBCE set is dense within the BCE set. $\square$

C. Proof of Theorem 3

The proof of Theorem 3 is divided in two parts. In the first part, we prove that the iBCE and sBCE sets are either dense or nowhere dense within the BCE set. In the second part, we demonstrate that, in generic settings, the sBCE set is dense within the BCE set.

C.1. Either dense or nowhere dense

As explained in the main text, this part follows directly from Theorem 2 and Lemma 1—provided that one can show the set of maximally differentiated BCEs is open and dense in the BCE set. Lemma 15 establishes the maximally differentiated BCE set is open. Thus, all that remains is to argue this set is also dense. We do so below.

Lemma 15. *The set of maximally differentiated BCEs is dense in the BCE set.*

Proof. Fix some $q \in \text{BCE}$. Take p to be a maximally differentiated BCE, which exists by Lemma 11. For every $t \in (0, 1)$, define $p^t = tp + (1 - t)q$. Because p has maximal support, the same is true for p^t for all $t \in (0, 1)$. Moreover, by Lemma 10, a finite set $T \subseteq (0, 1)$ exists such that for all $t \in (0, 1) \setminus T$, $i \in I$, and $a_i, b_i \in \text{supp}_i(p)$,

$$p_{a_i} \neq p_{b_i} \quad \text{implies} \quad p_{a_i}^t \neq p_{b_i}^t.$$

Thus, p^t is a maximally differentiated BCE for all $t \in (0, 1) \setminus T$. Thus, q is a limit point of $\{p^t : t \in (0, 1) \setminus T\}$. We conclude that the set of maximally differentiated BCEs is dense in the BCE set. $\square$

C.2. Dense, generically

In this section we demonstrate that, in generic settings, the sBCE set is dense within the BCE set. Since every sBCE is instrumental, this argument also proves generic denseness of the iBCE set.

Let $\|u\|$ be the Euclidean norm of $u \in \mathbb{R}^{I \times A \times \Theta}$. We begin with the following lemma.

Lemma 16. *For every $u \in \mathbb{R}^{I \times A \times \Theta}$, $p \in \text{BCE}(u)$, and $\epsilon > 0$, there exists $u' \in \mathbb{R}^{I \times A \times \Theta}$ such that $\|u - u'\| \leq \epsilon$ and $p \in \text{sBCE}(u')$.*

Proof. For each player $i \in I$, we consider a set $P_i \subseteq \Delta(A_{-i} \times \Theta)$ given by $P_i = \{p_{a_i} : a_i \in \text{supp}_i(p)\}$. Let n_i be the cardinality of P_i (of course, it could be that n_i is smaller than the cardinality of $\text{supp}_i(p)$). Reasoning inductively, we can find an enumeration $p_1, \dots, p_{n_i}$ of the elements of P_i such that, for every $m_i \in \{1, \dots, n_i\}$, p_{m_i} is an extreme point of the convex hull of $\{p_1, \dots, p_{m_i}\}$.

By an hyperplane separation theorem (e.g., Rockafellar, 1970, Corollary 11.4.2)

for every $m_i \in \{2, \dots, n_i\}$ we can find a function $f_{m_i} : A_{-i} \times \Theta \rightarrow \mathbb{R}$ such that

$$\sum_{a_{-i}, \theta} f_{m_i}(a_{-i}, \theta) p_{m_i}(a_{-i}, \theta) > 0 \geq \max_{l_i \in \{1, \dots, m_i-1\}} \sum_{a_{-i}, \theta} f_{m_i}(a_{-i}, \theta) p_{l_i}(a_{-i}, \theta). \quad (12)$$

For $m_i = 1$, we define $f_1(a_{-i}, \theta) = 1$ for all $a_{-i} \in A_{-i}$ and $\theta \in \Theta$.

For every $l_i \in \{1, \dots, n_i - 1\}$, we choose $t_{l_i} \in (0, 1]$ such that for every $m_i \in \{l_i + 1, \dots, n_i\}$,

$$\sum_{a_{-i}, \theta} f_{m_i}(a_{-i}, \theta) p_{m_i}(a_{-i}, \theta) > t_{l_i} \sum_{a_{-i}} f_{l_i}(a_{-i}) p_{m_i}(a_{-i}). \quad (13)$$

We can choose such a t_{l_i} because the left-hand side of (13) is positive—see (12). For $l_i = n_i$, we simply define $t_{n_i} = 1$.

For every $l_i \in \{1, \dots, n_i\}$, we define $s_{l_i} = \prod_{m_i=l_i}^{n_i} t_{m_i}$. Using (13), simple algebra shows that for every $l_i \in \{1, \dots, n_i - 1\}$ and $m_i \in \{l_i + 1, \dots, n_i\}$,

$$s_{m_i} \sum_{a_{-i}, \theta} f_{m_i}(a_{-i}, \theta) p_{m_i}(a_{-i}, \theta) > s_{l_i} \sum_{a_{-i}, \theta} f_{l_i}(a_{-i}, \theta) p_{m_i}(a_{-i}, \theta). \quad (14)$$

For $a_i \in \text{supp}_i(p)$, we define $g_{a_i} : A_{-i} \times \Theta \rightarrow \mathbb{R}$ by $g_{a_i}(a_{-i}, \theta) = s_{m_i} \cdot f_{m_i}(a_{-i}, \theta)$ where m_i is such that $p_{a_i} = p_{m_i}$. For $a_i \notin \text{supp}_i(p)$, we define $g_{a_i} = 0$.

Claim 4. For all $a_i \in \text{supp}_i(p)$ and $b_i \notin \{c_i \in \text{supp}_i(p) : p_{c_i} = p_{a_i}\}$,

$$\sum_{a_{-i}, \theta} g_{a_i}(a_{-i}, \theta) p_{a_i}(a_{-i}, \theta) > \sum_{a_{-i}, \theta} g_{b_i}(a_{-i}, \theta) p_{a_i}(a_{-i}, \theta). \quad (15)$$

Proof. Pick $m_i \in \{1, \dots, n_i\}$ such that $p_{a_i} = p_{m_i}$. From the left-hand side of (12) and the fact that $s_{m_i} > 0$, we obtain that

$$\sum_{a_{-i}, \theta} g_{a_i}(a_{-i}, \theta) p_{a_i}(a_{-i}, \theta) = s_{m_i} \sum_{a_{-i}, \theta} f_{m_i}(a_{-i}, \theta) p_{m_i}(a_{-i}, \theta) > 0. \quad (16)$$

Hence, for $b_i \notin \text{supp}_i(p)$, we have

$$\sum_{a_{-i}, \theta} g_{a_i}(a_{-i}, \theta) p_{a_i}(a_{-i}, \theta) > 0 = \sum_{a_{-i}, \theta} g_{b_i}(a_{-i}, \theta) p_{a_i}(a_{-i}, \theta),$$

where the equality follows from $g_{b_i} = 0$.

Assume now that $b_i \in \text{supp}_i(p)$. Choose l_i such that $p_{b_i} = p_{l_i}$. Since $p_{a_i} \neq p_{b_i}$, it must be that $m_i \neq l_i$. Suppose that $l_i > m_i$. It follows from the right-hand side of (12)—in (12) the roles of l_i and m_i are inverted—that $0 \geq \sum_{a_{-i}, \theta} f_{l_i}(a_{-i}, \theta) p_{m_i}(a_{-i}, \theta)$. Thus, given that $s_{l_i} > 0$, we deduce that

$$0 \geq \sum_{a_{-i}, \theta} g_{b_i}(a_{-i}, \theta) p_{a_i}(a_{-i}, \theta) = s_{l_i} \sum_{a_{-i}, \theta} f_{l_i}(a_{-i}, \theta) p_{m_i}(a_{-i}, \theta).$$

We obtain that $\sum_{a_{-i}, \theta} g_{a_i}(a_{-i}, \theta) p_{a_i}(a_{-i}, \theta) > 0 \geq \sum_{a_{-i}, \theta} g_{b_i}(a_{-i}, \theta) p_{a_i}(a_{-i}, \theta)$, where we use again (16). For the case $l_i < m_i$, the condition $\sum_{a_{-i}, \theta} g_{a_i}(a_{-i}, \theta) p_{a_i}(a_{-i}, \theta) > \sum_{a_{-i}, \theta} g_{b_i}(a_{-i}, \theta) p_{a_i}(a_{-i}, \theta)$ is equivalent to (14). We conclude that (15) holds. $\square$

To complete the proof of the lemma, for every $\delta > 0$ we define $u' = (u'_i)_{i \in I}$ by $u'_i(a, \theta) = u_i(a, \theta) + \delta g_{a_i}(a_{-i}, \theta)$. By choosing δ sufficiently small, we can be make sure that $\|u - u'\| \leq \epsilon$. Since $p \in BCE(u)$, it follows from (15) that for all $i \in I$ and $a_i \in \text{supp}_i(p)$, $a_i \in BR'(p_{a_i}) \subseteq \{b_i \in \text{supp}_i(p) : p_{a_i} = p_{b_i}\}$ where $BR'(p_{a_i})$ is the set of i 's best response to a belief p_{a_i} given utility function u'_i . Thus, $p \in sBCE(u')$. $\square$

A subset of a Euclidean space is **semi-algebraic** if it is defined by finite systems of polynomial inequalities. A correspondence between Euclidean spaces is semi-algebraic if its graph is semi-algebraic. The background knowledge on semi-algebraic sets that we use in this proof can be gathered from Blume and Zame (1994, Section 2).

To state the next result, let $\text{cl}(sBCE(u))$ be the closure of the sBCE set.

Lemma 17. *The correspondences $u \mapsto BCE(u)$, $u \mapsto sBCE(u)$, and $u \mapsto \text{cl}(sBCE(u))$ are semi-algebraic.*

Proof. The BCE correspondence is semi-algebraic: for all $u \in \mathbb{R}^{I \times A \times \Theta}$ and $p \in \mathbb{R}^{A \times \Theta}$, $p \in BCE(u)$ if and only if the pair (u, p) is a solution to the following finite system of polynomial inequalities: $p(a, \theta) \geq 0$ for all $a \in A$ and $\theta \in \Theta$; $\sum_a p(a, \theta) = \pi(\theta)$ for all $\theta \in \Theta$; and $\sum_{a_{-i}, \theta} (u(a, \theta) - u(b_i, a_{-i}, \theta)) p(a, \theta) \geq 0$ for all $i \in I$ and $a_i, b_i \in A_i$.

The sBCE correspondence is also semi-algebraic. To prove it, for $u \in \mathbb{R}^{I \times A \times \Theta}$, $p \in \mathbb{R}^{A \times \Theta}$, $i \in I$, and $a_i, b_i, c_i \in A_i$, we denote by $F(u, p, a_i, b_i, c_i)$ the quantity

$$\sum_{a_{-i}, \theta} (u(a_i, a_{-i}, \theta) - u(c_i, a_{-i}, \theta)) p(a_i, a_{-i}, \theta) + (u(b_i, a_{-i}, \theta) - u(c_i, a_{-i}, \theta)) p(b_i, a_{-i}, \theta).$$

We observe that $p \in sBCE(u)$ if and only if $p \in BCE(u)$ and for every $i \in I$ there is $T_i \subseteq A_i \times A_i$ such that the pair (u, p) is a solution of the following finite system of

polynomial inequalities:

$$\sum_{a_{-i}, \theta} (p(a_i, a_{-i}, \theta)p(b_i) - p(b_i, a_{-i}, \theta)p(a_i))^2 = 0 \quad \text{for all } (a_i, b_i) \in T_i,$$

$$F(u, p, a_i, b_i, c_i) > 0 \quad \text{for all } (a_i, b_i) \notin T_i \text{ and } c_i \in A_i.$$

Thus, $p \in sBCE(u)$ if and only if it is the solution of one of finitely many systems of polynomial inequalities; we conclude that the sBCE correspondence is semi-algebraic.

The correspondence $u \mapsto \text{cl}(sBCE(u))$ is also semi-algebraic. Indeed, we have $p \in \text{cl}(sBCE(u))$ if and only if for every $\epsilon > 0$ there exists $q \in \mathbb{R}^{A \times \Theta}$ such that $\|p - q\| \leq \epsilon$ and $q \in sBCE(u)$. Thus, since the sBCE correspondence is semi-algebraic, the graph of the correspondence $u \mapsto \text{cl}(sBCE(u))$ is defined by a first-order formula and therefore semi-algebraic by the Tarski-Seidenberg theorem (Blume and Zame, 1994, page 787). $\square$

We are ready to complete the proof of the theorem. By Lemma 17, the correspondence $u \mapsto \text{cl}(sBCE(u))$ is semi-algebraic. Hence, there is an open subsets U of $\mathbb{R}^{I \times A \times \Theta}$ such that the complement of U has Lebesgue measure zero, and $u \mapsto \text{cl}(sBCE(u))$ is continuous on U (Blume and Zame, 1994, page 786).

We claim that for all $u \in U$, $BCE(u) = \text{cl}(sBCE(u))$. To prove the claim, take any $u \in U$. Since $BCE(u)$ is closed and $sBCE(u) \subseteq BCE(u)$, $\text{cl}(sBCE(u)) \subseteq BCE(u)$. To verify the other inclusion, we use Lemma 16 to find a sequence of games $(u^n)_{n=1}^\infty$ such that $u^n \rightarrow u$ and, for every n , $p \in sBCE(u^n) \subseteq \text{cl}(sBCE(u^n))$. Since $\text{cl}(sBCE(u))$ is continuous at u , we have $p \in \text{cl}(sBCE(u))$; see Aliprantis and Border (2006, Theorem 17.16). Hence, $BCE(u) \subseteq \text{cl}(sBCE(u))$. We deduce that $BCE(u) = \text{cl}(sBCE(u))$, as desired.

D. Proof of Theorem 4

We show (i) implies (ii) using contraposition: Suppose there exists a player i who has no action that jeopardizes every other action. By the convexity of the BCE set, this implies the existence of a BCE p where player i has no single action that is best response to all of i 's interim beliefs, i.e., $\bigcap_{a_i \in \text{supp}_i(p)} BR(p_{a_i}) = \emptyset$. For such a BCE, player i 's uninformed value is strictly below their gross value, demonstrating that $E_{ex} \subset E_{ac}$.

The proof that (ii) implies (iii) also follows by contraposition: Suppose there is a social planner, represented by some $u_0 \in \mathbb{R}^{A \times \Theta}$ and $w \in \mathbb{R}_+^{I_0}$, for which exogenous and acquired information yield distinct policy evaluations, $W_{ex}(u_0, w) > W_{ac}(u_0, w)$. Letting p be the worst-case BCE under endogenous learning, there must exist a player i whose gross value associated with p is strictly exceeds their uninformed value associated with p . Consequently, $\bigcap_{a_i \in \text{supp}(p)} BR(p_{a_i}) = \emptyset$, which further implies $\bigcap_{a_i \in \text{supp}(p)} J(a_i) = \emptyset$.

It remains to show that (iii) $\implies$ (i). To show this, we suppose (i) fails, and argue (iii) must fail as well.

Hence, suppose (i) fails. Take any $(p^*, v^*) \in E_{ac} \setminus E_{ex}$ (which exists because $E_{ex} \subseteq E_{ac}$). Clearly, must have $v_j^* < \bar{v}_j(p^*)$ for some $j \in I$. Let

$$\begin{aligned} E_{ex}^j &= \{(p, y) \in BCE \times \mathbb{R} : y = \bar{v}_j(p)\}, \\ E_{ac}^j &= \{(p, y) \in BCE \times \mathbb{R} : y \in [\underline{v}_j(p), \bar{v}_j(p)]\}. \end{aligned}$$

Then $(p^*, v_j^*) \in E_{ac}^j \setminus E_{ex}^j$. Since E_{ex}^j is closed and convex, a separating hyperplane theorem delivers a $u_0 \in \mathbb{R}^{A \times \Theta}$ and $w_j \in \mathbb{R}$ such that

$$w_j v_j^* + \sum_{a, \theta} u_0(a, \theta) p^*(a, \theta) < w_j y + \sum_{a, \theta} u_0(a, \theta) p(a, \theta)$$

for all $(p, y) \in E_{ex}^j$. Since $(p^*, \bar{v}_j(p^*)) \in E_{ex}^j$ and $v_j^* \leq \bar{v}_j(p^*)$, we must have $w_j > 0$. To conclude the proof, note that if we set $w_0 = 1$, and $w_i = 0$ for all $i \neq j$, then

$$\begin{aligned} W_{ac}(u_0, w) &\leq W_{ac}(p^*, v^*; u_0, w) = w_j v_j^* + \sum_{a, \theta} u_0(a, \theta) p^*(a, \theta) \\ &< \min_{p \in BCE} \left[w_j \bar{v}_j(p) + \sum_{a, \theta} u_0(a, \theta) p(a, \theta) \right] = W_{ex}(u_0, w). \end{aligned}$$

The proof is now complete.

E. Proof of Proposition 1

Let $\mathcal{G}$ be a symmetric binary-action game. We begin our proof by showing that focusing on symmetric outcomes is without loss for welfare analysis in symmetric games (the assumption of binary actions has no role in this result).

Note it is without loss to set $w_i = 1$ for all $i \in I_0$. For an outcome p , set $w_{ex}(p) = \sum_{a,\theta} u_0(a, \theta)p(a, \theta) + \sum_{i \in I} \bar{v}_i(p)$, and $w_{ac}(p) = \sum_{a,\theta} u_0(a, \theta)p(a, \theta) + \sum_i \underline{v}_i(p)$. The worst-case welfare under exogenous and acquired information is respectively given by

$$W_{ex} = \min_{p \in BCE} w_{ex}(p) \quad \text{and} \quad W_{ac} = \min_{p \in BCE} w_{ac}(p).$$

For short, we denote by BCE^{sy} the set of symmetric BCEs.

Claim 5. For every BCE p , there is a symmetric BCE q such that $w_{ex}(q) = w_{ex}(p)$ and $w_{ac}(q) \leq w_{ac}(p)$.

Proof. Fix a BCE p . Let Φ be the set of permutations of I . For every permutation $\phi \in \Phi$, we define the outcome p_ϕ by $p_\phi(a, \theta) = p(a_\phi, \theta)$. Note that player i in p_ϕ behaves as player $j = \phi^{-1}(i)$ in p . One can verify that p_ϕ is a BCE because p is a BCE and the game is symmetric.

We define the outcome q by $q = \frac{1}{|\Phi|} \sum_{\phi \in \Phi} p_\phi$, where $|\Phi|$ is the cardinality of Φ . As noted above, each p_ϕ is a BCE. Since the BCE set is convex, q is a BCE.

The outcome q is symmetric. Indeed, $\Phi = \{\psi^{-1} \circ \phi : \phi \in \Phi\}$ for every permutation $\psi \in \Phi$. We deduce that

$$\begin{aligned} q(a_\psi, \theta) &= \frac{1}{|\Phi|} \sum_{\phi \in \Phi} p_\phi(a_\psi, \theta) = \frac{1}{|\Phi|} \sum_{\phi \in \Phi} p(\psi^{-1} \circ \phi)(a_\psi, \theta) \\ &= \frac{1}{|\Phi|} \sum_{\phi \in \Phi} p(a_\phi, \theta) = \frac{1}{|\Phi|} \sum_{\phi \in \Phi} p_\phi(a, \theta) = q(a, \theta). \end{aligned}$$

Hence, q is symmetric.

To conclude the proof, we observe $w_{ex}(q) = \frac{1}{|\Phi|} \sum_{\phi \in \Phi} w_{ex}(p_\phi) = \frac{1}{|\Phi|} \sum_{\phi \in \Phi} w_{ex}(p) = w_{ex}(p)$, where the first equality holds because $w_{ex}(p_\phi)$ is affine in p_ϕ , and the second equality because the game is symmetric. Finally, note that $w_{ac}(q) \leq \frac{1}{|\Phi|} \sum_{\phi \in \Phi} w_{ac}(p_\phi) = \frac{1}{|\Phi|} \sum_{\phi \in \Phi} w_{ac}(p) = w_{ac}(p)$, where the first inequality holds because $w_{ac}(p_\phi)$ is convex in p_ϕ , and the second equality because the game is symmetric. $\square$

By Claim 5, W_{ex} is the value of the optimization problem

$$\min_{p \in BCE^{sy}} w_{ex}(p), \tag{17}$$

and W_{ac} is the value of the optimization problem

$$\min_{p \in BCE^{sy}} w_{ac}(p). \quad (18)$$

We consider also the following optimization problem:

$$\min_{p \in \Delta_\pi^{sy}(A \times \Theta)} w_{ac}(p), \quad (19)$$

We proceed by successive claims.

Claim 6. The following conditions are equivalent: (i) $W_{ac} < W_{ex}$, and (ii) $\underline{v}_i(p) < \bar{v}_i(p)$ for all players i and optimal solutions p of (18).

Proof. We first prove that (i) implies (ii). Suppose $W_{ac} < W_{ex}$ and let p be an optimal solution of (18). Then, $w_{ac}(p) = W_{ac} < W_{ex} \leq w_{ex}(p)$. The inequality $w_{ac}(p) < w_{ex}(p)$ implies that $\underline{v}_i(p) < \bar{v}_i(p)$ for some player i . Since the game is symmetric and p is symmetric, $\underline{v}_i(p) < \bar{v}_i(p)$ for all players i .

We now prove that (ii) implies (i). Suppose $\underline{v}_i(p) < \bar{v}_i(p)$ for all players i and optimal solutions p of (18). Let $p \in BCE$ be an optimal solution of (17). If p is also an optimal solution of (18), then

$$w_{ac}(p) = \sum_{a, \theta} u_0(a, \theta) p(a, \theta) + \sum_i \underline{v}_i(p) < \sum_{a, \theta} u_0(a, \theta) p(a, \theta) + \sum_i \bar{v}_i(p) = w_{ex}(p)$$

by hypothesis; thus, $W_{ac} < W_{ex}$. If instead p is not an optimal solution of (18), then $W_{ac} < w_{ac}(p) \leq w_{ex}(p) = W_{ex}$; thus, $W_{ac} < W_{ex}$. $\square$

Claim 7. For every $p \in BCE$ and $i \in I$, the following conditions are equivalent: (i) $\underline{v}_i(p) < \bar{v}_i(p)$, and (ii) $a_i \in \text{supp}_i(p)$ and $BR(p_{a_i}) = \{a_i\}$ for all $a_i \in A_i$.

Proof. Condition (i) holds if and only if player i is strictly better by following the action recommendation of the mediator rather than best responding ex ante. In other terms, player i has no action a_i such that for all $b_i \in \text{supp}_i(p)$, $a_i \in BR(p_{b_i})$. Given that A_i has two elements, this is equivalent to condition (ii). $\square$

Claim 8. The following conditions are equivalent: (i) all optimal solutions of (18) are BCEs with elementary incentives, and (ii) all optimal solutions of (19) are BCEs with elementary incentives.

Proof. First we show that (i) implies (ii). Let p be an optimal solution of (18) and let q be an optimal solution of (19). For every $t \in [0, 1]$, define $p^t = (1 - t)p + tq$. Furthermore, set $s = \max\{t : p^t \in BCE^{sy}\}$. Note that s is well defined: the set BCE^{sy} is closed and $p^0 = p \in BCE^{sy}$.

We observe that p^s is an optimal solution of (18): since $w_{ac}(p^t)$ is convex in t , $w_{ac}(p^s) \leq (1 - s)w_{ac}(p) + sw_{ac}(q) \leq w_{ac}(p) = w_{ac}$. Thus, p^s must have elementary incentives. But this implies that $s = 1$; otherwise, one could find $\epsilon > 0$ sufficiently small so that $p^{s+\epsilon} \in BCE^{sy}$, contradicting the definition of p^s . This implies that $q = p^s$ is a BCE with elementary incentives.

Now we show that (ii) implies (i). Let p be an optimal solution of (18) and let q be an optimal solution of (19). By hypothesis, q is a BCE. Thus, q is an optimal solution of (18). This implies that p is an optimal solution of (19), and therefore has elementary incentives. $\square$

By combining the three claims above, we obtain Proposition 1.

F. Proof of Proposition 2

F.1. Proof of Proposition 2, part (i)

We begin with by characterizing the worst-case utilitarian welfare among all outcomes in $\Delta_\pi^{sy}(A \times \Theta)$, ignoring the obedience constraint.

Lemma 18. *In the regime change game, an outcome p^* minimizes $\sum_{i=1}^n v_i(p)$ over all $p \in \Delta_\pi^{sy}(A \times \Theta)$ if and only if the following conditions hold: for all $\theta \in \Theta$,*

$$p^* \left(\left\{ (a, \theta) : \theta - 1 \leq \sum_i a_i \leq \theta \right\} \right) = 0, \quad (20)$$

$$p^* \left(\left\{ (a, \theta) : \sum_i a_i > \theta \right\} \right) = \frac{k}{1+x}. \quad (21)$$

The two optimality conditions have simple interpretations. First, no investor is ever pivotal. And second, the attack succeeds with probability $k/(1+x)$.

Proposition 2, part (i), easily follows from Lemma 18. To see this, consider the following symmetric outcome p^* : investor behavior is independent of θ , with all investors attacking together with probability $k/(1+x)$ and no investor attacking with

the complementary probability. This outcome is clearly a BCE that satisfies (20) and (21). Lemma 18 then implies that p^* minimizes the sum of the uninformed values across all symmetric BCEs. Since the objective function is symmetric, it follows that p^* minimizes the sum of the uninformed values across all (possibly non-symmetric) BCEs. Proposition 2, part (i), then is obtained from the fact that p^* gives each player an uninformed value of $-xk/(1+x)$.

Proof of Lemma 18. For short, we define $w_{ac}(p) = \sum_i v_i(p)$.

We begin with a result that establishes a necessary condition for an outcome to solve the claim's program (which is the relaxed program from Proposition 1). To state the result, let $U_i(a_i, p)$ be player i 's payoff if she always takes action a_i while (a_{-i}, θ) is distributed according to p :

$$U_i(a_i, p) = \sum_{b_i, a_{-i}, \theta} u_i(a_i, a_{-i}, \theta) p(b_i, a_{-i}, \theta).$$

Note that for all $p \in \Delta_\pi^{sy}(A \times \Theta)$ and $i \in I$, $w_{ac}(p) = n \max\{U_i(0, p), U_i(1, p)\}$. Thus,

$$\operatorname{argmin}_{p \in \Delta_\pi^{sy}(A \times \Theta)} w_{ac}(p) = \operatorname{argmin}_{p \in \Delta_\pi^{sy}(A \times \Theta)} \max\{U_i(0, p), U_i(1, p)\}.$$

Claim 9. Every $p^* \in \operatorname{argmin}_{p \in \Delta_\pi^{sy}(A \times \Theta)} w_{ac}(p)$ has $U_i(0, p^*) = U_i(1, p^*)$ for all $i \in I$.

Proof. We prove the contrapositive: if $p^* \in \Delta_\pi^{sy}(A \times \Theta)$ has $U_i(0, p^*) \neq U_i(1, p^*)$, then $p^* \notin \operatorname{argmin}_{p \in \Delta_\pi^{sy}(A \times \Theta)} w_{ac}(p)$.

We first consider the case in which $U_i(0, p^*) > U_i(1, p^*)$. Let q be the outcome where all investors always attack. Observe that $q \in \operatorname{argmin}_{p \in \Delta_\pi^{sy}(A \times \Theta)} U_i(0, p)$, and that every $r \in \operatorname{argmin}_{p \in \Delta_\pi^{sy}(A \times \Theta)} U_i(0, p)$ has the speculative attack succeeding with probability one. Hence, every such r has $U_i(0, r) < U_i(1, r)$, which implies that $p^* \notin \operatorname{argmin}_{p \in \Delta_\pi^{sy}(A \times \Theta)} U_i(0, p)$. We deduce that $U_i(0, q) < U_i(0, p^*)$.

For every $\epsilon \in (0, 1)$, we define $q^\epsilon = \epsilon q + (1 - \epsilon)p^* \in \Delta_\pi^{sy}(A \times \Theta)$. Using the inequality $U_i(0, q) < U_i(0, p^*)$, we obtain that for all $\epsilon > 0$ small enough,

$$w_{ac}(q^\epsilon) = nU_i(0, p^\epsilon) = n(\epsilon U_i(0, q) + (1 - \epsilon)U_i(0, p^*)) < nU_i(0, p^*) = w_{ac}(p^*).$$

We conclude that $p^* \notin \operatorname{argmin}_{p \in \Delta_\pi^{sy}(A \times \Theta)} w_{ac}(p)$.

The argument for the case $U_i(0, p^*) < U_i(1, p^*)$ is similar, but with q being replaced

by the outcome where no one ever speculates. $\square$

Thanks to Claim 9, to determine $\operatorname{argmin}_{p \in \Delta_\pi^{sy}(A \times \Theta)} w_{ac}(p)$, we can study the following “simpler” optimization problem:

$$\min_{p \in \Delta_\pi^{sy}(A \times \Theta)} U_i(0, p) \quad \text{s.t.} \quad U_i(0, p) = U_i(1, p). \quad (22)$$

Claim 10. An outcome $p \in \Delta_\pi^{sy}(A \times \Theta)$ is an optimal solution of (22) if and only if

$$p \left(\sum_{j \neq i} a_j = \theta - 1 \right) = 0 \quad \text{and} \quad p \left(\sum_{j \neq i} a_j \geq \theta \right) = \frac{k}{1+x}.$$

Proof. Simple algebra shows that every p that satisfies $U_i(1, p) = U_i(0, p)$ must yield

$$U_i(0, p) = -xp \left(\sum_{j \neq i} a_j \geq \theta \right) = -x \left[\frac{k - p \left(\sum_{j \neq i} a_j = \theta - 1 \right)}{1+x} \right].$$

Therefore, we get that (22) is the same as

$$\min_{p \in \Delta_\pi^{sy}(A \times \Theta)} p \left(\sum_{j \neq i} a_j = \theta - 1 \right) \quad \text{s.t.} \quad p \left(\sum_{j \neq i} a_j \geq \theta \right) = \frac{k - p \left(\sum_{j \neq i} a_j = \theta - 1 \right)}{1+x}.$$

Hence, to complete the proof, we only need to be sure that there is $p \in \Delta_\pi^{sy}(A \times \Theta)$ such that

$$p \left(\sum_{j \neq i} a_j = \theta - 1 \right) = 0 \quad \text{and} \quad p \left(\sum_{j \neq i} a_j \geq \theta \right) = \frac{k}{1+x}.$$

Such an outcome is easy to construct: with probability $k/1+x$, all players attack; with the remaining probability, no player attacks. $\square$

The following result connects what we have just found with the conditions in the statement of Lemma 18.

Claim 11. For an outcome $p \in \Delta_\pi(A \times \Theta)$, the following conditions are equivalent:

(i) For all players i and payoff states θ ,

$$p\left(\sum_{j \neq i} a_j = \theta - 1\right) = 0, \quad (23)$$

$$p\left(\sum_{j \neq i} a_j \geq \theta\right) = \frac{k}{1+x}. \quad (24)$$

(ii) For all payoff states θ ,

$$p\left(\theta - 1 \leq \sum_i a_i \leq \theta\right) = 0, \quad (25)$$

$$p\left(\sum_i a_i > \theta\right) = \frac{k}{1+x}. \quad (26)$$

Proof. First we show that (i) implies (ii). Since $\max \Theta < n$,

$$p\left(\sum_i a_i = \theta - 1\right) = p\left(\sum_i a_i = \theta - 1, \text{ and } a_i = 0 \text{ for some } i\right).$$

Thus, $p(\sum_i a_i = \theta - 1) \leq \sum_i p(\sum_{j \neq i} a_j = \theta - 1, \text{ and } a_i = 0) = 0$, where the last equality follows from (23). Moreover, since $\min \Theta > 0$,

$$p\left(\sum_i a_i = \theta\right) = p\left(\sum_i a_i = \theta, \text{ and } a_i = 1 \text{ for some } i\right).$$

Thus, $p(\sum_i a_i = \theta) \leq \sum_i p(\sum_{j \neq i} a_j = \theta - 1, \text{ and } a_i = 1) = 0$, where the last equality follows from (23). We conclude that (25) holds.

To prove (26), notice that $p(\sum_i a_i > \theta) = p(\sum_i a_i \geq \theta)$, because we have just verified that $p(\sum_i a_i = \theta) = 0$. Then, fixing some player i^* ,

$$\begin{aligned} p\left(\sum_i a_i \geq \theta\right) &= p\left(\sum_{i \neq i^*} a_i \geq \theta, \text{ and } a_{i^*} = 0\right) + p\left(\sum_{i \neq i^*} a_i \geq \theta - 1, \text{ and } a_{i^*} = 1\right) \\ &= p\left(\sum_{i \neq i^*} a_i \geq \theta, \text{ and } a_{i^*} = 0\right) + p\left(\sum_{i \neq i^*} a_i \geq \theta, \text{ and } a_{i^*} = 1\right) \\ &= p\left(\sum_{i \neq i^*} a_i \geq \theta\right) = \frac{k}{1+x}, \end{aligned}$$

where the second equality holds by (23), and the last equality by (24). We deduce (26). This completes the proof that (i) implies (ii).

Now we show that (ii) implies (i). Observe that

$$p\left(\sum_{j \neq i} a_j = \theta - 1\right) = p\left(\sum_j a_j = \theta - 1, \text{ and } a_i = 0\right) + p\left(\sum_j a_j = \theta, \text{ and } a_i = 1\right).$$

By (25), the right-hand side is equal to zero: we deduce (23). We obtain (24) from the following chain of equalities:

$$\begin{aligned} p\left(\sum_{j \neq i} a_j \geq \theta\right) &= p\left(\sum_j a_j \geq \theta, \text{ and } a_i = 0\right) + p\left(\sum_j a_j > \theta, \text{ and } a_i = 1\right) \\ &= p\left(\sum_j a_j > \theta, \text{ and } a_i = 0\right) + p\left(\sum_j a_j > \theta, \text{ and } a_i = 1\right) \\ &= p\left(\sum_j a_j > \theta\right) = \frac{k}{1+x}, \end{aligned}$$

where the second equality follows from (25), and the last equality from (26). This completes the proof that (ii) implies (i). $\square$

Combining the three results above, we obtain Lemma 18.

F.2. Proof of Proposition 2, Part (ii)

For short, we write W_{ac} and W_{ex} instead of $W_{ac}(k, x, \pi)$ and $W_{ex}(k, x, \pi)$, respectively. To prove Proposition 2, part (ii), we first characterize when $W_{ac} < W_{ex}$ for an arbitrary prior π . We then specialize the characterization to priors with support of size 2 or less. The specialization immediately implies that $W_{ac} < W_{ex}$ whenever π is deterministic (i.e., has singleton support). When $n \geq 6$, it is also straightforward to use the characterization to find a π where $W_{ac} = W_{ex}$. The result then follows.

Claim 12. The inequality $W_{ac} < W_{ex}$ holds if and only if all symmetric outcomes p that satisfy (20) and (21), also satisfy

$$p_{a_i=1}\left(\sum_j a_j \geq \theta\right) > \frac{k}{1+x}, \quad (27)$$

where $p_{a_i=1}$ is the conditional probability of (a_{-i}, θ) given $a_i = 1$.

Proof. By Proposition 1, the inequality $W_{ac} < W_{ex}$ holds if and only if all optimal solutions of

$$\min_{p \in \Delta_{\pi^y}^{sy}(A \times \Theta)} \sum_i v_i(p)$$

have elementary incentives. By Lemma 18, the latter condition is equivalent to the following statement: all symmetric outcomes p that satisfy (20) and (21), also have elementary incentives. Next we verify that, for all symmetric outcomes p that satisfy (20) and (21), having elementary incentives is equivalent to (27).

Let p be a symmetric outcome that satisfy (20) and (21). First, note (21) implies the attack succeeds with a probability strictly between 0 and 1, and so players must both attack and not attack with positive probability due to symmetry. Hence $\text{supp}_i(p) = \{0, 1\} = A_i$.

Given $\text{supp}_i(p) = \{0, 1\}$, i 's obedience constraints are strict when

$$p_{a_i=1} \left(\sum_{j \neq i} a_j \geq \theta - 1 \right) - k > -x p_{a_i=1} \left(\sum_{j \neq i} a_j \geq \theta \right), \quad (28)$$

$$p_{a_i=0} \left(\sum_{j \neq i} a_j \geq \theta - 1 \right) - k < -x p_{a_i=0} \left(\sum_{j \neq i} a_j \geq \theta \right). \quad (29)$$

By (20)—see also Claim 11—

$$p_{a_i=1} \left(\sum_{j \neq i} a_j \geq \theta - 1 \right) = p_{a_i=1} \left(\sum_{j \neq i} a_j \geq \theta \right),$$

and

$$p_{a_i=0} \left(\sum_{j \neq i} a_j \geq \theta - 1 \right) = p_{a_i=0} \left(\sum_{j \neq i} a_j \geq \theta \right).$$

Thus, (28) and (29) hold if and only if

$$p_{a_i=1} \left(\sum_{j \neq i} a_j \geq \theta \right) > \frac{k}{1+x} > p_{a_i=0} \left(\sum_{j \neq i} a_j \geq \theta \right).$$

By (20) and (21)—see also Claim 11— $p \left(\sum_{j \neq i} a_j \geq \theta \right) = \frac{k}{1+x}$. Thus, by the law of

total probability, (28) and (29) hold if and only if

$$p_{a_i=1} \left(\sum_j a_j \geq \theta \right) = p_{a_i=1} \left(\sum_{j \neq i} a_j \geq \theta - 1 \right) = p_{a_i=1} \left(\sum_{j \neq i} a_j \geq \theta \right) > \frac{k}{1+x}.$$

Overall, we conclude that, for all symmetric outcomes p that satisfy (20) and (21), having elementary incentives is equivalent to (27). $\square$

Next we refine the characterization $W_{ac} < W_{ex}$ obtained in Claim 12. To state this refinement, denote the CDF of θ by $F(\theta) := \sum_{\theta' \leq \theta} \pi(\theta')$. Define also the cutoff θ^* by

$$\theta^* = \min \left\{ \theta \in \Theta : F(\theta) \geq \frac{k}{1+x} \right\}.$$

Claim 13. The inequality $W_{ex} > W_{ac}$ holds if and only if

$$F(\theta^*) (\theta^* - \mathbb{E}[\theta | \theta \leq \theta^*]) < \frac{k}{1+x} \left(3 - \frac{3k}{1+x} + \theta^* - \mathbb{E}[\theta] \right). \quad (30)$$

Proof. By Claim 12, $W_{ac} < W_{ex}$ is equivalent to

$$\begin{aligned} \frac{k}{1+x} &< \min_{p \in \Delta_{\pi}^{sy}(A \times \Theta)} p_{a_i=1} \left(\sum_{j \neq i} a_j \geq \theta - 1 \right) \\ &\text{s.t. (20) and (21).} \end{aligned} \quad (31)$$

Hence, showing (30) and (31) are equivalent is sufficient. To show this equivalence, we first characterize the unique solution to the program on the right hand side of (31). This solution gives the value of the program, which we then compare to $k/(1+x)$.

We begin with an alternative way of representing symmetric outcomes. This representation is based on the observation that an outcome $p \in \Delta_{\pi}(A \times \Theta)$ is symmetric if and only if, conditional on the state, all action profiles with the same number of attackers have the same probability. Consequently, $p \in \Delta_{\pi}^{sy}(A \times \Theta)$ if and only if there is $Q : \Theta \rightarrow \Delta(\{0, \dots, n\})$ such that

$$p(a, \theta) = \binom{n}{\sum_j a_j} Q \left(\sum_j a_j \middle| \theta \right) \pi(\theta),$$

where $\binom{n}{\sum_j a_j}$ is the binomial coefficient. Thus, one can write

$$p(a_i = 1) = \sum_{\theta} \pi(\theta) \sum_{m=1}^n \frac{m}{n} Q(m|\theta).$$

Moreover, condition (20) is equivalent to $Q(\theta - 1|\theta) = Q(\theta|\theta) = 0$. Therefore,

$$p\left(\sum_{j \neq i} a_j \geq \theta - 1 \text{ and } a_i = 1\right) = \sum_{\theta} \pi(\theta) \sum_{m \geq \theta+1} \frac{m}{n} Q(m|\theta),$$

and

$$p\left(\sum_j a_j \geq \theta\right) = \sum_{\theta} \pi(\theta) \sum_{m \geq \theta+1} Q(m|\theta).$$

Hence, letting

$$f(Q) = \frac{\sum_{\theta} \pi(\theta) \sum_{m \geq \theta+1} m Q(m|\theta)}{\sum_{\theta} \pi(\theta) \sum_{m=1}^n m Q(m|\theta)},$$

we can write the program on the right hand side of (30) as

$$\begin{aligned} \min_{Q: \Theta \rightarrow \Delta(\{0, \dots, n\})} & f(Q) \\ \text{s.t.} \quad & \sum_{\theta} \pi(\theta) \sum_{m \geq \theta+1} Q(m|\theta) = \frac{k}{1+x}, \\ & Q(\theta - 1|\theta) = Q(\theta|\theta) = 0 \text{ for all } \theta. \end{aligned} \tag{32}$$

Since the constraint set is compact and the objective continuous, the above program admits a solution, Q^* . We now use perturbation-based arguments to show Q^* must satisfy a few properties:

1. $Q^*(m|\theta) = 0$ whenever $m \notin \{\theta - 2, \theta + 1\}$: if $Q^*(m|\theta) > 0$ for $m > \theta + 1$ (resp., $m < \theta - 2$), one can reduce the objective without violating the constraints by moving $\epsilon > 0$ mass from $Q^*(m|\theta)$ to $Q^*(\theta + 1|\theta)$ (resp., $Q^*(\theta - 2|\theta)$).
2. If $Q^*(\theta + 1|\theta) > 0$, then $Q^*(\theta' + 1|\theta') = 1$ for all $\theta' < \theta$: For a contradiction, suppose $Q^*(\theta + 1|\theta) > 0$, but $Q^*(\theta' + 1|\theta') < 1$ for some $\theta' < \theta$. For every $\epsilon > 0$,

define the following perturbation Q^ϵ of Q :

$$Q^\epsilon(m|\hat{\theta}) = \begin{cases} Q^*(\theta + 1|\theta) - \epsilon & \text{if } m = \theta + 1, \hat{\theta} = \theta, \\ Q^*(\theta - 2|\theta) + \epsilon & \text{if } m = \theta - 2, \hat{\theta} = \theta, \\ Q^*(\theta' + 1|\theta') + \epsilon \frac{\pi(\theta)}{\pi(\theta')} & \text{if } m = \theta' + 1, \hat{\theta} = \theta', \\ Q^*(\theta' - 2|\theta') - \epsilon \frac{\pi(\theta)}{\pi(\theta')} & \text{if } m = \theta' - 2, \hat{\theta} = \theta', \\ Q^*(m|\hat{\theta}) & \text{otherwise.} \end{cases}$$

The contradiction assumption means Q^ϵ is feasible for all sufficiently small $\epsilon > 0$.

Direct computation shows

$$\lim_{\epsilon \searrow 0} \frac{1}{\epsilon} (f(Q^\epsilon) - f(Q^*)) = \frac{\pi(\theta)(\theta' - \theta)}{\sum_{\hat{\theta}} \pi(\hat{\theta}) \sum_{m=1}^n m Q^*(m|\hat{\theta})} < 0,$$

contradicting the optimality of Q .

3. If $Q^*(\theta - 2|\theta) > 0$, then $Q^*(\theta' - 2|\theta') = 1$ for all $\theta' > \theta$: For a contradiction, suppose $Q^*(\theta - 2|\theta) > 0$, but $Q^*(\theta' - 2|\theta') < 1$ for some $\theta' > \theta$. For every $\epsilon > 0$, define the following perturbation Q^ϵ of Q :

$$Q^\epsilon(m|\hat{\theta}) = \begin{cases} Q^*(\theta - 2|\theta) - \epsilon & \text{if } m = \theta - 2, \hat{\theta} = \theta, \\ Q^*(\theta + 1|\theta) + \epsilon & \text{if } m = \theta + 1, \hat{\theta} = \theta, \\ Q^*(\theta' - 2|\theta') + \epsilon \frac{\pi(\theta)}{\pi(\theta')} & \text{if } m = \theta' - 2, \hat{\theta} = \theta', \\ Q^*(\theta' + 1|\theta') - \epsilon \frac{\pi(\theta)}{\pi(\theta')} & \text{if } m = \theta' + 1, \hat{\theta} = \theta', \\ Q^*(m|\hat{\theta}) & \text{otherwise.} \end{cases}$$

The contradiction assumption means Q^ϵ is feasible for all sufficiently small $\epsilon > 0$.

Direct computation shows

$$\lim_{\epsilon \searrow 0} \frac{1}{\epsilon} (f(Q^\epsilon) - f(Q^*)) = \frac{\pi(\theta)(\theta - \theta')}{\sum_{\hat{\theta}} \pi(\hat{\theta}) \sum_{m=1}^n m Q(m|\hat{\theta})} < 0,$$

contradicting the optimality of Q^* .

The above conditions imply the optimal Q^* admits a cutoff $\tilde{\theta}$ such that $Q^*(\theta + 1|\theta) = 1$ for all $\theta < \tilde{\theta}$, $Q^*(\theta - 2|\theta) = 1$ for all $\theta > \tilde{\theta}$, and $Q^*(\{\tilde{\theta} + 1, \tilde{\theta} - 2\}|\tilde{\theta}) = 1$. Then, the

constraint

$$\sum_{\theta} \pi(\theta) \sum_{m \geq \theta+1} Q(m|\theta) = \frac{k}{1+x}$$

pins down the optimum: we must have $\tilde{\theta} = \theta^*$, and

$$Q^*(\theta^* + 1|\theta^*) = \frac{1}{\pi(\theta^*)} \left(\frac{k}{1+x} - F(\theta^* - 1) \right).$$

Therefore, the inequality (31) becomes

$$\begin{aligned} \frac{k}{1+x} < f(Q^*) &= \frac{\sum_{\theta} \pi(\theta) \sum_{m \geq \theta+1} m Q^*(m|\theta)}{\sum_{\theta} \pi(\theta) \sum_{m=1}^n m Q^*(m|\theta)} \\ &= \frac{F(\theta^*) \mathbb{E}[\theta + 1|\theta \leq \theta^*] - \left(F(\theta^*) - \frac{k}{1+x}\right) (\theta^* + 1)}{\mathbb{E}[\theta] + \frac{k}{1+x} - 2 \left(1 - \frac{k}{1+x}\right)}. \end{aligned}$$

Rearranging the above equation gives (30). $\square$

Next, we specialize Claim 13 to two states.

Claim 14. Suppose $\text{supp}(\pi) = \{\underline{\theta}, \bar{\theta}\}$, where $\bar{\theta} \geq \underline{\theta}$. Then $W_{ac} = W_{ex}$ if and only if $\bar{\theta} - \underline{\theta} \geq 3$ and

$$1 - \frac{1}{3} (\bar{\theta} - \underline{\theta}) \pi(\bar{\theta}) \leq \frac{k}{1+x} \leq \frac{1}{3} (\bar{\theta} - \underline{\theta}) (1 - \pi(\bar{\theta})). \quad (33)$$

Proof. We begin the proof by explicitly stating the implication of (30) for the binary case. In particular, we show $W_{ac} < W_{ex}$ if and only if one of the following two conditions hold:

- (i) $\frac{k}{1+x} > \pi(\underline{\theta})$ and $\frac{k}{1+x} > \frac{1}{3} \pi(\underline{\theta}) (\bar{\theta} - \underline{\theta})$.
- (ii) $\frac{k}{1+x} \leq \pi(\underline{\theta})$ and $\frac{k}{1+x} < 1 - \frac{1}{3} \pi(\bar{\theta}) (\bar{\theta} - \underline{\theta})$.

To prove the above, we consider two cases, depending on the value of θ^* :

- *Case 1:* $k/(1+x) > \pi(\underline{\theta})$. Then $\theta^* = \bar{\theta}$, and the inequality (30) specializes to

$$\bar{\theta} - \mathbb{E}[\theta] < \frac{k}{1+x} \left(3 - \frac{3k}{1+x} + \bar{\theta} - \mathbb{E}[\theta] \right).$$

Substituting $\bar{\theta} - \mathbb{E}[\theta] = \pi(\underline{\theta})(\bar{\theta} - \underline{\theta})$ and rearranging gives

$$\left(1 - \frac{k}{1+x}\right) \pi(\underline{\theta})(\bar{\theta} - \underline{\theta}) < 3 \frac{k}{1+x} \left(1 - \frac{k}{1+x}\right),$$

which is equivalent to

$$\frac{1}{3} \pi(\underline{\theta})(\bar{\theta} - \underline{\theta}) < \frac{k}{1+x}.$$

Thus, we have established (i) is sufficient for $W_{ac} < W_{ex}$, and necessary if $\pi(\underline{\theta}) \geq \frac{k}{1+x}$.

- *Case 2:* Suppose now $k/(1+x) \leq \pi(\underline{\theta})$. Then $\theta^* = \underline{\theta}$. Thus, the inequality (30) is now

$$0 < \frac{k}{1+x} \left(3 - \frac{3k}{1+x} + \underline{\theta} - \mathbb{E}[\theta]\right).$$

Note $\underline{\theta} - \mathbb{E}[\theta] = -\pi(\bar{\theta})(\bar{\theta} - \underline{\theta})$. Therefore, the above inequality is equivalent to

$$\frac{k}{1+x} < 1 - \frac{1}{3} \pi(\bar{\theta})(\bar{\theta} - \underline{\theta}).$$

Hence, (ii) is sufficient for $W_{ac} < W_{ex}$, and necessary if $\pi(\underline{\theta}) \leq \frac{k}{1+x}$.

Next, we argue that a violation of one of the claim's conditions implies that either (i) or (ii) above hold. Suppose first $\bar{\theta} - \underline{\theta} < 3$. In this case, $\frac{1}{3} \pi(\underline{\theta})(\bar{\theta} - \underline{\theta}) < \pi(\underline{\theta})$, and so (i) holds whenever $\frac{k}{1+x} > \pi(\underline{\theta})$. If $\frac{k}{1+x} \leq \pi(\underline{\theta})$, then (ii) holds, because

$$1 - \frac{1}{3} \pi(\bar{\theta})(\bar{\theta} - \underline{\theta}) > 1 - \pi(\bar{\theta}) = \pi(\underline{\theta}) \geq \frac{k}{1+x}.$$

Suppose now $\bar{\theta} - \underline{\theta} \geq 3$, but (33) fails. Then one of the following inequality chains must hold: either

$$\frac{k}{1+x} > \frac{1}{3}(\bar{\theta} - \underline{\theta})(1 - \pi(\bar{\theta})) = \frac{1}{3}(\bar{\theta} - \underline{\theta})\pi(\underline{\theta}) \geq \pi(\underline{\theta}),$$

or

$$\frac{k}{1+x} < 1 - \frac{1}{3}(\bar{\theta} - \underline{\theta})\pi(\bar{\theta}) \leq 1 - \pi(\bar{\theta}) = \pi(\underline{\theta}).$$

Either way, $W_{ac} < W_{ex}$ holds: the first inequality chain implies (i), whereas the second inequality chain implies (ii).

To conclude the proof, we show that the claim's condition must hold if neither (i) nor (ii) hold. Suppose first that $\frac{k}{1+x} > \pi(\underline{\theta})$, but (i) fails. Then

$$\frac{1}{3}(\bar{\theta} - \underline{\theta})(1 - \pi(\bar{\theta})) = \frac{1}{3}(\bar{\theta} - \underline{\theta})\pi(\underline{\theta}) \geq \frac{k}{1+x} > \pi(\underline{\theta}),$$

meaning $\bar{\theta} - \underline{\theta} \geq 3$, and the right inequality in (33) holds. For the left inequality, note that

$$1 - \frac{1}{3}(\bar{\theta} - \underline{\theta})\pi(\bar{\theta}) \leq 1 - \pi(\bar{\theta}) = \pi(\underline{\theta}) < \frac{k}{1+x}.$$

Suppose now $\frac{k}{1+x} \leq \pi(\underline{\theta})$, but (ii) fails. Then,

$$1 - \pi(\bar{\theta}) \geq \frac{k}{1+x} \geq 1 - \frac{1}{3}(\bar{\theta} - \underline{\theta})\pi(\bar{\theta}).$$

The right inequality above delivers the left inequality in (33). Moreover, the implied inequality between the left most expression and the right most expression implies

$$\frac{1}{3}(\bar{\theta} - \underline{\theta})\pi(\bar{\theta}) \geq \pi(\bar{\theta}),$$

and so $\bar{\theta} - \underline{\theta} \geq 3$. Finally, to get the right inequality in (33), notice that

$$\frac{k}{1+x} \leq 1 - \pi(\bar{\theta}) \leq \frac{1}{3}(\bar{\theta} - \underline{\theta})(1 - \pi(\bar{\theta})),$$

where the last inequality holds because $\bar{\theta} - \underline{\theta} \geq 3$. □

We are now ready to prove Proposition 2, Part (ii). Take π' to be any distribution with singleton support. Note that (33) does not hold for any x and k , meaning $W_{ac}(k, x, \pi') < W_{ex}(k, x, \pi')$.

Next, set $\pi(2) = k/(1+x)$, and $\pi(5) = 1 - \pi'$ (notice $\{2, 5\} \subseteq \Theta$ because $n \geq 6$). Direct computation shows (33) holds, and so $W_{ac}(k, x, \pi) = W_{ex}(k, x, \pi)$, as desired.

G. Proof of Proposition 3

Let us briefly outline the proof of the proposition. The implication from (ii) to (iii) is straightforward: if a player has an action that is always optimal, a BCE can be separated only if that player's beliefs about (a_{-i}, θ) remain unchanged regardless

of the recommendation they receive. Similarly, the implication from (iii) to (i) is immediate, as a player who remains uninformed cannot learn anything about others' payoffs.

The key step is showing that (i) implies (ii). We establish it in two parts. First, we negate (ii) and construct a (potentially informationally simple) sBCE in which some learning occurs for some player i . Then, leveraging the separability of the game, we introduce a small amount of θ_j -dependent noise into i 's signal, thereby generating a new sBCE that is no longer informationally simple—contradicting (i).

We now proceed with the formal proof.

(i) implies (ii). We proceed by contradiction. Suppose an $p \in sBCE(\mathcal{G})$ exists that involves some learning: an $i \in I$ and $a_i^*, b_i^* \in \text{supp}_i(p)$ such that $p_{a_i^*} \neq p_{b_i^*}$. If p is not informationally simple, then we are done. Assume, therefore, that $p = \prod_{j \in I} p_j$ is informationally simple. We will use p to build an outcome $q \in sBCE(\mathcal{G})$ that is not informationally simple.

By hypothesis, there is a player $j \neq i$ such that $\text{supp}(\pi_j)$ has at least two distinct elements. Thus, let E_j be a subset of Θ_j such that both E_j and its complement, E_j^c , have positive probability, defined as $\pi_j(E_j)$ and $\pi_j(E_j^c)$, respectively.

For all $a_i, b_i \in \text{supp}_i(p)$, we write $a_i \sim b_i$ if $p_{a_i} = p_{b_i}$. Notice that $\sim$ is an equivalence relation; let X_i be the set of equivalence classes: $X_i = \{[a_i] : a_i \in \text{supp}_i(p)\}$. For every $x_i \in X_i$ and $\theta_i \in \Theta_i$, let $p_i(x_i|\theta_i)$ be the conditional probability of x_i given θ_i :

$$p_i(x_i|\theta_i) = \sum_{a_i \in x_i} \frac{p_i(a_i, \theta_i)}{\pi_i(\theta_i)}.$$

Given $\epsilon \in (0, 1)$, we define an experiment for player i , $\xi_i : \Theta_i \times \Theta_j \rightarrow \Delta(X_i)$, as follows:

$$\xi_i(x_i|\theta_i, \theta_j) = \begin{cases} (1 - \epsilon)p_i(x_i|\theta_i) + \epsilon & \text{if } x_i = [a_i^*] \text{ and } \theta_j \in E_j, \\ (1 - \epsilon)p_i(x_i|\theta_i) & \text{if } x_i \neq [a_i^*] \text{ and } \theta_j \in E_j, \\ (1 - \epsilon)p_i(x_i|\theta_i) + \epsilon & \text{if } x_i = [b_i^*] \text{ and } \theta_j \notin E_j, \\ (1 - \epsilon)p_i(x_i|\theta_i) & \text{if } x_i \neq [b_i^*] \text{ and } \theta_j \notin E_j. \end{cases}$$

The experiment ξ_i is well defined because $p_{a_i^*} \neq p_{b_i^*}$ implies $[a_i^*] \neq [b_i^*]$. For ϵ small, one can think of $(\theta_i, \theta_j) \mapsto \xi_i(\cdot|\theta_i, \theta_j)$ as a perturbation of $\theta_i \mapsto p_i(\cdot|\theta_i)$.

Let $\sigma_i : X_i \rightarrow \Delta(A_i)$ be an action plan for player i such that, for each $x_i \in X_i$,

there is $a_i \in A_i$ such that $\sigma_i(a_i|x_i) = 1$ and

$$a_i \in \arg \max_{b_i \in B_i} \sum_{\theta_i, \theta_j} \bar{u}_i(b_i, \theta_i) \xi_i(x_i|\theta_i, \theta_j) \pi(\theta_i) \pi(\theta_j).$$

Let $q : \Delta(A \times \Theta)$ be the outcome defined by

$$q(a, \theta) = \left(\sum_{x_i \in X_i} \sigma_i(a_i|x_i) \xi_i(x_i|\theta_i, \theta_j) \pi_i(\theta_i) \right) \prod_{j \neq i} p_j(a_j, \theta_j).$$

We show that, for ϵ small enough, q is a separable BCE but not informationally simple.

For every $x_i \in X_i$, let p_{x_i} be conditional distribution of (a_{-i}, θ) given x_i , according to p . It is easy to verify that $p_{x_i} = p_{a_i}$ for all $a_i \in x_i$. Let $q_{x_i} \in \Delta(A_{-i} \times \Theta)$ be conditional distribution of (a_{-i}, θ) given x_i , according to q . Denote by $\bar{p}_{x_i}$ and $\bar{q}_{x_i}$ the marginals of p_{x_i} and q_{x_i} on Θ_i , respectively. The probability distributions $\bar{p}_{x_i}$ and $\bar{q}_{x_i}$ are related as follows. If $x_i = [a_i^*]$, then

$$\bar{q}_{x_i} = \frac{(1 - \epsilon)p(x_i)}{(1 - \epsilon)p(x_i) + \epsilon\pi_j(E_j)} p_{x_i} + \frac{\epsilon\pi_j(E_j)}{(1 - \epsilon)p(x_i) + \epsilon\pi_j(E_j)} \pi$$

where $p(x_i) = \sum_{a_i \in x_i} p(a_i)$. If $x_i = [b_i^*]$, then

$$\bar{q}_{x_i} = \frac{(1 - \epsilon)p(x_i)}{(1 - \epsilon)p(x_i) + \epsilon\pi_j(E_j^c)} p_{x_i} + \frac{\epsilon\pi_j(E_j^c)}{(1 - \epsilon)p(x_i) + \epsilon\pi_j(E_j^c)} \pi$$

where $p(x_i) = \sum_{a_i \in x_i} p(a_i)$. If $x_i \neq [a_i^*]$ and $x_i \neq [b_i^*]$, then $\bar{q}_{x_i} = \bar{p}_{x_i}$. Notice that, for all $x_i \in X_i$, $\bar{q}_{x_i}$ approaches $\bar{p}_{x_i}$ as ϵ approaches zero. As a result, since the best-response correspondence is upper hemicontinuous and the game is separable, we can choose ϵ sufficiently small so that

$$BR(q_{x_i}) \subseteq BR(p_{x_i}) \quad \text{for all } x_i \in X_i. \quad (34)$$

Claim 15. If $\sigma_i(a_i|x_i) > 0$ and $\sigma_i(a_i|y_i) > 0$, then $x_i = y_i$.

Proof. By construction, $a_i \in BR(q_{x_i}) \cap BR(q_{y_i})$. It follows from (34) that $BR(p_{x_i}) \cap BR(p_{y_i}) \neq \emptyset$. Take $b_i, c_i \in A_i$ such that $b_i \in x_i$ and $c_i \in y_i$. As noted above, $p_{x_i} = p_{b_i}$ and $p_{y_i} = p_{c_i}$. As a result, $BR(p_{a_i}) \cap BR(p_{b_i}) \neq \emptyset$. Since p is separated, $p_{a_i} = p_{b_i}$.

$x_i = y_i$. □

Claim 16. Given outcome q , the obedience and separation constraints hold for player i .

Proof. Take $a_i \in \text{supp}_i(q)$. By Claim 15, there exists a unique $x_i \in X_i$ such that $\sigma_i(a_i|x_i) > 0$. Thus, $q_{a_i} = q_{x_i}$. By definition of σ_i , $a_i \in BR(q_{x_i})$. This demonstrates that the obedience constraint is satisfied for player i .

To verify the separation constraint, take $a_i, b_i \in \text{supp}_i(q)$ such that $q_{a_i} \neq q_{b_i}$. Take $x_i, y_i \in X_i$ such that $\sigma_i(a_i|x_i) > 0$ and $\sigma_i(b_i|y_i) > 0$. Since $q_{a_i} \neq q_{b_i}$, the signals x_i and y_i must be distinct (Claim 15). Take $c_i \in x_i$ and $d_i \in y_i$. As noted above, $p_{x_i} = p_{c_i}$ and $p_{y_i} = p_{d_i}$. It follows from $x_i \neq y_i$ that $p_{c_i} \neq p_{d_i}$. Since p is separated, $BR(p_{c_i}) \cap BR(p_{d_i}) = \emptyset$. It follows from (34) that $BR(p_{x_i}) \cap BR(q_{x_i}) = \emptyset$. We conclude that the separation constraint is satisfied for player i . □

Claim 17. The outcome q is not informationally simple.

Proof. We show i 's action is not independent of j 's state. Given $x_i = [a_i^*]$, take a_i so that $\sigma_i(a_i|x_i) = 1$. Let $q_{a_i}(E_j)$ and $q_{x_i}(E_j)$ be the conditional probabilities of event E_j given a_i and x_i , respectively. By Claim 15, $q_{a_i}(E_j) = q_{x_i}(E_j)$. Thus,

$$q_{a_i}(E_j) = q_{x_i}(E_j) = \frac{(1 - \epsilon)p(x_i)\pi_j(E_j) + \epsilon\pi_j(E_j)}{(1 - \epsilon)p(x_i) + \epsilon\pi_j(E_j)}$$

where $p(x_i) = \sum_{a_i \in x_i} p(a_i)$. We conclude that $q_{a_i}(E_j) > \pi_j(E_j)$. This shows that i 's action is not independent of j 's state. □

Claim 18. Given outcome q , obedience and separation constraints hold for every player $j \neq i$.

Proof. Fix a player $j \neq i$. Notice that p and q induce the same marginal distributions on $A_j \times \Theta_j$. Thus, since the base game is separable, we have that

$$BR(p_{a_j}) = BR(q_{a_j}) \quad \text{for all } a_j \in \text{supp}_j(p) = \text{supp}_j(q).$$

In particular, since p is a BCE, the obedience constraint is satisfied for player j given outcome q . To verify separation, take $a_j, b_j \in \text{supp}_j(q)$ such that $q_{a_j} \neq q_{b_j}$. Note that

for all $a_{-j} \in A_{-j}$ and $\theta \in \Theta$,

$$q_{a_j}(a_{-j}, \theta) - q_{b_j}(a_{-j}, \theta) = \left(\frac{p_j(a_j, \theta_j)}{p_j(a_j)} - \frac{p_j(b_j, \theta_j)}{p(b_j)} \right) \sum_{c_j, \theta'_j} q(c_j, a_{-j}, \theta'_j, \theta_{-j}).$$

Thus, $q_{a_j} \neq q_{b_j}$ implies $p_{a_j} \neq p_{b_j}$. Since p is separated,

$$BR_j(q_{a_j}) = BR_j(p_{a_j}) \cap BR_j(q_{b_j}) = BR_j(p_{b_j}) = \emptyset.$$

This shows that the separation constraint holds for player j given outcome q . $\square$

This concludes the proof that (i) implies (ii).

(ii) implies (iii). We proceed by contraposition. Suppose there is a player i who has no action that is a best response to all beliefs. We will construct a separable BCE that involves some learning for player i .

For every $a_i \in A_i$, let $\Theta_i(a_i)$ be the set of i 's states for which a_i is optimal:

$$\Theta_i(a_i) = \{\theta_i \in \Theta_i : \bar{u}_i(a_i, \theta_i) \geq \bar{u}_i(b_i, \theta_i) \text{ for all } b_i \in A_i\}.$$

We say that an action a_i is *maximal* if there is no $b_i \in A_i$ such that $\Theta_i(a_i) \subset \Theta_i(b_i)$ (strict inclusion). Since A_i is finite, a maximal action exists. In addition, for every $\theta_i \in \Theta_i$, there exists a maximal action a_i such that $\theta_i \in \Theta_i(a_i)$.

Let $X_i \subseteq A_i$ be a set of maximal actions such that $\cup_{x_i \in X_i} \Theta_i(x_i) = \Theta_i$. We can choose X_i so that $\Theta_i(x_i) \neq \Theta_i(y_i)$ for all $x_i, y_i \in X_i$ such that $x_i \neq y_i$. Since no action is a best response to all beliefs, $\Theta_i(x_i) \subset \Theta_i$ for all $x_i \in X_i$. Thus, $|X_i| \geq 2$.

We consider an experiment for player i , denoted as $\xi_i : \Theta_i \rightarrow \Delta(A_i)$, given by

$$\xi_i(a_i | \theta_i) = \begin{cases} \frac{1}{|\{x_i \in X_i : \theta_i \in \Theta_i(x_i)\}|} & a_i \in X_i \text{ and } \theta_i \in \Theta_i(a_i), \\ 0 & \text{otherwise.} \end{cases}$$

With a slight abuse of notation, we denote by $BR(x_i)$ the set of i 's actions that are optimal following signal $x_i \in X_i$:

$$BR(x_i) = \arg \max_{a_i \in A_i} \sum_{\theta_i} \bar{u}_i(a_i, \theta_i) \xi_i(x_i | \theta_i) \pi_i(\theta_i).$$

Claim 19. For every $x_i \in X_i$,

$$BR(x_i) = \{a_i \in A_i : \Theta_i(a_i) = \Theta_i(x_i)\}.$$

Proof. Action x_i is optimal at every state $\theta_i \in \Theta_i(x_i)$. In addition, $\xi_i(x_i|\theta_i) > 0$ if and only if $\theta_i \in \Theta_i(x_i)$. It follows that $x_i \in BR(x_i)$. Moreover, each $a_i \in BR(x_i)$ must also be optimal at every state $\theta_i \in \Theta_i(x_i)$. Thus, $\Theta_i(x_i) \subseteq \Theta_i(a_i)$. Since x_i is a maximal action, $\Theta_i(x_i) = \Theta_i(a_i)$. $\square$

Claim 20. For every $x_i, y_i \in X_i$, if $x_i \neq y_i$ then $BR(x_i) \cap BR(y_i) = \emptyset$.

Proof. By construction, if $x_i \neq y_i$ then $\Theta_i(x_i) \neq \Theta_i(y_i)$. It follows from Claim 19 that $BR(x_i) \cap BR(y_i) = \emptyset$. $\square$

We now use ξ_i to construct a separable BCE that involves some learning for player i . For every player $j \neq i$, let $\alpha_j \in \Delta(A_j)$ be an action distribution supported by

$$\arg \max_{a_j \in A_j} \sum_{\theta_j} \bar{u}_j(a_j, \theta_j) \pi_j(\theta_j).$$

Define an outcome $p \in \Delta(A \times \Theta)$ by

$$p(a, \theta) = \xi_i(a_i|\theta_i) \pi_i(\theta_i) \prod_{j \neq i} \alpha_j(a_j) \pi_j(\theta_j).$$

It follows from Claims 19 and 20 that p is a separable BCE. In addition, each action recommendation $x_i \in X_i$ allows player i to rule out any state $\theta_i \notin \Theta_i(x_i)$. Hence, p involves some learning for player i .

(iii) implies (i). Let p be a separable BCE. For every $i \in I$ and $a_i, b_i \in \text{supp}_i(p)$, it follows from (iii) that $BR_i(p_{a_i}) \cap BR_i(p_{b_i}) \neq \emptyset$. The separation constraint implies that $p_{a_i} = p_{b_i}$. This shows that p involves no learning, and therefore p is informationally simple.

H. Proof of Proposition 4

We begin with a preliminary lemma.

Lemma 19. Suppose $|I| = 2$ and $|\text{supp } \pi| \geq 2$. Let $u = (u_1, u_2)$ be a utility profile, and let B_1 and B_2 be the set of actions that are not strictly dominated for player 1 and player 2, respectively, where $|B_1| \geq |B_2| \geq 2$. A BCE $p \in \text{BCE}(u)$ exists such that $p(a_1, \theta) = 0$ for some $a_1 \in B_1$ and $\theta \in \text{supp } \pi$.

Proof. Without loss of generality, assume $\Theta = \text{supp } \pi$. Let $p \in \text{BCE}(u)$ be induced by a Bayes-Nash equilibrium with the following (exogenous) information technology: player 1 sees the state and player 2 gets a degenerate signal. Let $p_1 \in \Delta(B_1 \times \Theta)$ be the marginal of p on $B_1 \times \Theta$, and let $p_2 \in \Delta(B_2)$ be the marginal of p on B_2 . Since player 2's action is independent of the state and player 1's action, we can write $p(a_1, a_2, \theta) = p_1(a_1, \theta)p_2(a_2)$ for all $a_1 \in B_1$, $a_2 \in B_2$, and $\theta \in \Theta$.

Without loss, we focus on the case where $p_1(a_1, \theta) > 0$ for all $a_1 \in B_1$ and $\theta \in \Theta$; otherwise, we are done. Since player 1 sees the state, we get that player 2's behavior must make player 1 indifferent between all (undominated) actions in all states. On the other hand, player 2's incentive compatibility constraints can be expressed as follows: fixing some $\hat{a}_2 \in \text{supp}(p_2)$, p_1 is a solution of the linear system

$$\sum_{a_1, \theta} [u_2(a_1, a_2, \theta) - u_2(a_1, \hat{a}_2, \theta)] p_1(a_1, \theta) = 0 \quad \forall a_2 \in \text{supp}(p_2) \setminus \{\hat{a}_2\}, \quad (35)$$

$$\sum_{a_1, \theta} [u_2(a_1, a_2, \theta) - u_2(a_1, \hat{a}_2, \theta)] p_1(a_1, \theta) \leq 0 \quad \forall a_2 \in B_2 \setminus \text{supp}_2(p_2), \quad (36)$$

$$\sum_{a_1} p_1(a_1, \theta) = \pi(\theta) \quad \forall \theta \in \Theta. \quad (37)$$

Because p_2 results in player 1 being indifferent between all actions in all states, and because π sums up to 1, any $q_1 \in \mathbb{R}_+^{B_1 \times \Theta}$ that solves the above system induces a BCE q via $q(a_1, a_2, \theta) = q_1(a_1, \theta)p_2(a_2)$. Thus, to prove the lemma, it is sufficient to show that the system admits a non-negative solution with at least one zero entry. We next verify it.

Let M be the coefficient matrix of the system (35)–(37). The matrix has $|B_2| + |\Theta| - 1$ rows and $|B_1| \cdot |\Theta|$ columns. Since $|B_1| \geq |B_2| \geq 2$ and $|\Theta| \geq 2$, the columns are strictly more than the rows. Thus, M cannot have full column rank. By the rank-nullity theorem, there is a non-zero vector $r_1 \in \mathbb{R}^{B_1 \times \Theta}$ such that $Mr_1 = 0$. For each $\gamma \in \mathbb{R}$, the vector $q_1 = p_1 + \gamma r_1$ solves the system (35)–(37). Now we show that we can choose γ so that q_1 is non-negative with at least one zero entry.

Let $\hat{\gamma}$ be the supremum of the set of all γ such that $p_1(a_1, \theta) + \gamma r_1(a_1, \theta) \geq 0$ for all

$a_1 \in A_1$ and $\theta \in \Theta$. Since p is non-negative, it is feasible to choose $\gamma = 0$; thus, $\hat{\gamma} \geq 0$. Moreover, $\hat{\gamma} < +\infty$; this is because $r_1(a_1, \theta) < 0$ for some $a_1 \in A_1$ and $\theta \in \Theta$, which follows from (37) and $r_1 \neq 0$. Finally, $p_1(a_1, \theta) + \hat{\gamma}r_1(a_1, \theta) = 0$ for some $a_1 \in B_1$ and $\theta \in \Theta$; otherwise, it would be possible to raise $\hat{\gamma}$ and still satisfy the non-negativity constraints. Setting $q_1 = p_1 + \hat{\gamma}r_1$, we obtain the desired result. $\square$

We now proceed to proving Proposition 4. Throughout, we assume there are exactly two players, $I = \{1, 2\}$. This assumption is without loss: one can always focus on utility functions whereby all other players have a strictly dominant action. Similarly, it is without loss to assume that both players have exactly two available actions, (again, one can make all other actions strictly dominated). We also assume π has full support.

Let $\bar{u} \in \mathbb{R}^{A_1 \times A_2}$ be the utilities associated with the matching pennies game. Let $q \in \Delta(A)$ be the game's unique correlated equilibrium (i.e., the Nash equilibrium where each player chooses each action with probability 0.5 independently of the other player). Let $u^* = (u_1^*, u_2^*)$ be a utility profile for the ambient incomplete information game such that $u^*(a, \theta) = \bar{u}(a)$ for all $a \in A$ and $\theta \in \Theta$. It is easy to verify that $p \in \Delta(A \times \Theta)$ is a BCE of the game induced by u^* if and only if (i) $\sum_{\theta} p(a, \theta) = q(a)$ for all action profiles a , and (ii) $\sum_a p(a, \theta) = \pi(\theta)$ for all states θ .

For every $\epsilon > 0$, let $B_{\epsilon}(u^*)$ be the set of utility profiles u that are within ϵ of u^* , where the distance between u and u^* is measured by the sup-norm. Below we show that an $\epsilon > 0$ exists such that

$$\inf_{u \in B_{\epsilon}(u^*)} \sup_{p \in BCE(u)} \mathbb{I}(p) > 0. \quad (38)$$

Before proving (38) holds for some $\epsilon > 0$, we argue this inequality is sufficient for proving the proposition. For this purpose, take any $b < \epsilon/2$, and consider the set

$$V_b = \{u \in B_{2b}(u^*) : u_i^*(a, \theta) - u_i(a, \theta) \in (0, b) \ \forall (a, \theta), \ i\}.$$

Notice that V_b is open, and that every $u \in V_b$ and every feasible transfer t satisfy $u + t \in B_{\epsilon}(u^*)$. Therefore, if $\text{cl}(iBCE(u + t)) = BCE(u + t)$, we have by (38) that $\sup_{p \in iBCE(u+t)} \mathbb{I}(p) > 0$. But setting $t^* = u^* - u$ gives

$$\sup_{p \in iBCE(u+t^*)} \mathbb{I}(p) = \sup_{p \in iBCE(u^*)} \mathbb{I}(p) = 0.$$

The proposition obviously follows.

Thus, all that remains is to show (38) holds for some $\epsilon > 0$. By way of contradiction, suppose a sequence of utility profiles $(u^n)_{n=1}^\infty$ exists such that

$$\lim_{n \rightarrow \infty} u^n = u^* \quad \text{and} \quad \lim_{n \rightarrow \infty} \sup_{p \in BCE(u^n)} \mathbf{I}(p) = 0.$$

By Lemma 19, for every n , we can choose $p^n \in BCE(u^n)$ such that p_1^n , the marginal of p^n on $A_1 \times \Theta$, does not have full support. Since the sets A_1 and Θ are finite, by passing to a subsequence we may assume that there are $a_1 \in A_1$ and $\theta \in \Theta$ such that $p^n(a_1, \theta) = 0$ for all n . Since the set $\Delta(A \times \Theta)$ is compact, by passing to a further subsequence we may assume that the sequence $(p^n)_{n=1}^\infty$ converges to some p . Since the BCE correspondence is upper hemicontinuous, $p \in BCE(u^*)$.

A fortiori, $\text{marg}_{A_1 \times \Theta} p(a_1, \theta) = 0$. Since $p \in BCE(u^*)$ and q has full support, $p(a_1) > 0$. This shows that $\mathbb{I}(p) > 0$. At the same time, however, since $\mathbb{I}$ is lower semicontinuous, we have

$$0 = \lim_{n \rightarrow \infty} \mathbb{I}(p^n) \geq \mathbb{I}(p) > 0.$$

We reached a contradiction.